

Customary Tenure and Agricultural Investment in Uganda

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Abstract

Customary tenure in Uganda often features land rights being held by different individuals in a community, creating complex incentives for long-term agricultural investment. I introduce a theoretical model of these incentives, exploring farmers' strategic responses to changing land pressures, which may change the social determinants of tenure insecurity. Empirically, I find results consistent with the model in Uganda. As land pressures rise, long-term input use responds more on freehold than on customary parcels. This is consistent with elite incentives for expropriation, which intensify with external demand for land, creating insecurity on customary parcels. This study offers a nuanced conceptualization of the relationship between tenure systems and agricultural investment, incorporating social dynamics (such as elite expropriation) in both theory and empirics.

1 Introduction

Secure property rights are fundamental to investment and prosperity, and land tenure in particular is critical for poor farmers to make long-term investments and improve agricultural productivity (Place, 2009; Aberra and Chemin, 2021). In much of Sub-Saharan Africa, customary tenure, a socially-created and enforced, typically undocumented form of land rights, predominates (Alden Wily, 2011). Economists have traditionally focused on the undocumented and therefore theoretically less secure nature of customary tenure when compared with freehold land, where the owner enjoys all rights in perpetuity (Fenske, 2011). Large-scale

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formalization policies across much of the continent have aimed to address this insecurity to allow farmers to make transformative investments in their own farms (Ali et al., 2016; Lawry et al., 2023). However, empirical work has been more mixed than theory; customary land is not always less secure in farmers’ own estimation (nor under-invested relative to freehold land), and formalization has not always unlocked investment (Bambio and Bouayad Agha, 2018; Lawry et al., 2017; Fenske, 2011; Higgins et al., 2018; Boxho et al., 2022; Bellemare, 2013). By contrast, qualitative work has emphasized the socially-constructed nature of customary tenure, where multiple individuals may hold overlapping rights over a single piece of land, and in particular, local elites such as chiefs or lineage heads hold transfer rights (Cotula et al., 2007; Ensminger, 1997; Alden Wily, 2011; Van Leeuwen, 2014; Barry and Danso, 2014).¹ The distribution of rights can therefore influence the (perceived) security of tenure (Goldstein and Udry, 2008), as other rightsholders may claim the land, although the structure of rights are distinct from the security of those rights (Le Rossignol et al., 2024). Often framed around issues of elite capture, these patterns have been difficult to document in quantitative survey data, although their importance is highlighted in qualitative work on land disputes and dispossession (Kandel, 2022).²

I incorporate the social dynamics of land rights, where multiple rightsholders interact strategically, into a theoretical model of agricultural investment and empirical tests using data from Uganda. My model’s primary contribution lies in capturing not just different investment incentives under different tenure regimes, but how these incentives change in response to external land pressures. As land pressures increase, the relative output price of (input-intensive) cash crops should increase, increasing investment, but local elites may face more incentive to expropriate customary land for sale to outsiders (Cotula et al., 2007; Ubink, 2008; Berry, 2018; Bambio and Bouayad Agha, 2018).³ If farmers begin to worry about elite expropriation,⁴ this should impact their investment decisions. In particular, I find that farmers under-invest in long-term

¹A hallmark of the customary is its very localized, negotiated nature; this enables its adaptability to changing circumstances but can also marginalize those in weaker bargaining positions (Cotula et al., 2007; Lawry et al., 2023; Chimhowu, 2019). Ensminger (1997) argues that “a common characteristic in almost all African customary systems is for use rights to be assigned at the household level, whereas transfer rights are assigned at a higher level such as the lineage, clan, or chiefdom.” This means that although individuals may be quite secure in their ability to use a particular plot of land in their lifetime, adaptive reallocation within the community could occur in response to changing circumstances. Customary leaders also are able to allocate (often unused) land to migrants into an area as well as adjudicate any conflicts that may occur between members of the community. Although the particular ways these rights are divided may vary considerably across the region (Hasanbasri et al., 2022), this ‘dis-bundling’ of property rights is common and adds a strategic dimension to any decisions about land. At the same time, however, the dis-bundling does not closely mirror a sharecropping contract, as the elite holding transfer rights does not hold residual profits and there is no formal contract between multiple rightsholders; a classic principal-agent model with potential holdup problems is therefore of limited relevance (Aghion and Tirole, 1997; Goetz et al., 2019).

²This more localized dispossession is distinct from the large scale land acquisitions that have received international attention, although the phrase ‘land grabs’ is used as a catch-all term (Van Leeuwen et al., 2023).

³Kandel (2022) documents that those with off-farm salaried income are “increasingly acquiring rural land in order to invest in small-scale commercial agriculture or as speculative investments,” a phenomenon termed ‘emergent farmers’ by Sitko and Jayne (2014). Indeed, Lawry et al. (2023) describes greater demand for land in response to population growth as one of the primary threats to customary tenure systems. This ability for local land elites to take advantage of new economic situations and their better connections to outsiders has been documented in other contexts, including Pakistan (Beg, 2021).

⁴Traveling around Uganda, it is common to see signs on properties especially agricultural land stating “this land is not for sale”: an indication that farmers are indeed worried about elite expropriation for sale to outsiders. Newspapers also report on traditional leaders registering customary land for their own private profit (Observer Staff, 2019).

agricultural inputs on customary parcels (when compared with freehold) as land pressures rise.

I begin with an endogenous model of customary tenure and agricultural investment where farmers can make socially-recognized investments in their land to enhance their tenure security, with inputs applied to both staple and cash crops. Long-term investments are weakly lower on (relatively insecure) customary land than on freehold land, although the customary system may provide sufficient investment incentives (Brasselle et al., 2002; Lawry et al., 2023). This connects with empirical work linking customary tenure arrangements to a (lack of) adoption of cash crops, including Deininger et al. (2021)⁵ and Goldstein et al. (2018)⁶. New pressures on land values, which often emerge as a result of increased access to urban centers, also raise the relative price of input-intensive cash crops. This implies that optimal investment is an (increasing) function of land pressures on freehold land. However, I explicitly consider a case common in customary tenure systems: one household or individual may hold primary use rights, while another, such as a lineage elder, may hold rights of transfer (Ensminger, 1997; Cotula et al., 2007).⁷ This partition of rights means that outsiders may struggle to identify land ‘owners’, finding local tenure systems illegible (difficult to parse without deep knowledge of the area and people nor the socially-recognized investments that demonstrate and secure tenure locally). Elites are then able to sell land used by other members of the community for their own private benefit (Ubink, 2008; Sitko et al., 2014). Their incentives to do so are greater in contexts with rapidly increasing land values (Barry and Danso, 2014; Cotula et al., 2007). This means that rising land values, driven by sales options to outsiders, may lead local elites to assert their historic right to sell land with little or no advance notice, creating insecurity for smallholders (Perego, 2019). This expropriation is a real threat for Ugandans: the Prindex survey of Uganda finds that 22.4% of respondents report personally experiencing any involuntary land loss (Prindex, 2020), and qualitative work finds that dispossession by local elites is more likely for the poor and socially vulnerable (Kandel, 2022).⁸ The model predicts that the farmer, anticipating elite expropriation, may actually feel more tenure insecure and therefore make fewer long-term investments on customary land as land values rise, in contrast to the freehold case.

In order to empirically test the implications of this model, I use data from a survey of 2,189 farmer households in Uganda. My primary empirical specification looks at heterogeneous responses to rising land pressures between customary and freehold land for a variety of investments of long and short duration.

⁵Who find that bequest and sales rights, particularly those held by men, are strongly related to cash crop adoption; these long-term rights are more important than short-term decisionmaking rights.

⁶Who find that households who received demarcation of their customary parcels in Benin were 23-48 percent more likely to grow perennial cash crops.

⁷I focus on these two rights given their first-order importance for investment incentives, despite the plethora of other secondary rights that may be held by various individuals.

⁸The public Prindex data does not allow us to empirically examine where these expropriations happen, however, nor the kind of land which was expropriated. Nevertheless, it is a valuable resource for understanding the prevalence of tenure insecurity and land disputes in Uganda: 16.6% of respondents report having experienced a dispute over land in the past, and 26% of the adult population report feeling insecure in their land rights.

In order to capture an exogenous component of land pressures, I use two spatial measures derived from Muller-Crepon (2021). The first is the inverse travel time to Kampala, the national capital and source of much outsider demand for land, which incentivizes elite expropriation but also increases the relative price of input-intensive cash crops. The second is the historical decrease in travel time to Kampala due to new or improved roads, relative to the 2005 travel time. This isolates new incentives to expropriate customary land, which play a role in uncertainty and tenure insecurity. These measures should capture the degree to which outsiders are interested in purchasing local land, and therefore the drivers of elite expropriation, but abstract away from other pathways by which land values impact agricultural investment. I also use estimated per-acre rental prices for parcels; with this measure, I control for farmer group fixed-effects to account for unobserved farmer quality characteristics.⁹ The empirical approach makes every effort to identify causal impacts of exogenous land pressures on customary and freehold parcels within the limitations of the cross-sectional data, but the most conservative interpretation is that I find conditional correlations that are consistent with the detailed predictions of my theoretical model.

Empirically, I find that long-term input use responds to rising land values more strongly on freehold land than on customary land. Elite incentives to expropriate rise along with land pressures, causing tenure insecurity on customary land (seen empirically)¹⁰, and this attenuates the impacts of a shift to input-intensive cash crops. This divergence in input use between freehold and customary parcels (the novel prediction of my model) is smaller for short-term inputs, perhaps due to additional demands for cash crops or frictions in land and labor markets (Adamopoulos et al., 2022). Nevertheless, the difference in responses is stronger for long-term inputs, which can be attributed to the tenure-security effect. Additionally, the divergence in investment is more responsive to rising land pressures closer to Kampala.

By incorporating the social nature of customary land rights in a model of agricultural investment, this paper avoids the common conflation of incompleteness of rights with insecurity (Doss and Meinzen-Dick, 2020; Hasanbasri et al., 2022; Place, 2009; Le Rossignol et al., 2024).¹¹ Instead, it considers how multiple rights-holders interact in different land value environments to make more nuanced predictions about investment

⁹These farmer groups were part of the sampling strategy in the data collection process. They extend the intuition of household fixed-effects to control for unobserved heterogeneity as farmer groups are self-selected, but allow for sufficient identifying variation. A household fixed effects specification looking at differences across tenure types (particularly in West Africa) may identify from households with the greatest differences in tenure security (and thus households most likely to invest differently across plots), as only those households who feel insecure about land allocated through customary mechanisms will purchase other plots (Fenske, 2011).

¹⁰I am unable to distinguish sources of insecurity in the data, but other sources, such as inheritance disputes, impact freehold parcels as well.

¹¹'Security' could be composed of multiple elements of duration of tenure, assurance, and completeness of rights. Despite the theoretic clarity of tenure security, "there is no agreed upon way to measure tenure security and results may be related to choice of proxy" (Place, 2009). For example, Bellemare (2013) finds in Madagascar that one measure of tenure security, the presence of title, seems unimportant for investment, while beliefs about rights do matter, while Deininger et al. (2021) in matrilineal areas of Malawi that rights to bequeath and sell land are more important than short-term management rights. Gumucio et al. (2025) differentiates robustness, completeness, and durability of rights; in this paper, I consider how changing external circumstances such as land pressures can shape the robustness of customary rights, and thereby reduce their durability.

on customary land, reconciling the mixed evidence in the literature (Fenske, 2011). In addition to bringing qualitative insights to an economic framework, I use economic modeling to explore how farmers would strategically respond to changing incentives. This model captures how the potential for elite capture affects individual farmers, similarly to Aberra and Chemin (2021). It thereby returns the focus to the welfare of smallholders, and by examining a broader population allows me to quantitatively document the effects of (potential) elite capture (which historically has been difficult due to the rarity of observed cases). Unlike many previous papers which have conceptually modeled the impacts of tenure *security* but then empirically used tenure *type* as a (poor) proxy, I consider the particular institutional arrangements of tenure type and the incentives it creates throughout my model, therefore linking more closely to my empirical tests (Fenske, 2011; Abdulai et al., 2011; Linkow, 2019). Uganda’s tenure context, where customary and freehold parcels exist in the same local areas due to exogenous historical factors (Deininger and Ali, 2008), also allows me to isolate the impact of land pressures on tenure insecurity from structural issues such as land and labor market frictions leading to misallocation.¹² These potentially missing markets are common across parcels in a given location, and therefore are unlikely to drive divergence between parcels of different tenure types in an area.

The remainder of this paper consists of a description of the context of customary tenure and agricultural investment in Sub-Saharan Africa, with a particular focus on Uganda, followed by a model which captures many of the insights explored. I then discuss the data used, as well as an empirical strategy with several hypotheses laid out. I present results for a variety of short- and long-term inputs, and conclude by discussing their implications for the model’s relevance to the Ugandan context.

2 Context

Importantly, the dispersion of rights in many customary tenure systems across the continent is not in itself a cause of instability or under-investment (Brasselle et al., 2002). Within a local community, farmers are able to demonstrate their stewardship of the land via socially-recognized investments, such planting trees, that signal socially that the land is being used (Awanyo, 2009). However, the continual adaptation of customary institutions means that such systems are dynamic and see considerable changes over time. My model explores how outside, market-based pressures on land shape customary tenure security and investment incentives.

In particular, I focus on the role of elites, such as lineage heads, (land) chiefs, or elders. In customary

¹²The restrictions on land transferability, as in Adamopoulos et al. (2022) and Adamopoulos and Restuccia (2019), or the sectoral misallocation of labor, as in Adamopoulos et al. (2024), do lead to underinvestment in the land and lower agricultural productivity, and could explain some of the gap I observe for all investments between customary and freehold parcels. However, these market frictions do not explain a differential gap between short and long-term inputs, nor that differential gap widening with increasing land pressures, which I document and which is predicted by my model as stemming from insecurity.

systems across Uganda and the continent, elites, representing the family or lineage writ large, serve as trustees of land owned by the family group (Ensminger, 1997). Often, they hold transfer rights in order to (re)allocate land within the community in response to shocks (such as the death of a large landholder or an influx of new migrants to an area), as well as to prevent the alienation of a communal resource.

However, states have attempted to fit notions of ownership based on freehold tenure onto customary systems, and this institutional mismatch has created ambiguity about who the ‘owners’ of land actually are. This ambiguity can be exploited by local elites: “There is a fine line between chiefs as (often self-declared) owners of all land in customary laws, and chiefs as trustee administrators” (Alden Wily, 2011). Elites are able to respond as individuals rather than as guardians of the corporate group, and can thus assert their rights to outsiders unfamiliar with the complexities of local tenure arrangements.

The qualitative literature has documented tensions and conflicts that emerge as customary systems adapt to pressures such as rising land values.¹³ “As land values rise, farmers may be forced or tempted to sell their land. Where land is still under customary chiefs, these may be tempted to sell off lands ... regardless of the views of those actually farming this land” (Cotula et al., 2007). In eastern Uganda, Kandel (2022) documents a land dispute case: “Social differentiation in Teso has increased since the culmination of conflict and local elite land accumulation is a key aspect to this process. Often drawing on off-farm salaried income, local elites are increasingly acquiring rural land in order to invest in small-scale commercial agriculture or as speculative investments... Certain villages also report that wealthier and politically connected people are coercively acquiring land and dispossessing poorer and more vulnerable residents in the process.” Despite abundant stories of how “local elites have been able to use their position and the ambiguities of customary law to appropriate land to further their own economic and political interests” (Ubink, 2008), this particular facet of how customary tenure adapts to external pressures has been little studied by economists, with the notable exception of Berry (2018)’s work on land governance in these areas.

2.1 Land Rights in Uganda

Between 12-20% of land in Uganda is subject to formal title, so customary tenure remains significant throughout the country (Alden Wily, 2011; Goldstein et al., 2016), and customary law is recognized under the Constitution (Foote, 2025).¹⁴ Customary authorities still play a major role in Uganda: Gochberg (2021) argues

¹³Most cases have considered land values rising due to *non-agricultural* uses, such as urban and peri-urban expansion, or natural resource extraction; this also includes the ‘emergent farmers’ documented by Sitko and Jayne (2014) who are more commercially-oriented but no more productive than smallholders. Some large scale land acquisitions have been for agricultural purposes, although generally requiring scale or technologies such that local smallholders are unable to participate (Smalley and Corbera, 2012; Purdon, 2014). Therefore, this should be thought of as an increase in the *marketable* value of land rather than its value to smallholder agriculture.

¹⁴This is slightly higher than for the continent as a whole, in part due to colonial administration which focused on documenting land allocation in Uganda.

individuals may choose not to title their land given the social costs of leaving the customary system (echoing Honig (2022)’s work in other countries on the continent).¹⁵ This customary land is subject to the kind of elite expropriation I model: Adoko and Akin (2011) frames the likelihood of elite land grabbing as a function of physical or ‘social strength’ in the community and family, and Gochberg (2021)’s qualitative work relates elites can “go so far as to survey another individual’s land at night.”

Even within Uganda, there is some variation in the relative strength and allocation of bundles of rights. In the west, where population pressures on the land are higher, there is more individualized customary tenure, while the relatively land-abundant north retains stronger rights for customary authorities (Van Leeuwen, 2014), in part due to historical agropastoral systems and collective rights¹⁶. In the historic Kingdom of Buganda, in the center of the country, the *mailo* system is key, a remnant of British indirect colonial rule.¹⁷ *Mailo* land only exists in the historic kingdom of Buganda, however, and even there, customary land is still substantial.¹⁸ A given smallholder’s likelihood of having their land expropriated also varies, particularly with their social status within the community (Goldstein and Udry, 2008; Cherchi et al., 2019; Poulin and Pierotti, 2024; Linkow, 2019).

Qualitative work has documented throughout the country that rights to the land in perpetuity are differentiated from responsibility for managing the land (Adoko and Akin, 2011). Specific land rights regimes can be incredibly precise: Howard and Nabanoga (2007) document that rights are determined for individual people over individual plants on different types of land under different circumstances.¹⁹ I focus on the first-order rights of land use/management and transfer rights, which have the most direct impacts on agricultural investment. The precise institutional details shape the identities of the different rightsholders in a region, how elites are selected (hereditary or elected), and in what other domains they have power. Nevertheless, the fundamental distinction I model is common across the country: on customary land, although the farmer has use rights, a customary elite will hold transfer rights.

Deininger and Castagnini (2004) estimate that up to 5% of the rural population in Uganda is involved in a

¹⁵In lab-in-the-field experiments, references to traditional authorities such as the *kabaka* (the king of Buganda, one of the major regions in Uganda) induce higher contributions to public goods games (Goist and Kern, 2018).

¹⁶Post-war resettlement and return additionally complicates customary systems in the north (Foote, 2025; Sjögren, 2014) and southwest (Mathenge, 2013)

¹⁷Much of the academic literature on land rights in Uganda has looked at the *mailo* system. In 1900, the British signed a treaty with the Kingdom of Buganda which allocated square-mile tracts in the center of the country to Buganda elites for their ‘ownership’ as absentee landlords. These tracts were then sub-leased to the actual inhabitants and land users (Deininger and Castagnini, 2004). Ever since, there has been a significant tension over government policy towards *mailo* owners and tenants and how to balance the two parties’ competing interests (Coldham, 2000). The 1998 Land Act attempted to balance these by giving *mailo* owners the powers of a freehold owner, while still recognizing the rights of ‘lawful occupants’ of the land who had used it for more than 12 years (Coldham, 2000). *Mailo* tenants still face substantial tenure insecurity, particularly regarding expropriation by owners.

¹⁸My survey includes one district with a substantial share of *mailo* land; 20% of parcels there are reported as *mailo*, with less than 1% reported similarly in any other surveyed district. However, a nearly identical share of parcels in that district are reported as under customary tenure. For this analysis, I categorized *mailo* parcels (reported by tenants) as customary land, although results are robust to their exclusion from the analysis entirely, as seen in appendix C.8.2.

¹⁹For instance, that a woman who is pregnant or sick is allowed to pick certain medicinal plants from someone else’s river plot, but not from a home garden.

pending land-related conflict at any time, mostly between neighbors about boundaries. More recent estimates from Prindex find that 16.6% of the population report ever having experienced a dispute, with 22.4% having personally experienced the involuntary loss of land (Prindex, 2020). The prevalence of boundary disputes has led to many farmers planting trees to demarcate their land; “boundary trees are useful evidence when the land can be visited during a case by customary authorities, but they are less useful when the case is heard in a court of law far away from the land” (Adoko and Akin, 2011). There has been a push to incorporate customary leaders into statutory court systems, with mixed success (Van Leeuwen, 2014).

The law in Uganda recognizes four tenure types (freehold, leasehold, customary, and *mailo*) (Joireman, 2007) and provides for an extensive and decentralized land administration (Tripp, 2004) that could issue titles as well as ‘certificates of customary ownership.’ These certificates voluntarily registered rights held by multiple people and could, in time, be converted into freehold titles (Coldham, 2000; Perego, 2019). Coldham (2000) observes that “where the grant of certificates of title is based on individual applications, there is always a risk of land-grabbing, that is, that an applicant may lay claim to a larger area of land than that to which he is customarily entitled.” Furthermore, the widely-lauded land reform has been largely unfunded and thus little-implemented (Deininger and Castagnini, 2004; Joireman, 2007).²⁰

Two previous economic analyses have looked at the impacts of tenure on agricultural investment in Uganda. Deininger and Castagnini (2004) focus on land conflicts²¹; Deininger and Ali (2008) find that households invest in both long-term and short-term inputs significantly more on plots they own outright rather than plots they have only customary rights over. Furthermore, they find that registering customary rights (by acquiring a certificate of customary ownership) has little impact on investment, but legal efforts to strengthen occupancy rights do increase investment.²²

3 A model of tenure and investment

The simplest theoretical models of land tenure and investment attempt to capture tenure as an exogenous probability of losing the land, along with any fixed investments, before the profits of those investments can be fully realized.²³ However, much qualitative evidence on customary tenure regimes in Sub-Saharan

²⁰Individuals do seem receptive to titling: Cherchi et al. (2019) found in Uganda that 91% of households chose to title their land when offered a fully-subsidized freehold title, and Perego (2019) finds that farmers choose to title land in response to agricultural price shocks, particularly with greater market access and tenure insecurity. Perego (2019) argues that this demand for titling is driven by improved liquidity to pay titling fees from higher crop revenues as well as a desire to protect newly more valuable land from potential land grabs.

²¹They argue that the intensive multi-cropping system prevalent in much of Uganda implies individual crop production functions are inappropriate for productivity measures or investment.

²²They find that investment increases with length of occupancy, with a discontinuity at the 12 year mark when rights become recognized by law.

²³These models often make this assumption to instead focus attention on other features of the investment context (Jacoby and Minten, 2006), such as intra-household bargaining (Dillon and Voena, 2015).

Africa has stressed the endogeneity of tenure security, which has been incorporated into quantitative models. Actions taken by the farmer, particularly certain investments in the land, demonstrate responsible use and defend against expropriation by others in the community who recognize these investments as land stewardship (Awanyo, 2009; Bambio and Bouayad Agha, 2018).²⁴ The visibility of investment to others in the community is key, for tenure security is fundamentally about social recognition of rights (Deininger and Jin, 2006). Place and Otsuka (2002) contrast three possible investments: planting trees, which is visible and thus reduces the probability of losing the land; management effort, which is invisible and thus has no effect on tenure security; and fallowing, which in a tenure system predicated on land *use*, may actually increase the risk of expropriation.²⁵

I now turn to my own theoretical model of tenure and investment, explicitly incorporating changing incentives for expropriation on land under different tenure systems. This model will allow me to derive reduced-form comparative statics, and from those, empirical predictions; I leave the agronomic production function general rather than formally characterizing it.

Consider a two-period model. In the first period, a farmer chooses to apply a short-term input, fertilizer, F_1 , and a long-term one, trees, T ,²⁶ separately to a staple crop s and cash crop c planted on his or her exogenously given land L under tenure system h in order to produce according to $f_s(F^s, T^s)$ and $f_c(F^c, T^c)$.²⁷ A fraction $0 \leq \alpha \leq 1$ of the land is devoted to the staple crop, with the remainder $(1 - \alpha)$ used for the cash crop; this share is taken as given for the input use decisions.²⁸ The farmer also has some initial wealth endowment w_1 . The farmer has a discount rate β for second period returns (the optimized expected second-period total profit is given by π_2^*). There is some probability that the farmer’s land will be expropriated before the second period, in which case fixed investments (trees) would also be lost. This probability is given by $(1 - \phi(T, h, vL))$ (where v is the per-unit value of the land) and may be a function of tree investment, tenure systems, and the value of the land. For convenience, I use the word ‘expropriate’ to capture both

²⁴Indeed, Zamchiya and Musa (2023) find in Zambia that a majority of investments undertaken were made in order to secure the land they were on.

²⁵Goldstein and Udry (2008) focus on this latter effect in their work on tenure security in Ghana, where fallowing is the primary investment in land productivity. They focus on how an individual’s social status interacts with these incentives: “farmers who lack political power are not confident of maintaining their land rights over a long fallow. As a consequence, they fallow their land for much shorter durations than would be technically optimal, at the cost of a large proportion of their potential farm output.” This effect is driven by non-elite households on their land obtained through customary tenure. Social elites do not face the same disincentives to fallow, as they would be the ones to transfer unused land, so empirically they fallow land similarly regardless of how a parcel was acquired.

²⁶These inputs stand in for bundles of short and long-term investments, although trees may have particular tenure-enhancing effects not seen for other long-term inputs.

²⁷I leave this production function general, only assuming diminishing marginal returns to each input ($\frac{\partial^2 f}{\partial F^2} < 0$, $\frac{\partial^2 f}{\partial T^2} < 0$). These assumptions allow me to argue that in cases where the marginal productivity of a given input are lower, the level of that input used is higher, and therefore to predict how investment will respond to changing land values, which is the empirically relevant hypothesis, even without estimating a full agronomic production function which is beyond the scope of this model. The production functions for each crop may differ, and in particular may have different marginal productivities of inputs. This means, however, that the production-side mechanisms should not be interpreted literally.

²⁸The model could be extended to a three period model, where α is chosen in the first stage prior to input use decisions, and retain its primary results.

‘horizontal’ expropriation (where others in the community take over the farmer’s use rights), as well as the exertion of sales rights by local elites without the permission of the user (‘vertical’ expropriation). ϕ is written as a function of vL as well as h , the tenure system, and T , tree investment. In particular, $\phi(T, h, vL)$ could have the following properties (laid out in more detail in appendix A): while freehold land is perfectly secure (and not responsive to either tree planting or land values), customary tenure security is not. The security of customary parcels is increasing in tree planting (which demonstrates responsible stewardship of the land, guarding against horizontal expropriation by neighbors or family), and decreasing in land values (which heighten incentives for elite expropriation for sale to outsiders).

After working through the main features of the model primarily graphically (a fuller, more formal treatment of the model, including assumptions and full solution, is presented in appendix A), I consider what different forms of this ϕ function could imply for the farmer’s decisions as land pressures increase.

The first period optimization problem faced by the farmer is therefore

$$\begin{aligned} \max_{F_1^s, F_1^c, T^s, T^c} \quad & \alpha p_s f_s(F_1^s, T^s) + (1 - \alpha) p_c f_c(F_1^c, T^c) - p_F(F_1^s + F_1^c) - p_T(T^s + T^c) + \beta \phi(T, h, vL) \pi_2^*(T^s, T^c) \\ \text{subject to:} \quad & p_F(F_1^s + F_1^c) + p_T(T^s + T^c) \leq w_1 \end{aligned}$$

In the second period, the farmer only chooses fertilizer (F_2^s and F_2^c) and therefore maximizes according to the following:

$$\begin{aligned} \max_{F_2^s, F_2^c} \quad & \alpha p_s f_s(F_2^s, T^s) + (1 - \alpha) p_c f_c(F_2^c, T^c) - p_F(F_2^s + F_2^c) \\ \text{subject to:} \quad & p_F(F_2^s + F_2^c) \leq w_2 \\ & w_2 = w_1 + \alpha p_s f_s(F_1^s, T^s) + (1 - \alpha) p_c f_c(F_1^c, T^c) - p_F(F_1^s + F_1^c) - p_T(T^s + T^c) \end{aligned}$$

Figure 1 depicts the equilibria conditions for the case where the budget constraint does not bind (that is, the farmer can invest optimally in all inputs in both periods). γ and τ represent the total expected marginal productivity of inputs F and T , respectively; these marginal productivity curves differ for the staple s and cash crop c . The unconstrained farmer will choose inputs F and T for each crop such that the total expected marginal productivity is equal to the marginal cost of the input: at the intersection of the marginal productivity curve with the price of the input.²⁹ For fertilizer, depicted in the two left diagrams, optimal application per unit of land is the same between freehold land (above) and customary land (below); that is, $F^{s1} = F^{s0}$ and $F^{c1} = F^{c0}$.

²⁹If output prices are low enough, or if input prices are high enough, optimal application may be zero, particularly for lumpy investments such as trees. However, the impact of changing land pressures that we will examine should still hold, even if farmers do not cross the threshold to non-zero investment.

3.1 Customary Tenure and Investment

The threat of future expropriation of the land is real for many farmers, particularly on customary land. This expropriation can happen with little to no advance notice, as it takes the form of the sale or physical expropriation of the land by someone other than the use rights holder, rather than the non-renewal of a contract.³⁰ Customary parcels can therefore be thought of as having long duration of tenure (households routinely keep parcels for generations), but that duration is uncertain; the likelihood of losing the land is therefore crucial for long-term investment decisions.

The model captures this risk in two ways. First, customary land has a higher risk of alienation than freehold land: $\phi(T, vL|h=0) < \phi(T, vL|h=1) \forall T, v, L$. This decreases the expected total marginal returns to investment in trees, as shown by a shift towards the origin of τ , decreasing optimal investment in trees.

Drawing upon qualitative understandings of customary tenure, security on customary parcels is endogenous: the probability of losing land is a decreasing function of planting trees (a visible investment) (Place and Otsuka, 2002). If another member of the community tries to encroach on the plot, visible investment decisions can be used as evidence of responsible land use, and thus make it more likely that disputes within the community will be resolved in the farmer's favor. In my model, this implies $\frac{\partial \phi}{\partial T} > 0$, attenuating the leftward shift of the τ curve (and could perhaps even lead to an 'overinvestment' in trees relative to the unconstrained optimum if the tenure-enhancing effect is large, although this is rarely documented).

The equilibrium determined by the marginal productivity of trees under (less-secure) customary tenure, then, likely sees under-investment in trees relative to the allocation under freehold tenure ($T^{s0} < T^{s1}$ and $T^{c0} < T^{c1}$). I will empirically test this using data from Uganda.

3.2 Expropriation Risk and Changing Land Pressures

The model so far has ignored the particular bundles of rights and their distribution among individuals that are hallmarks of customary tenure systems. Primary use rights may be held by one household, while local elites (such as lineage heads or traditional chiefs) often hold transfer rights over large areas of customary land. These decentralized rights do not in themselves cause major tenure insecurity (Bruce and Migot-Adholla, 1994). Mechanisms, such as planting trees, have developed to allow use-holders to demonstrate tenure against other claimants, with local elites serving as arbiters and rights-holders for the community.

In the context of this model, increasing land pressures has two main impacts. First, the relative output

³⁰As Aberra and Chemin (2021) note, the power differentials inherent in elite expropriation also mean that expropriators may not face any sanctions for their behavior. A useful analogy is that of the roving bandit (Olson, 1993): local elites had been functioning as stationary bandits, in an ongoing relationship with the primary use rights holder. The new temptation of sales to outsiders, however, may shift their role to that of a roving bandit who expropriates.

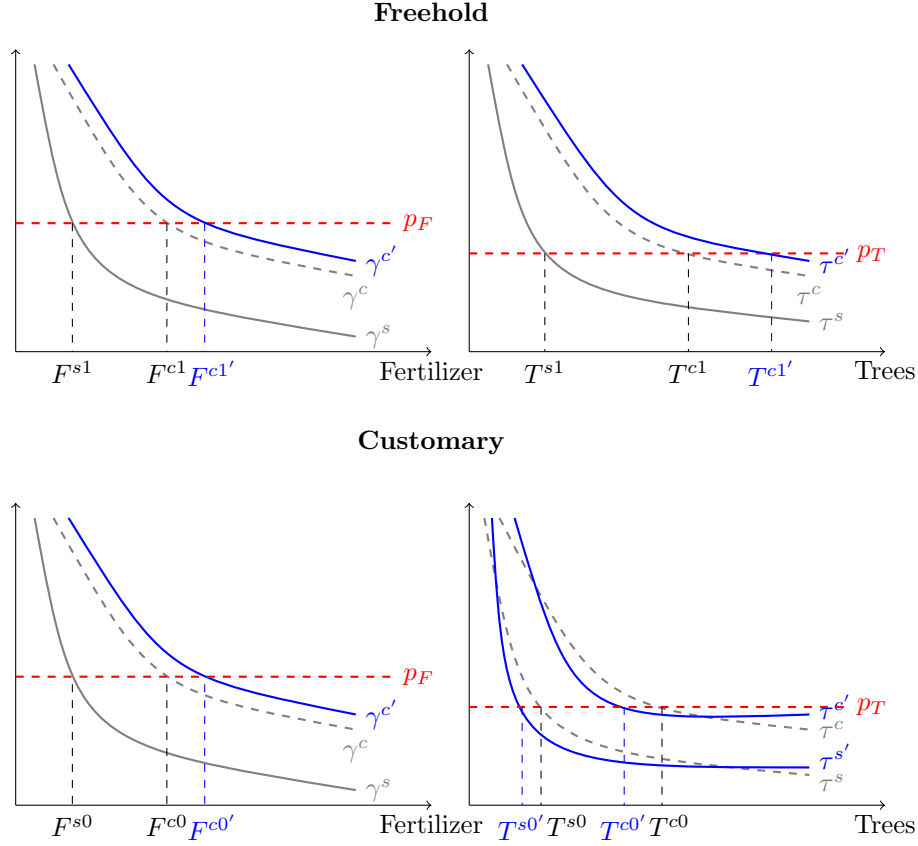


Figure 1: Equilibria conditions for fertilizer (left) and tree (right) investment on freehold (top) vs customary (bottom) land, for staple crops s and cash crops c . Expected total marginal productivity of fertilizer depicted by γ , and of trees depicted by τ , with low land pressure environments in grey and high land pressure environments in blue. Optimal investment is where each curve crosses the dashed line, representing the price of the input. As land pressures increase, the relative output price of cash crops increases, raising the expected total marginal productivity of inputs on the cash crop from the dashed line to $\gamma^{c'}$ or $\tau^{c'}$, and thereby increasing optimal fertilizer and tree application to the cash crop on freehold land and fertilizer application to the cash crop on customary land. On customary land, however, increasing land pressures also increase the likelihood of expropriation, decreasing the expected total marginal productivity of trees on customary land. This results in the changed shape of the marginal productivity curves in the bottom right subfigure, with a lower optimal allocation of trees to both crops on customary land. This means that although optimal fertilizer use on customary parcels increases to $F^{c0'}$, and optimal tree planting for cash crops on freehold parcels increases to $T^{c1'}$, optimal tree planting on customary parcels does not increase with higher land values and is instead depicted by $T^{s0'}$ and $T^{c0'}$.

price of cash crops increases with greater access to urban markets.³¹ Secondly, the increased access to urban markets also brings in outsiders who may seek to acquire land locally. These outsiders may find it difficult to determine who owns undocumented customary land. Local elites may not have found it profitable to exercise their traditional rights in an inactive land market, preferring instead to maintain the status quo. However, as land values rise (whether due to offers from outside investors, population pressures from growing urban areas, or other exogenous forces that introduce potential buyers who are less familiar with the customary) they may be more likely to expropriate from the land user in order to alienate the land to others.³²

The land user (farmer/decision-maker) can anticipate these changing incentives for their co-rights-holders, captured by $\frac{\partial \phi}{\partial v} < 0$. Rising land values may mean that no matter the level of their own investment in production, the risk of expropriation increases.³³ Figure 1 illustrates this: the higher risk of expropriation could outweigh the increased output price impact of increasing land values, and perhaps even on net decrease longer-term investments in customary land. The shift from T^{c1} to $T^{c1'}$ and F^{c1} to $F^{c1'}$ reflects the effect of increasing land values on investment on freehold land: investment in both fertilizer and trees increases, due to higher prices for cash crops. For customary land in Figure 1, increasing the value of the land does not simply shift the marginal productivity of trees on cash crops rightward, but changes the shape due to the effect of $\frac{\partial \phi}{\partial v}$ discussed above. In sum, we would expect a smaller increase in tree planting on customary land due to an increase in land pressures than on freehold land.

4 Data & Empirical Strategy

To test this model empirically, I use data from the baseline survey I conducted for a proposed impact evaluation of Uganda’s Agriculture Cluster Development Project (ACDP) to examine differential patterns of short- and long-term agricultural investments on parcels under different tenure regimes as land pressures exogenously increase.³⁴ This baseline survey was conducted in early 2019 in four districts throughout Uganda.

³¹Although greater access to markets may also increase the price of the staple crop, and perhaps decrease input prices of fertilizer and trees (as well as improve their availability or quality), in relative terms the largest effect is the increase in the price of the cash crop. On the other hand, greater access to urban markets may increase the opportunity cost of on-farm labor (but also may increase the supply of hired labor), which could be thought of as an change in the shadow price of labor-intensive inputs such as trees. This effect should be common for all parcels in an area, though, and therefore should not differentially impact customary vs. freehold parcels within the same geography, the comparison I am making.

³²An alternate, complementary, understanding of this process (explored in Honig (2022)) is that in isolated communities, returns to elite status are primarily social – respect and prestige from their community. But as outsiders penetrate the community, that social prestige may be less valuable or obtainable from a community in flux. This may make the monetary benefits of expropriation seem even more valuable.

³³Indeed, these rising land pressures may also weaken the relationship between tree planting and tenure security, $\frac{\partial \phi}{\partial T}$, as community norms around responsible land stewardship may not outweigh individual expropriation incentives of more valuable land. This would increase the numerator of the equilibria conditions for tree planting, and therefore mean yet lower optimal investment in trees on customary land.

³⁴This project, supported by the Government of Uganda and the World Bank, aims to provide farmers in five crop-specific ‘clusters’ (maize, rice, beans, cassava, and coffee) with the resources needed to move from subsistence farmers to commercialized producers. To that end, the project provides subsidized farm inputs to selected farmers, improves agricultural infrastructure, and supports post-harvest handling technologies. The impact evaluation was planned as part of the pilot phase of the project,

Respondents were chosen in a two-stage process: first, we randomly selected 133 registered farmer organizations that had at least 20 members in the target districts, and membership rosters were drawn up. Member households were then randomly selected to be interviewed, with a total of 2,189 interviews in the baseline survey used here. The districts selected for the impact evaluation are located throughout the country and thus encompass some of the diversity of customary tenure arrangements in Uganda.³⁵

As part of the impact evaluation, the Ministry of Agriculture, Animal Industry and Fisheries and a team from UC Davis conducted an extensive baseline survey,³⁶ which has data well-suited to the structure of this model. Given that the program was designed to induce investment in agriculture, the survey has an exhaustive plot-level elicitation of all inputs and outputs. The survey asks detailed questions about the tenure status of each agricultural parcel farmed by the respondent household, land rental prices in the region, and contains GPS coordinates for the household, allowing me to match households to other geospatial datasets. For most specifications, I use a dataset of 3,076 parcels; most were observed over two seasons, for a total of 5,760 parcel-season observations which I pool.

To some extent, this sample is positively selected in comparison to all farmers in Uganda: they are members of farmer groups, indicating a higher likelihood of participating in output markets and perhaps higher farming skill, although farmer groups are relatively common in Uganda.³⁷ This positive selection can be seen in rates of input use in the ACDP data when compared with the nationally-representative Living Standards Measurement Survey (LSMS) conducted by the Uganda Bureau of Statistics and the World Bank: only 4.6% of parcels have organic fertilizer applied and 2.6% inorganic fertilizer in the 2019 LSMS, as compared with 11.5% and 10% of parcels in the ACDP data, respectively. This higher rate of input use provides a key advantage over the more widely used LSMS data: I can estimate the interacted models that emerge from my model due to variation in the outcomes.

The theoretical model explored above shows how using tenure *type* as a rough proxy of tenure *security* may be insufficient: in low land-value environments, customary land may be equally secure as freehold, but as land values increase, customary land may become less secure.³⁸ My model demonstrates how overlapping

focusing on the provision of electronic vouchers for subsidized inputs that are redeemable at certified local agro-input dealers (Bird et al., 2026). The impact evaluation was planned in 4 districts throughout Uganda, each assigned one of 4 target crops (maize, rice, beans, and coffee; cassava was omitted as it has a longer growing season). The impact evaluation randomly assigned farmer organizations eligible for the ACDP subsidy to one of two treatment groups (defined by the timing of benefits) or to a ‘downstream’ control group. Approximately 36 farmer organizations were assigned to each treatment arm, from each of which an average of 20 members were selected to be surveyed throughout the four districts.

³⁵Selected districts are located in the east, southwest, center (where *mailo* land predominates), and north (where instability has led to many internally displaced people).

³⁶Subsequent mid-line and end-line surveys were planned but were cancelled along with the implementation of the Impact Evaluation in November 2020, due to compliance issues and COVID-19. The baseline data I use was entirely collected pre-intervention, so there should be no impacts of the intervention on input use.

³⁷In the 2019 Living Standards Measurement Survey, 28% of communities report having a farmers’ group; 43% report a savings & loan group that was identified by subcounty extension workers as serving as a farmers’ group for the purposes of ACDP targeting.

³⁸The lack of results in previous studies which elide tenure security and tenure type may be understood, then, as an issue of this poor proxy.

rights-holders respond to rising land pressures, and therefore investment incentives change differently on freehold and customary land.

Consider the following cross-sectional empirical specification which closely follows the model above:

$$I_i^Q = \gamma_0 + \gamma_1^Q D_i + \gamma_2^Q \bar{v} + \gamma_3^Q D_i * \bar{v} + \beta' X_j + \eta_f + \epsilon_{fi} \quad (1)$$

where I_i^Q is a measure of investment of type Q (either short- or long-term) by household j (a member of farmer group f) on parcel i ; D_i is a dummy equal to one if the parcel is customary; $\bar{v}$ is the exogenous component of land pressures in the locality; X_j is a vector of household and parcel characteristics (a probability of poverty wealth index, total household landholdings, the first year the parcel was farmed, whether a cash crop is grown on the parcel, and dummies for how the parcel was acquired); and η_f controls for farmer-group fixed effects. This equation is separately estimated for long and short-term inputs.

The particular hypotheses predicted by the model are derived in the mathematical appendix. Here I summarize proposed tests of the predicted signs of the following in cross-sectional data:

1. $\left. \frac{\partial F}{\partial v} \right|_{h=1}$: How does short-term input use vary with land pressures on freehold land? The model would predict that *ceteris paribus*, farmers would use more short-term inputs as land values increase: $\gamma_2^S > 0$.
2. $\left. \frac{\partial T}{\partial v} \right|_{h=1}$: How does long-term input use vary with land pressures on freehold land? The model would predict that *ceteris paribus*, farmers would use more long-term inputs as land values increase: $\gamma_2^L > 0$.
3. $\left. \frac{\partial F}{\partial v} \right|_{h=0}$: How does short-term input use vary with land pressures on customary land? The model would predict that *ceteris paribus*, farmers would use more short-term inputs as land values increase: $\gamma_2^S + \gamma_3^S > 0$.
4. $\left. \frac{\partial T}{\partial v} \right|_{h=0}$: How does long-term input use vary with land pressures on customary land? The model would predict that *ceteris paribus*, by contrast to freehold parcels, farmers may not use more long-term inputs as land values increase: $\gamma_3^L < 0$.

with the last of these being the primary innovation of this model. Without considering the changing incentives of local elites, the impact of rising land values on customary land would parallel that on freehold.

One key limitation of the ACDP data is its cross-sectional, rather than panel, structure. I am only able to observe the stock of trees (a long-duration investment) on a parcel, rather than the flow of investment in trees modeled. Trees planted in the distant past will be counted as investment on a parcel, even if they would not be planted in the contemporary land pressure environment. Appendix C.7.5 considers the potential bias from this limitation in more detail, but the surplus stock of trees should make it more difficult to detect

differential responses in tree investment.³⁹ Other shorter-term investments, including organic fertilizer, inorganic fertilizer, and pesticides, are captured as a binary variable equal to one if the investment was made on the parcel in a given season.

The theoretical model takes land values to be an exogenous force shaping the incentives faced by farmers. This was driven by qualitative literature, which has focused on the breakdown of customary institutions caused by outside pressures, such as rapid urban expansion in much of Africa. In reality, the value of an individual parcel of land is a product of its particular features, including any investments made and tenure type.⁴⁰ Empirically, I proxy for land pressures in three ways. First, I use standardized per-acre land rental values reported in the survey (either actual paid rental price or estimated potential rent if the parcel were rented out, and predicted using similar nearby parcels if the respondent did not report a rental price).⁴¹ These results carry the crucial caveat that land prices are likely a result of land quality, which may change optimal investment behavior through pathways other than those modeled. That is, although the model captures the effect of land prices on tenure security, tenure security may impact rental prices of individual parcels through investments made in the land.

To avoid this circularity, I also present results for all outcomes where I proxy for the exogenous pressures on land values with two geospatial measures adapted from Muller-Crepon et al. (2021) and Muller-Crepon (2021). These measures, constructed from historical road maps from Michelin, estimate the shortest travel time via roads of different qualities to the national capital⁴² for each 5x5km grid cell across the continent.⁴³ I match households to a grid cell using GPS locations collected during the survey⁴⁴. Following Muller-Crepon et al. (2021), I compute the household’s ‘access’ to Kampala in 2015 as $(1 + time_j)^{-1}$.⁴⁵ The inverse of travel time to the capital represents the degree to which outsiders (such as urban residents of the capital) may be interested in purchasing local land: precisely the concern of customary landowners that creates insecurity.⁴⁶

³⁹Further research, employing panel data with more granular measures of tree investment, would be valuable in confirming this hypothesis.

⁴⁰The ACDP survey also asks for self-reported rental values, including estimated values for parcels not actually rented, which could be influenced by farmer awareness of land market conditions (and their human capital).

⁴¹I standardize rental prices by first assigning a parcel the median per-acre rental price in its district if it is missing a rental price, then subtracting the full sample mean per-acre rental price and dividing by the full sample standard deviation. In appendix C.6.2, I also use alternate measures of land prices, including predicted prices for all parcels, median village prices, and non-standardized rental value.

⁴²I focus on travel time to Kampala rather than to regional capitals in this analysis, as I want to isolate away from the politically-driven administrative unit proliferation that has occurred in Uganda in the recent past. Although a new district capital may be identified close to respondents, these new centers do not yet have the population to support substantial outsider demand for land.

⁴³This is approximately 2.5 arc-minutes at the equator, which Uganda spans.

⁴⁴Taken at the household residence, not at agricultural fields which could be some distance away.

⁴⁵This represents the most recent year for which Muller-Crepon (2021) has data on travel time to national capitals, regional capitals, and distance to markets, and thus the closest to the ACDP baseline survey in 2019.

⁴⁶The correlation between per-acre shilling rental prices and access to Kampala is low (.08) but increases to .21 when I winsorize several outliers in estimated rental prices (among those who own their land and are estimating how much they could rent it out for). In most empirical specifications using rental prices, I standardize to deal with outliers rather than a more ad-hoc approach such as winsorizing. For this correlational check with shilling prices, I assign all per-acre rental prices below 10,000 shillings (85 parcels) a value of 10,000 shillings per acre (lower than the 5th percentile) and the top 1.6 percent of prices (the 98.4th percentile value) and to a very high .93 when I consider district median rental prices, suggesting that it captures

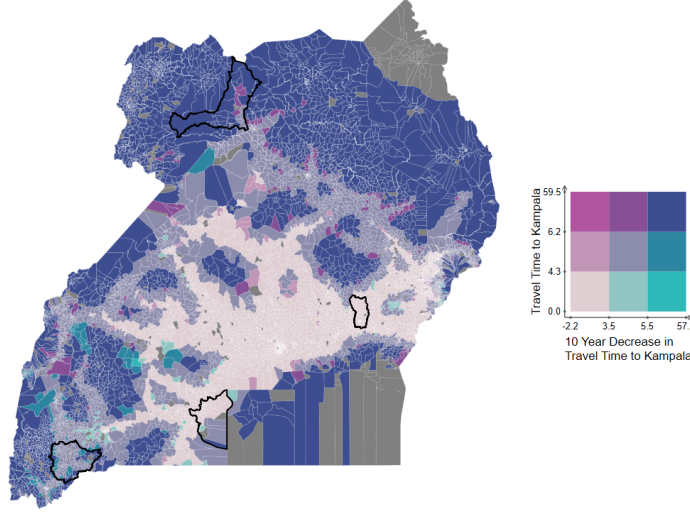


Figure 2: Map of the average travel time in 2015 to Kampala, as well as the 10-year decrease in that travel time, for each Parish in Uganda, with ACDP districts highlighted in black.

The third measure I use attempts to isolate *changing* pressures on local land: the historical (10-year) decrease in travel time to Kampala, due to the construction or upgrading of roads, relative to the 2005 travel time to Kampala. This measure captures the extent to which elites face *new* opportunities to expropriate. I match this measure spatially in the same way as the inverse travel time variable. Both travel time measures should impact agricultural investment through the pathways described in the model: certainly, more (peri-)urban areas have more active cash crop output markets, which would increase the relative price faced by the household, and a pool of urban buyers is the canonical situation incentivizing local elites to expropriate customary land. The process of urbanization, though, should not impact agricultural investment through other channels, such as via an increase in total factor productivity. These geospatial measures of land pressures should not be endogenous to the tenure or other characteristics of a particular parcel, and therefore should not suffer from the same endogeneity bias. Figure 2 plots both the average travel time to Kampala, as well as the 10 year decrease in that time, for each parish of the country, with ACDP districts highlighted in black (maps of each measure individually can be found in figures 7 and 8).

These measures could potentially be used as an instrument for local land values; however, I choose to treat them as proxies for the underlying phenomenon of interest, external pressures and demand for land. Rental values for individual parcels are perhaps an even more problematic proxy for the forces modeled, so I focus on these more exogenous measures which should be better-identified. Nevertheless, these results may not be strictly causal; however, they can be understood as descriptive of patterns consistent with my theoretical model. I have constructed all measures of land pressures such that higher values represent greater broader pressures than individual prices which may be influenced by soil quality or other agricultural value considerations.

pressure on land: higher standardized rental prices, larger inverse travel times (equivalent to shorter travel times to Kampala), or greater decreases in travel time over the past 10 years relative to 2005 times.

Several other concerns are apparent with this empirical strategy. First, the tenure regime of particular parcels may not be exogenous (for example, households may be more likely to pursue titling for higher-quality plots). Deininger and Ali (2008) argue that Uganda has what amounts to an “exogenous historical assignment of land rights... and the absence of readily available opportunities to change the tenure status of occupied land to full ownership imply that the case at hand can be considered akin to a natural experiment.”⁴⁷ In the appendix, I probe this exogeneity by controlling for any observable differences in parcel characteristics by tenure type (Appendix C.7.1; results are robust to these controls) and separately estimating responses for inherited parcels (for which tenure status is more likely to be exogenous) and purchased ones (where farmers have more ability to choose tenure type) (Appendix C.8.1; results do change, but do not raise obvious issues with the exogeneity of tenure status). Nevertheless, a more conservative interpretation of my empirical evidence is that it provides correlations (conditional on observables) that are consistent with the predictions of a theoretical model.

Second, there may be unobserved heterogeneity in household farming decisions caused by factors such as household shadow prices or farming skill, which may be correlated with the tenure of parcels farmed by households.⁴⁸ Ideally, this would be addressed using a household fixed-effects strategy. Unfortunately, of the 2,189 households surveyed, only 6.5% of respondents reported having parcels under both freehold and customary tenure, which is much lower than the rates observed in the LSMS-ISA (Deininger and Ali, 2008); these households are concentrated in two of the four surveyed districts. This limits the power of a household fixed-effects model to detect changing investment patterns as land pressures increase.

Despite this, the ACDP data sampling strategy involving farmer groups allows me to extend the intuition of household fixed effects to farmer group fixed effects. These farmer groups are co-located, self-selected groups of farmers who can purchase inputs in bulk, share farming methods, and aggregate outputs in order to receive better output prices, in addition to meeting regularly for mutual support. Therefore, it seems plausible that many unobservable characteristics are shared by members of the same farmer group. By including a farmer group fixed effect, I am able to control for a portion of the unobserved heterogeneity while still retaining some identifying variation. However, this does restrict the identifying variation almost to zero for regressions using geospatial measures of land pressures: most farmer groups are all within the same grid square or parish. Therefore, my preferred specification includes farmer group fixed effects for regressions

⁴⁷Perego (2019) describes the costly and onerous process required to register customary land, which provides substantial barriers to titling.

⁴⁸In Appendix C.7.4, I control for the household dependency ratio as well as the number of adults in the household, to explicitly address these shadow prices; results are robust to their inclusion, suggesting the labor channel does not drive my results.

using the standardized rental price as the measure of land pressures (column (2) in results tables 1 - 2) but without the fixed effects for regressions using the inverse travel time or changes in travel time (columns (3) - (4) in tables 1 - 2). In interpreting coefficients on rental price with the farmer group fixed effects, I identify off of within-group differences in land pressures; for other measures, the results are between-groups or regions. All specifications cluster standard errors at the larger parish level, to account for geographic relationships between households. In appendix C.5, I present results with a more traditional parish fixed effects specification with land values (table A19) for robustness, as well as computing spatially dependent standard errors as in Conley (1999) (tables A5 - A9 in appendix C.2).

The primary results that I present have land pressures entering linearly, with a constant marginal pressure on either customary or freehold land. This may not be the appropriate specification: it may be that elite incentives to expropriate only begin at intermediate or relatively high land pressures, or alternately that in high land pressure environments, the likelihood of expropriation of customary land has been ‘priced in’ to investment behavior, meaning the marginal effect of increasing pressure is minimal. In order to explore this, I examine whether the relationship differs on either side of 2 hours travel time from Kampala, a threshold identified in interviews as being relevant for external land demand. I separately estimate the responsiveness of investment to land pressures on customary and freehold land on either side of the threshold.

5 Results

In this section, I present results for a variety of agricultural investments, following the empirical strategy in equation (1). The model predicts that long-term inputs respond to rising land values more strongly on freehold land than on customary land due to the attenuating effect of increased insecurity on customary land induced by elite incentives to expropriate.

Before turning to the regressions with individual agricultural inputs, consider figure 3. Each subfigure presents the difference in the coefficients (essentially, γ_3) for (binary) input use between freehold and customary parcels as a function of land values in the area (measured by rental values on the top left, the 10-year relative decrease in travel time on the top right, and the inverse travel time to Kampala below).⁴⁹ This divergence in responses to land pressures between customary and freehold parcels is the primary focus of the model: many sources of tenure insecurity may rise with land pressures, including border encroachment, inheritance disputes, and government expropriation. In addition to all these sources, customary tenure also faces insecurity from elite expropriation, driving differential responses. For short-term inputs such as pes-

⁴⁹I separately regressed input use on the measure of land pressures for customary and freehold parcels, and then plot the difference in coefficients from each of these regressions. This is in contrast to the regression tables below, where I include all parcels and use interactions to model the relationship between land pressures, tenure type, and input use.

ticides and inorganic fertilizer, the model would predict equivalent responses to rising land pressures on all parcels, which would appear as a flat line on these graphs. As can be seen, however, there is a slight divergence in short-term input use on freehold vs. customary parcels as rental prices or the accessibility to Kampala rise, reflected by a positive (although shallow) slope in these figures for short-term inputs. This result, explored further below, can be explained by a shift toward input-intensive cash crops (both short- and long-term) near urban markets.⁵⁰ As urban markets expand, so does the local demand for cash crops (such as vegetables and fruits which are difficult to transport long distances). Farmers facing this increased demand therefore shift their production away from staples and towards cash crops. This shift is stronger on freehold parcels, perhaps because such parcels remain secure. Cash crops, in turn, may have a higher return to inputs, and particularly to pesticides, than staple crops (Riwthong et al., 2017). This shift towards cash crops with increasing land pressures is not captured in my model, where the share of land devoted to cash crops, α , was fixed for simplicity.

If the model holds, the effect of changing crop choices on long-term inputs should be compounded by the direct impact of additional insecurity on customary land prompted by rising land pressures. In figure 3, there is a stronger response for trees, the most long-term, than for short-term inputs. Visually, the insecurity effect can be seen in the difference in slopes between the short- and long-term inputs in these figures.

Figure 3 also depicts the difference in reported insecurity between freehold and customary land. This measure is discussed in further detail below, but note that at low land pressures, parcels under customary and freehold tenure are perceived as roughly equivalently secure. As pressures rise, respondents report higher perceived likelihoods of losing their customary parcels without an equivalent rise on freehold parcels.

In each table below, I present several different specifications; additional robustness checks are provided in appendix C.⁵¹ I include three distinct measures of land pressures. First, I use the standardized per-acre

⁵⁰An alternate explanation could be different collateralizability of customary and freehold parcels. Farmers may be less able to internalize rising land values to relax their liquidity constraint on customary parcels. This would be less concerning if farmers held parcels under different tenure systems, due to the fungibility of loans. When I examine household credit usage in my sample, those in areas closer to Kampala are actually slightly less likely to take out loans; the relationship between credit usage and normalized individual rental prices is insignificant. This suggests that collateral is not driving the results for short-term inputs. Yet another, empirically indistinguishable, mechanism would be sectoral misallocation as a result of land and labor market restrictions (Adamopoulos et al., 2022), which could act more strongly in more remote areas; in my data, however, hired labor usage is near-universal, suggesting labor market restrictions do not bind. A third potential channel, emerging from the structural transformation literature, is that less resource-constrained farmers may differentially sort onto freehold parcels as land pressures increase and are able to make more investments in their land. This channel should operate both on short- and long-term inputs, which both require cash investment.

⁵¹These include exploring tenure security's relationship to land pressures across an expanded geographic sample that was part of a related survey, using Conley (1999)'s spatially dependent standard errors, splitting the sample with binary measures of land pressures rather than continuous measures, and including parish fixed effects. In Appendix C.6, I explore alternate measures of land values either derived from the survey, looking at regional market access, or using a predicted probability of urbanization from Seto et al. (2012). In C.7, I include various additional controls, such as parcel characteristics, excluding the possibly endogenous control of whether a cash crop was grown on the parcel, and controlling for reported individual access to markets: the reported one-way travel cost to either the nearest agroinput dealer or the nearest village in which crops can be sold. These controls do not meaningfully change the results of the paper, although they do result in a substantial loss of sample size, hence their relegation to the appendix. I also control for measures of the household labor supply and look at responses to land pressures on more recently farmed parcels, to capture more recent investments in the stock of trees. In appendix C.8, I explore robustness of results when restricting to various subsamples, including separately estimating results for parcels

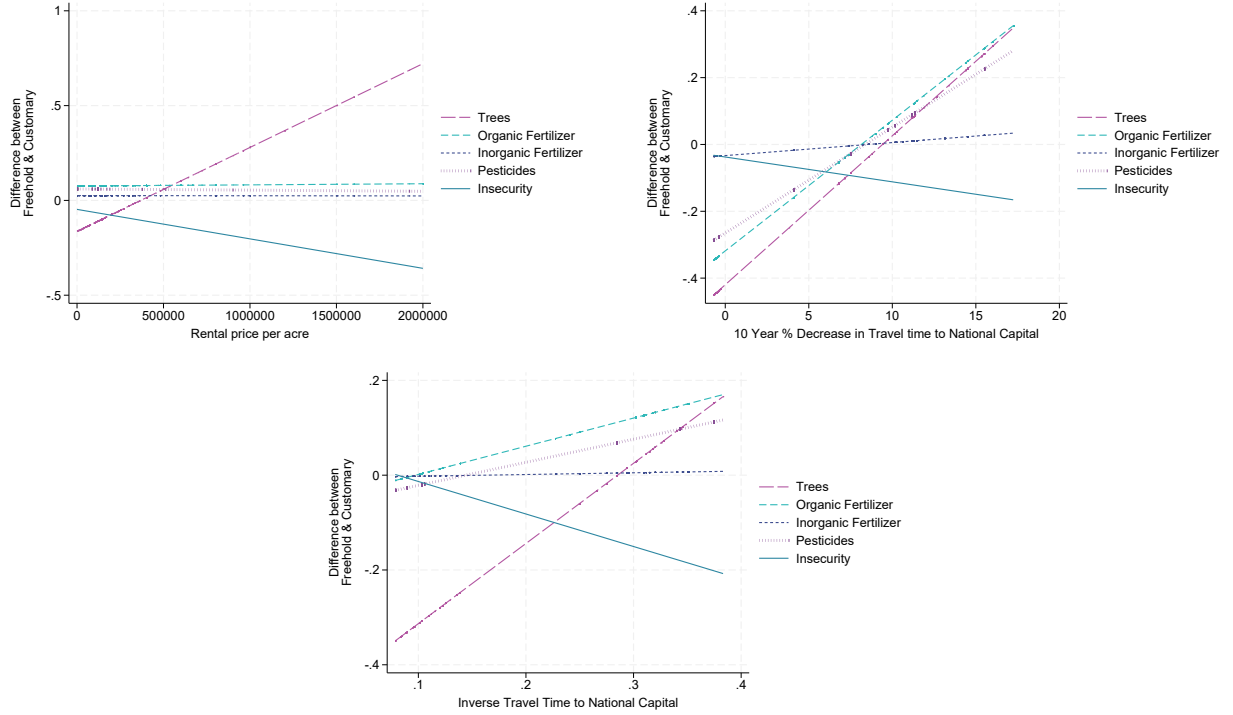


Figure 3: Disparities of investment across tenure types increase with rental values (top left), the percent decrease in travel time to Kampala (top right), and the inverse travel time to Kampala (below) more strongly for long-term than short-term inputs.

rental price in shillings as reported for a given parcel.⁵² I also show results using the inverse travel time to the national capital and 10-year decreases in travel time to Kampala (proportional to 2005 travel time) as proxies for underlying pressures, particularly outsider demand for land. These measures of land values are interacted with a dummy for the tenure status of a parcel, so the coefficient on ‘Land Pressure’ represents the responsiveness to land pressures on freehold land, while ‘Customary*Land Pressure’ represents the difference in that responsiveness for customary parcels.⁵³

In all regressions shown, I control for household characteristics: the Probability of Poverty Index wealth index (PPI) as well as a household’s total landholdings in acres.⁵⁴ I also control for how the parcel was acquired (inherited, purchased, assigned by authorities, loaned, etc), to support the conditional exogeneity purchased versus inherited, and excluding mailo parcels.

⁵²As discussed above, this measure is not solely driven by exogenous land market pressures, and it could be capturing unobserved land quality (or even influenced by prior long-term investments in the land). In appendix C.6.2, I show that results are similar when using predicted rental prices for all parcels, predicted using the probability of urbanization, latitude and longitude of the household, tenure status, existence of trees on the parcel, and village fixed effects.

⁵³Although conceptually similar to the slopes in figure 3, they are computed differently, include the addition of controls, and use clustered standard errors.

⁵⁴The PPI uses low-cost survey indicators to estimate the likelihood that a household has consumption below a given poverty line, and it can be interpreted approximately as a wealth index (with higher values indicating a lower likelihood of poverty) (Schreiner, 2012). Encouragingly, the coefficient on PPI accords with intuition: wealthier farmers are generally more likely to use agricultural inputs, with the exception of trees, which may be more important for their security-enhancing functions for poorer households.

of land tenure status, and for the first year in which the parcel was farmed, to account for duration of tenure effects. In table 2, I control for whether or not any plot on the parcel is primarily devoted to a cash crop, to isolate the insecurity effect. However, there may be some concern that this specification suffers from ‘bad control’ problems, as crop choice is potentially an outcome of land pressures. Controlling for crop choice would therefore absorb part of the treatment effect and understate the effect of interest; omitting it, however, risks confounding. The main results I present include it as a control to be conservative, but in Appendix C.7.2, I present regressions that omit this control and show results that are extremely close to those in the main text, although marginally larger in magnitude.

In columns (1), (3), and (4) I run pooled regressions, clustering standard errors at the parish level. In column (2), farmer-group fixed effects are added to the clustered model. This model is the most conservative, but because of relatively low input use and the geographic clustering of farmer groups (restricting the identifying variation somewhat) results are at times insignificant (although patterns are generally consistent with the pooled models).

5.1 Tenure Security

Before turning to agricultural inputs used on parcels, recall that exogenous land pressures should influence input decisions through two pathways: increasing optimal input usage by increasing the relative price of cash crops (which should be equally true for all parcels), and increased tenure insecurity (attenuating the cropping response, but only for long-term inputs under customary tenure). Customary tenure can become less secure if local elites face incentives to expropriate customary land. These incentives, I argue, increase with local land values.

Actual elite expropriation of customary land to sell to outsiders is relatively rare and therefore difficult to detect statistically in random-sample surveys such as this one. However, even if respondents have not personally experienced elite expropriation, they may worry about it, particularly as their community experiences the external pressures that have led to elite expropriation elsewhere. If this pattern holds, I would expect respondents to express greater concern about losing customary parcels as land pressures increase. These land pressures should not affect the perceived security of freehold land to the same extent, as titles provide some legal recourse in case of expropriation.

Panel A of table 1 presents regression results using the average perceived insecurity of a parcel as the outcome. That is, for each parcel, respondents were asked first who in the household had rights to that parcel, and what the likelihood was that each rights-holder would involuntarily lose the land (on a scale of 1 to 5, with higher numbers representing a greater chance of losing it). These responses were averaged

across all rights-holders to each parcel to measure household-level insecurity about that parcel. Respondents reported for 21% of parcels that it was ‘slightly likely’ or greater that they would involuntarily lose rights to the parcel, so this is a meaningful concern in our sample.

In table 1, tenure insecurity responds weakly or even negatively to higher land pressures on freehold land, but demonstrates a statistically significant positive relationship with land pressures (measured by rental prices) on customary land. This implies that customary land becomes more insecure than freehold as land pressures rise. These results suggest that the hypothesized mechanism for agricultural investment (rising insecurity on customary land) does indeed hold in Uganda.⁵⁵

5.2 Crop Choice

In the model, the share of land devoted to cash crops, $(1 - \alpha)$, is taken as exogenous for simplicity. In reality, however, this parameter is a choice made by farmers. If output prices for cash crops increase along with land pressures (given better access to urban markets), we would expect a shift into cash crops in higher land pressure environments. Therefore, in panel B of table 1, I regress an indicator for at least one plot on the parcel being primarily planted to a cash crop on my three measures of land pressures. Table A41 instead looks at an analogous measure for staple crops, with roughly the inverse pattern.

Overall, farmers are more likely to plant cash crops in higher land pressure environments, and this relationship holds equally for both freehold and customary parcels. This is unsurprising: nearby urban markets increase demand locally for fragile cash crops such as fruits and vegetables. The model demonstrates how this higher demand for input-intensive cash crops should increase optimal input use for both long-term inputs such as trees as well as short-term ones such as pesticides and inorganic fertilizer.⁵⁶ I control for planting a cash crop on the parcel in all input use regressions, so that the divergence in long-term input application on customary versus freehold land with increasing land pressures can be attributed directly to insecurity. Unfortunately, the ACDP survey does not indicate the specific crop on a plot to which an input was applied if multiple crops are grown (and intercropping is very common in this context), so I cannot separately estimate input use responses on cash crops and staples.

⁵⁵In table A4, I conduct a similar exercise using a larger sample of data collected for monitoring & evaluation purposes of ACDP across an additional 30 districts (this data does not contain consistent measures of input use, so cannot be used for the full analysis). Results are consistent, even stronger, when using this more geographically diverse sample.

⁵⁶This crop choice effect, where inputs are more productive on newly more profitable cash crops, would only be confounded by increasing access to the city relaxing both price and non-price constraints on those inputs, such as availability or quality. Changing input markets, like changing output markets, should be equivalently available to customary and freehold parcels in the same area, and therefore not drive the divergence I explore.

VARIABLES	(1)	(2)	(3)	(4)
<i>Panel A:</i>		Average Insecurity		
Land Pressure	0.0213 (0.0217)	0.0247 (0.0156)	-1.397*** (0.466)	-2.651* (1.326)
Customary $\times$ Land Pressure	0.106*** (0.0137)	0.120*** (0.0186)	0.767 (0.501)	-0.186 (1.140)
Customary	0.103 (0.0781)	0.0759 (0.0747)	-0.499** (0.222)	-0.149 (0.199)
R-squared	0.261	0.382	0.259	0.253
<i>Panel B:</i>		Planted Cash Crop		
Land Pressure	0.0125* (0.00646)	0.0103** (0.00446)	0.750* (0.394)	5.452*** (0.741)
Customary $\times$ Land Pressure	0.0179 (0.0160)	-0.000699 (0.00283)	0.327 (0.297)	2.551*** (0.844)
Customary	-0.129*** (0.0313)	-0.0152 (0.0140)	-0.000478 (0.0531)	0.218*** (0.0623)
R-squared	0.094	0.389	0.106	0.189
Observations	3,240	3,240	3,222	3,222
Land Pressure Measure	Rental Price	Rental Price	Inverse Travel Time	Decrease in Travel Time
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 1: OLS results of average tenure insecurity across all rightsholders (scale of 1-5, with higher values indicating more insecure) on a parcel in panel A and a dummy equal to one for parcels with at least one plot primarily devoted to a cash crop in panel B on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Land pressures are proxied by the standardized rental price in columns (1) and (2), the inverse travel time to Kampala in column (3), and the 10-year relative decrease in travel time to Kampala in column (4). Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

5.3 Investment

Planting trees in Uganda represents a long-term investment in agricultural productivity. In an inter-cropped system, trees provide structure for climbing crops (increasing the productivity of those crops), prevent soil erosion, and produce fruit or coffee themselves (Deininger and Ali, 2008), though their value takes some time to be realized. Therefore, the expected benefits of tree planting are lessened as the risk of expropriation increases, as modeled above. Additionally, the visible nature of trees means they can actually reduce the risk of expropriation (particularly by neighbors who understand the social context of customary tenure); in low land-pressure areas, tree planting may actually be more valuable on customary land than freehold. As a point of interest, note that in figure 3, tree planting is actually more common on customary plots than freehold at low land values. As external land pressures increase and the primary security concern shifts to elite expropriation to sell to outsiders, this security-enhancing effect may not be as strong. Therefore, the model would predict that as external land pressures increase, tree-planting on customary land may increase; on freehold land, tree-planting would increase substantially. In the ACDP data, I observe whether or not there is a tree on the parcel; given the long duration of trees, this stock measure may not perfectly capture recent tree-planting investment. That is, trees planted in the more distant past on customary parcels may not have been planted in a new, higher land pressure environment. This is a fundamental limitation of the cross-sectional nature of the data; to truly test the model, I would need a long panel that contains up-to-date measures of land pressures and investment in new trees. However, if capturing stocks rather than flows does bias my results, it should make it harder to detect a difference in responses between freehold and customary land empirically than is predicted by the model.⁵⁷

When looking at panel A of table 2, this pattern does largely hold. When land pressures are measured using standardized rental prices or relative decrease in travel time, there is a small, statistically insignificant increase in tree-planting on freehold land; when using the inverse travel time to Kampala to proxy these pressures, freehold land sees more trees as external pressures mount. With all measures, however, the response is (often statistically significantly) attenuated on customary parcels, as predicted by the model and seen in negative coefficients on the interaction terms.

Organic fertilizer use, in panel B, both improves agricultural productivity in the season in which it is applied, and can also have medium-term effects by improving soil organic content (Johansen et al., 2015). Indeed, Deininger et al. (2021) follow a long literature in treating organic fertilizer application as a key long-

⁵⁷For customary parcels in high-land pressure environments, the observed stock of trees may be higher than predicted by the model, if they were planted in a prior low-land pressure context. In appendix C.7.5, I explore this attenuation by separating the sample into more recently-farmed and older parcels. Parcels that were first farmed more recently show a stronger differential in tree planting responses to land pressures between customary and freehold parcels than longer-farmed parcels. This suggests that the results in the full sample are an understatement of the true differential in tree planting due to the fact I only observe the stock of trees.

VARIABLES	(1)	(2)	(3)	(4)
<i>Panel A:</i>			Trees	
Land Pressure	0.0127 (0.00817)	0.00844 (0.00954)	0.886** (0.396)	1.007 (0.885)
Customary $\times$ Land Pressure	-0.0435*** (0.0116)	-0.0377*** (0.00841)	-7.85e-05 (0.432)	-1.703** (0.809)
Customary	0.0283 (0.0433)	0.0233 (0.0507)	0.276** (0.109)	0.299*** (0.0905)
R-squared	0.061	0.242	0.071	0.066
<i>Panel B:</i>			Used Organic Fertilizer	
Land Pressure	-0.000986 (0.00129)	-0.00120 (0.000964)	0.541*** (0.199)	3.411*** (0.445)
Customary $\times$ Land Pressure	0.0137*** (0.00508)	0.00307 (0.00199)	0.297*** (0.105)	1.192*** (0.310)
Customary	-0.0661** (0.0249)	-0.0184 (0.0210)	0.000714 (0.0419)	0.170*** (0.0452)
R-squared	0.169	0.322	0.175	0.224
<i>Panel C:</i>			Used Inorganic Fertilizer	
Land Pressure	0.00330 (0.00267)	0.00497** (0.00186)	0.750*** (0.0846)	1.396*** (0.426)
Customary $\times$ Land Pressure	0.00894 (0.00865)	0.00422 (0.00263)	0.588*** (0.108)	0.658** (0.323)
Customary	0.00844 (0.0363)	0.00489 (0.0405)	0.0528 (0.0476)	0.0867** (0.0336)
R-squared	0.029	0.150	0.057	0.041
<i>Panel D:</i>			Used Pesticides	
Land Pressure	-0.00142 (0.00342)	-0.00205 (0.00253)	0.855*** (0.118)	2.155*** (0.502)
Customary $\times$ Land Pressure	0.0116 (0.00812)	-0.0124*** (0.00333)	0.584** (0.227)	0.00778 (0.544)
Customary	-0.0315 (0.0657)	-0.0441 (0.0672)	0.0455 (0.0497)	0.192*** (0.0425)
R-squared	0.079	0.203	0.103	0.096
Observations	3,250	3,250	3,232	3,232
Land Pressure Measure	Rental Price	Rental Price	Inverse Travel Time	Decrease in Travel Time
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 2: OLS results of a dummy equal to one for parcels with a given investment made on measures of land pressures, controlling for a dummy equal to one if the primary crop on any plot on the parcel is a cash crop, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Land pressures are proxied by the standardized rental price in columns (1) and (2), the inverse travel time to Kampala in column (3), and the 10-year relative decrease in travel time to Kampala in column (4). Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

term investment in land productivity in the Sub-Saharan African smallholder context, which can be affected by tenure security (Jacoby et al., 2002; Jacoby and Mansuri, 2008). The model would predict, then, that organic fertilizer use would increase along with land values on freehold land, but the effect will be attenuated on customary land.

When looking at a binary indicator for the application of organic fertilizer in panel B of table 2, the relationships are not fully consistent with the model's predictions. Organic fertilizer use does seem to respond positively to land pressures on both freehold and customary land (especially as measured by travel time; the relationship is less statistically significant for standardized rental prices). On customary land, the response is actually stronger than for freehold, although this is not always true for rental prices. This is surprising if we strictly consider organic fertilizer as an intermediate between trees (long-term) and inorganic fertilizer (short-term) in the context of the model. However, organic fertilizer may have properties outside its duration. First, organic fertilizer application is less visible to other potential claimants (in contrast to trees). This may make application of organic fertilizer less responsive to the social dimensions of tenure insecurity. Furthermore, organic fertilizer may serve as a substitute for tree-planting in a way inorganic fertilizer does not, by amending soil carbon and soil structure. Finally, farmers may have access to organic fertilizer even when liquidity constrained, such as through own animal rearing or green composting, making it an investment strategy available to farmers facing other constraints.

Inorganic fertilizer, by contrast, largely exhibits short-term returns to agricultural land. The model predicts that for short-term inputs, rising land values will increase the relative price of cash crops and thus increase optimal short-term input use equivalently on freehold and customary land. In panel C of table 2, inorganic fertilizer use responds more to land pressures on customary than freehold land, although the differences are small. This is despite the fact that I am controlling for whether or not there is a cash crop on the parcel, and so can be interpreted as isolating the tenure security channel.

Pesticides, another short-term input, are generally used as-needed in Uganda rather than being applied proactively. The model would predict that use of pesticides responds similarly to land pressures on freehold and customary parcels; however, in panel D of table 2 there is a mixed pattern on differential responsiveness by tenure type, depending on the measure of land pressures. In column (2), looking within farmer groups, pesticide use actually diminishes with rental prices on customary parcels; in columns (3) and (4) it increases with inverse travel time or the decrease in travel time on both freehold and customary parcels, and perhaps even more so on customary parcels. These differences are generally smaller than those observed for long-term inputs, as the increasing insecurity on customary land does not impact decisions about pesticide use.

There are several potential explanations for this small divergence in short-term input use between customary and freehold parcels that was not predicted by the model. First, if customary parcels are less able to

be used as collateral, farmers may not be able to borrow against rising land values and relax their liquidity constraint on customary parcels. However, households closer to Kampala are actually less likely to have loans than those further away; credit usage is unrelated to normalized individual rental prices, suggesting collateral is not the primary driver of the observed wedge. A second mechanism would be sectoral misallocation emerging from joint land and labor market restrictions binding in more remote areas (Adamopoulos et al., 2022); this seems less relevant in the Ugandan case, as hired labor in the ACDP data is near-universal. A third potential channel is differential sorting out of agriculture in high land pressure environments by the type of tenure a household owns. That is, if the farmers who remain on customary parcels are more resource-constrained, they may not be able to invest optimally in any inputs (short or long-term) when compared with freehold farmers.

5.4 Thresholds

Above, I have modeled the relationship between land pressures and agricultural investment as linear on both customary and freehold land. It could be, however, that there are multiple regimes of land pressures where the relationship differs. To explore this empirically, I use a threshold of 2 hours travel time to Kampala; in interviews, this was identified as the distance within which most urban residents are seeking agricultural land, and therefore there should be higher likelihood of elite expropriation of customary land in these areas.⁵⁸

Beyond this question of whether the levels of investment differ more or less than 2 hours travel time from Kampala, it may be that the responsiveness of investment to changes in land pressures differ. That is, do the slopes in figure 3 differ on either side of a two-hour travel time cutoff? It could be that within this 2-hour commute of Kampala, the insecurity stemming from outside pressures has already been ‘priced in’ to farmers’ investment decisions, and therefore investment behavior on customary vs. freehold parcels is relatively unchanging as land pressures increase within this high-pressure environment. Alternately, in more remote areas, outsider demand for land may not be salient, so customary land is not perceived as any less secure even as land pressures ramp up. To explore this, I separately run the regressions on either side of the cutoff (regression tables in appendix C.4).

Figure 4 shows, like the bottom of figure 3, the difference between freehold and customary parcels as land pressures increase, but models this relationship separately on either side of the two-hour travel time threshold.⁵⁹ Investments on customary vs. freehold land diverge more sharply as land pressures increase close to Kampala: note the steeper slopes on the right-hand side of Figure 4, within 2 hours of Kampala.

⁵⁸In appendix C.3, I compare investment outcomes on customary vs. freehold parcels in low vs. high pressure environments using dummy variables and their interactions, for a parsimonious representation of my results. Consistent with the model, I find lower input use on customary parcels overall; there is higher input use closer to Kampala, but input use differs significantly across tenure types close to Kampala.

⁵⁹Recall that the inverse travel time measure is $(1 + time_j)^{-1}$, so two hours is located at $1/3$ on the x-axis.

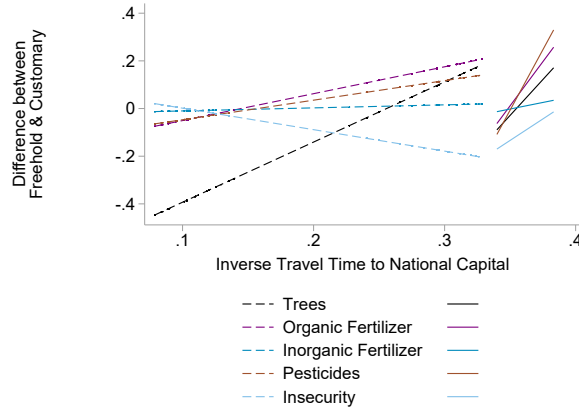


Figure 4: The divergence in input use across customary and freehold parcels is more responsive to land pressures closer to Kampala.

In more remote areas, outsider demand shaping insecurity seems to translate less intensely into patterns of agricultural investment.⁶⁰

This result, in conjunction with the prior linear specifications, has policy implications for where titling may have the greatest benefits. In general, customary tenure is less secure close to Kampala, where outside demand for land leaves customary parcels vulnerable to elite expropriation. This leads to an underinvestment in long-term agricultural inputs on customary parcels in these areas. From the threshold analysis, I conclude that investment on customary parcels is more sensitive to land pressures closer to Kampala. Therefore, securing rights to those parcels, such as by titling them and bringing them from the customary realm into freehold, should have larger benefits (such as unlocking agricultural productivity) close to Kampala.

This implies that the government should target titling activities in high land pressure environments, such as those easily accessed from Kampala. Not only are the benefits of clarifying customary rights and making them legally enforceable higher in areas with more demand from outsiders, the costs of titling, including specialized staff such as surveyors, are likely lower in more easily accessible areas. This pattern, where the net benefits of clarifying the potential ambiguity of customary land rights through formal documentation are higher in environments with higher land pressures, likely holds across a variety of customary tenure contexts on the continent. Indeed, there is evidence that formalization has been more successful in unlocking agricultural investment (and faced higher demand) in near-urban contexts in Burkina Faso (Huntington et al., 2023). Other constraints to agricultural investment, such as capital access, input availability, and output market access are also less likely to bind in these high-pressure areas. At the same time, the higher stakes of

⁶⁰Interestingly, relative tenure insecurity, which in the previous figures was downwards-sloping (customary parcels became relatively more insecure as land pressures increased) is here upwards-sloping although negative close to Kampala (implying that although customary land is less secure than freehold, the difference narrows closer to Kampala), but downwards-sloping further away. This is in contrast to the various investments, which retain their direction on both sides of the threshold. Perhaps perceptions of tenure insecurity have adapted to external threats close to Kampala.

documenting rights in locations with ready demand for clearly documented land may lead to elite capture of the titling process (Lawry et al., 2023).⁶¹

6 Conclusion

In this paper, I developed a theoretical model of short- and long-term agricultural investments on customary and freehold land. This model incorporates how expropriation incentives change as land pressures increase. Local elites, observing outsiders' new demand for land and inability to distinguish between overlapping customary rights, may choose to expropriate land used by non-elites in the community to sell. Even if uncommon in practice, the perception and fear of elite expropriation reduces tenure security on customary tenure parcels (in addition to other sources of tenure insecurity, which operate on all land). This, in turn, reduces the expected returns to long-term inputs.

The model predicts that long-term input use will diverge between customary and freehold parcels as land pressures increase in an area. Indeed, when I test the model's implications using survey data from four regions of Uganda, I find that long-term inputs (such as trees and organic fertilizer) are used more often on freehold land as land pressures increase, while on customary land this relationship is attenuated by tenure insecurity. Changing cropping patterns due to the proximity of urban markets appears to also influence short-term input use, but the effects on long-term inputs are larger, suggesting an independent effect of tenure security on long-term inputs.

The theoretical model and empirical results together suggest the importance of expanding beyond a simplification of customary tenure as less secure to thinking carefully about the social nature of land rights and the interacting incentives of multiple stakeholders. This insight allows researchers to make more detailed predictions about agricultural investment and may help explain the mixed results in much of the empirical literature. Secure customary tenure may become less secure as tenure systems intersect. This also suggests targeted policy to document and clarify rights in areas vulnerable to elite expropriation as land pressures mount to protect the rights of the most vulnerable. In particular, this work suggests that the net benefits of expensive documentation and formalization of rights will be highest in areas such as those accessible to cities that face high land pressures from outsiders, as the risk of expropriation of customary land is a more binding constraint on investment in those regions. Tenure insecurity is but one constraint on agricultural investment, and titling alone may not raise investment from the low levels seen in Uganda without complementary efforts to improve markets.

⁶¹For example, in Burkina Faso, local elections faced more political competition in anticipation of land documentation in municipalities closer to urban centers, consistent with political attempts to control the documentation process (Meinzen-Dick, 2026).

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A General Analytical Solution

Broadly, land tenure has been thought to encourage investment through three possible mechanisms, going back as far as the discussion in Feder (1988, cited in Place (2009)). These have been termed the assurance, collateralizability, and realizability mechanisms: stronger land rights should provide assurance that the farmer will be able to reap gains from investment, increase access to capital by leveraging the land as collateral, and allow the farmer to sell the land and realize gains sooner (as well as transferring land to those most able or likely to invest) (Besley, 1995, cited in Fenske (2011)).

Each of these mechanisms is the subject of an extensive theoretical and empirical literature, and their relevance to customary tenure in Africa has been long debated. Credit markets have been shown to be thin, farmers may be credit rationed for other reasons, and land markets are often missing or face other restrictions (Tabetando et al., 2023; Higgins et al., 2018).⁶² Lawry et al. (2017), reviewing a number of studies on property rights interventions, find no consistent evidence on the credit channel. My discussion will focus on the first of these mechanisms, which does not rely on the existence of functioning complementary markets.

Consider a two-period model, where the farmer has allocated a (fixed) share $0 \leq \alpha \leq 1$ of their exogenously-given land endowment L under tenure system h to staple crops s and the remainder $(1 - \alpha)$ to cash crops c . In the first period, the farmer choose to apply fertilizer, F_1^s and F_1^c , and trees, T^s and T^c , to the staple and cash crops respectively; each crop has its own production function and may have different marginal productivities of inputs. I leave these production functions general, only assuming diminishing marginal returns to each input ($\frac{\partial^2 f}{\partial F^2} < 0$, $\frac{\partial^2 f}{\partial T^2} < 0$). These assumptions allow me to argue that in cases where the marginal productivity of a given input are lower, the level of that input used is higher, and therefore to predict how investment will respond to changing land values, which is the empirically relevant hypothesis, even without estimating a full agronomic production function which is beyond the scope of this model. The production functions for each crop may differ, and in particular may have different marginal productivities of inputs. The farmer also has some wealth endowment, w_1 , and cannot borrow. When considering second period returns, the farmer has a discount rate of β and faces some probability that their land will be expropriated before the second period, in which case fixed investments (trees) would be lost as well. This probability is given by $(1 - \phi(T, h, vL))$ and could be a function of tree investment, tenure system, as well as the value of the land. In particular, $\phi(T, h, vL)$ could have the following properties:

⁶²In China, Li and Zhu (2023) for example find that a legal reform recognizing transfer rights increases investment, although land concentration seems to be an important mechanism, which may be less possible in Africa.

$$\phi(T, h, vL) = \begin{cases} 1 & \text{if } h = 1 \text{ (Freehold),} \\ g(T, vL) & \text{if } h = 0 \text{ (Customary).} \end{cases}$$

where $\frac{\partial g}{\partial T} > 0$, as in Place and Otsuka (2002), and $\frac{\partial g}{\partial v} < 0$. Intuitively, this implies that freehold land is perfectly secure (and not responsive to either tree planting or land values). Customary tenure security, by contrast, is increasing in tree planting (which demonstrates responsible stewardship of the land and thus guards against horizontal expropriation by neighbors or family members), and decreasing in land values (which heighten incentives for elite expropriation for sale to outsiders).

After working through the main features of the model, I will consider what different forms of this ϕ function could imply for the farmer's decisions.

The model captures the risk of expropriation on customary parcels in two ways. Simple models of customary tenure and investment consider customary land to have a higher risk of alienation than freehold land: $\phi(T, vL|h = 0) < \phi(T, vL|h = 1) \forall T, v, L$, with $\frac{\partial \phi}{\partial T} = 0$. That is, for any level of investment in trees or land values, the risk of expropriation is higher on customary land, and this risk does not change in response to investment in the land. This decreases the expected total marginal returns to investment in trees, as shown by a shift towards the origin of τ , decreasing the optimal investment in trees. Considering the equilibria conditions, the risk of expropriation $\phi < 1$ deflates the denominator of $\frac{\partial f}{\partial T}$, meaning the marginal productivity is higher, which occurs at lower levels of investment in trees given diminishing marginal returns.

Drawing upon qualitative understandings of customary tenure, though, more subtle models have sought to capture the endogenous nature of tenure security on customary land. Place and Otsuka (2002), for instance, model the probability of losing land as a decreasing function of planting trees, a visible investment in the land. In my model, this implies that $\frac{\partial \phi}{\partial T} > 0$, which would attenuate the leftward shift of the τ curve⁶³ (and could perhaps even lead to an 'overinvestment' in trees relative to the unconstrained optimum if the tenure-enhancing effect is large, although empirically this is rarely documented).

The equilibrium determined by the marginal productivity of trees under (less-secure) customary tenure, then, likely sees under-investment in trees relative to the allocation under freehold tenure ($T^{s0} < T^{s1}$ and $T^{c0} < T^{c1}$). I will empirically test this using data from Uganda.

⁶³Equivalently, it decreases the numerator of $\frac{\partial f}{\partial T}$, meaning the marginal productivity of trees are lower, occurring at higher levels of tree investment – attenuating the effect of ϕ on the denominator discussed above

The first period problem faced by the farmer is then to:

$$\max_{F_1^s, F_1^c, T^s, T^c} \alpha p_s f_s(F_1^s, T^s) + (1 - \alpha) p_c f_c(F_1^c, T^c) - p_F(F_1^s + F_1^c) - p_T(T^s + T^c) + \beta \phi(T^s + T^c, h, vL) \pi_2^*(F_2^s, F_2^c, T^s, T^c) \quad (2)$$

$$\text{subject to: } p_F(F_1^s + F_1^c) + p_T(T^s + T^c) \leq w_1 \quad (3)$$

While in the second period, the farmer only chooses fertilizer (F_2^s and F_2^c):

$$\max_{F_2^s, F_2^c} \alpha p_s f_s(F_2^s, T^s) + (1 - \alpha) p_c f_c(F_2^c, T^c) - p_F(F_2^s + F_2^c) \quad (4)$$

$$\begin{aligned} & p_F(F_2^s + F_2^c) \leq w_2 \\ \text{subject to: } & w_2 = w_1 + \alpha p_s f_s(F_1^s, T^s) + (1 - \alpha) p_c f_c(F_1^c, T^c) - p_F(F_1^s + F_1^c) - p_T(T^s + T^c) \end{aligned} \quad (5)$$

The First Order Conditions of this second stage are:

$$\alpha p_s \frac{\partial f_s}{\partial F_2^s} - p_F - \lambda_1 p_F = 0 \quad (6)$$

$$(1 - \alpha) p_c \frac{\partial f_c}{\partial F_2^c} - p_F - \lambda_1 p_F = 0 \quad (7)$$

$$\lambda_1 [p_F(F_2^s + F_2^c) - w_1 - \alpha p_s f_s(F_1^s, T^s) - (1 - \alpha) p_c f_c(F_1^c, T^c) + p_F(F_1^s + F_1^c) + p_T(T^s + T^c)] = 0 \quad (8)$$

Consider the case where the budget constraint does not bind in the second stage, as the constraint binding does not add substantively to understanding the mechanism at play. Therefore, $\lambda_1 = 0$.

The equilibrium conditions are then:

F_2^{s*} such that:

$$\frac{\partial f_s}{\partial F_2^s} = \frac{p_F}{\alpha p_s} \quad (9)$$

and F_2^{c*} such that:

$$\frac{\partial f_c}{\partial F_2^c} = \frac{p_F}{(1 - \alpha) p_c} \quad (10)$$

Therefore the value function of second period profit is:

$$\pi_2^*(T^s, T^c) = \alpha p_s f_s(F_2^{s*}, T^s) + (1 - \alpha) p_c f_c(F_2^{c*}, T^c) - p_F(F_2^{s*} + F_2^{c*}) \quad (11)$$

Which I write omitting fertilizer because at the optimum, the derivatives with respect to the first period choice variables are:

$$\frac{\partial \pi_2^*}{\partial F_1^s} = 0 \quad (12)$$

$$\frac{\partial \pi_2^*}{\partial F_1^c} = 0 \quad (13)$$

$$\frac{\partial \pi_2^*}{\partial T^s} = \alpha p_s \frac{\partial f_s}{\partial T^s} \Big|_{F_2^{s*}} \quad (14)$$

$$\frac{\partial \pi_2^*}{\partial T^c} = (1 - \alpha) p_c \frac{\partial f_c}{\partial T^c} \Big|_{F_2^{c*}} \quad (15)$$

Turning to the first stage, then:

$$\max_{F_1^s, F_1^c, T^s, T^c} \alpha p_s f_s(F_1^s, T^s) + (1 - \alpha) p_c f_c(F_1^c, T^c) - p_F(F_1^s + F_1^c) - p_T(T^s + T^c) + \beta \phi(T^s, T^c, h, vL) \pi_2^*(T^s, T^c) \quad (16)$$

$$\text{subject to: } p_F(F_1^s + F_1^c) + p_T(T^s + T^c) \leq w_1 \quad (17)$$

The First Order Conditions are then:

$$\alpha p_s \frac{\partial f_s}{\partial F_1^s} - p_F + \beta \phi(T, h, vL) \frac{\partial \pi_2^*}{\partial F_1^s} - \lambda_2 p_F = 0 \quad (18)$$

$$(1 - \alpha) p_c \frac{\partial f_c}{\partial F_1^c} - p_F + \beta \phi(T, h, vL) \frac{\partial \pi_2^*}{\partial F_1^c} - \lambda_2 p_F = 0 \quad (19)$$

$$\alpha p_s \frac{\partial f_s}{\partial T^s} - p_T + \beta \frac{\partial \phi}{\partial T} \pi_2^* + \beta \phi(T, h, vL) \frac{\partial \pi_2^*}{\partial T^s} - \lambda_2 p_T = 0 \quad (20)$$

$$(1 - \alpha) p_c \frac{\partial f_c}{\partial T^c} - p_T + \beta \frac{\partial \phi}{\partial T} \pi_2^* + \beta \phi(T, h, vL) \frac{\partial \pi_2^*}{\partial T^c} - \lambda_2 p_T = 0 \quad (21)$$

$$\lambda_2 [p_F(F_1^s + F_1^c) + p_T(T^s + T^c) - w_1] = 0 \quad (22)$$

Which simplify to:

$$\alpha p_s \frac{\partial f_s}{\partial F_1^s} - p_F - \lambda_2 p_F = 0 \quad (23)$$

$$(1 - \alpha) p_c \frac{\partial f_c}{\partial F_1^c} - p_F - \lambda_2 p_F = 0 \quad (24)$$

$$\alpha p_s \frac{\partial f_s}{\partial T^s} - p_T + \beta \frac{\partial \phi}{\partial T} \pi_2^* + \beta \phi(T, h, vL) \alpha p_s \frac{\partial f_s}{\partial T^s} \Big|_{F_2^{s*}} - \lambda_2 p_T = 0 \quad (25)$$

$$(1 - \alpha) p_c \frac{\partial f_c}{\partial T^c} - p_T + \beta \frac{\partial \phi}{\partial T} \pi_2^* + \beta \phi(T, h, vL) (1 - \alpha) p_c \frac{\partial f_c}{\partial T^c} \Big|_{F_2^{c*}} - \lambda_2 p_T = 0 \quad (26)$$

$$\lambda_2 [p_F(F_1^s + F_1^c) + p_T(T^s + T^c) - w_1] = 0 \quad (27)$$

Consider a farmer who has a sufficient initial endowment of wealth w_1 that the budget constraint does not bind in either period ($\lambda_1 = \lambda_2 = 0$)

The first-period FOCs are then:

$$\alpha p_s \frac{\partial f_s}{\partial F_1^s} - p_F = 0 \quad (28)$$

$$(1 - \alpha) p_c \frac{\partial f_c}{\partial F_1^c} - p_F = 0 \quad (29)$$

$$\alpha p_s \frac{\partial f_s}{\partial T^s} - p_T + \beta \frac{\partial \phi}{\partial T} \pi_2^* + \beta \phi(T, h, vL) \alpha p_s \frac{\partial f_s}{\partial T^s} \Big|_{F_2^{s*}} = 0 \quad (30)$$

$$(1 - \alpha) p_c \frac{\partial f_c}{\partial T^c} - p_T + \beta \frac{\partial \phi}{\partial T} \pi_2^* + \beta \phi(T, h, vL) (1 - \alpha) p_c \frac{\partial f_c}{\partial T^c} \Big|_{F_2^{c*}} = 0 \quad (31)$$

This then gives us:

$$\frac{\partial f_s}{\partial F_1^s} = \frac{p_F}{\alpha p_s} \quad (32)$$

$$\frac{\partial f_c}{\partial F_1^c} = \frac{p_F}{(1 - \alpha) p_c} \quad (33)$$

Which are the same conditions as in the second period. Therefore, in the totally unconstrained case, $F_1^{s*} = F_2^{s*}$ and $F_1^{c*} = F_2^{c*}$. Furthermore, if I assume that under freehold tenure, $\phi = 1$ and $\frac{\partial \phi}{\partial T} = 0$ (tenure is perfectly secure and planting trees has no effect on tenure security), then

$$\alpha p_s \frac{\partial f_s}{\partial T^s} - p_T + \beta \alpha p_s \frac{\partial f_s}{\partial T^s} \Big|_{F_2^{s*}} = 0 \quad (34)$$

$$(1 - \alpha) p_c \frac{\partial f_c}{\partial T^c} - p_T + \beta (1 - \alpha) p_c \frac{\partial f_c}{\partial T^c} \Big|_{F_2^{c*}} = 0 \quad (35)$$

Because $F_1^{s*} = F_2^{s*}$ and $F_1^{c*} = F_2^{c*}$ in this totally unconstrained case, $\frac{\partial f}{\partial T} = \frac{\partial f}{\partial T} \Big|_{F_2^*}$. The equilibrium conditions are then

$$\frac{\partial f_s}{\partial T^s} = \frac{p_T}{\alpha p_s(1 + \beta)} \quad (36)$$

$$\frac{\partial f_c}{\partial T^c} = \frac{p_T}{(1 - \alpha)p_c(1 + \beta)} \quad (37)$$

However, if $\phi < 1$, such as is possible under customary tenure, and $\frac{\partial \phi}{\partial T} > 0$ then the equilibrium conditions are:

$$\frac{\partial f_s}{\partial T^s} = \frac{p_T - \beta \frac{\partial \phi}{\partial T} \pi_2^*}{\alpha p_s(1 + \beta \phi)} \quad (38)$$

$$\frac{\partial f_c}{\partial T^c} = \frac{p_T - \beta \frac{\partial \phi}{\partial T} \pi_2^*}{(1 - \alpha)p_c(1 + \beta \phi)} \quad (39)$$

Compared to the freehold case, the inclusion of $\phi < 1$ increases the marginal productivity of trees, lowering optimal investment in this long term input; this effect is attenuated by the security-enhancing effect of tree planting ($\frac{\partial \phi}{\partial T} > 0$, therefore $p_T > p_T - \beta \frac{\partial \phi}{\partial T} \pi_2^*$). The model therefore accords with most theoretical models of tenure and long-term investments.

Let us now consider how optimal allocation would respond to a change in land pressures. As land pressures increase, particularly if this is driven by greater access to urban markets, we would expect the relative price of the cash crop p_c to increase. That is, compared to the prices of inputs and the staple crop output price, the price farmers can receive for their cash crops such as fruits and vegetables is higher with new access to relatively wealthy urban markets.

As p_c increases, $\frac{\partial f_c}{\partial F_1^c}$ decreases, which implies the optimal fertilizer use on cash crops increases. Furthermore, on freehold parcels, $\frac{\partial f_c}{\partial T^c}$ also would decrease as seen in equation (37), so optimal tree planting on cash crops increases. Optimal input use on staple crops will not changed for farmers whose wealth constraint does not bind.

On customary parcels, in equation (39), p_c will increase; importantly, however, the likelihood of expropriation from local elites who hold transfer rights to customary parcels also increases. In the model, this can be captured as $\frac{\partial \phi}{\partial v} < 0$: ϕ , the expected likelihood of retaining a parcel and its returns in the second period, decreases with rising land pressures. This ‘vertical’ expropriation does not recognize investments made in the land such as trees; assume for simplicity that ‘horizontal’ expropriation, such as border incursions, is

constant in its responsiveness to tree investment, so $\frac{\partial \phi}{\partial T}$ is constant as land pressures rise. Therefore, referring to the equilibria conditions for customary parcels (equations (38) and (39), repeated here):

$$\frac{\partial f_s}{\partial T^s} = \frac{p_T - \beta \frac{\partial \phi}{\partial T} \pi_2^*}{\alpha p_s (1 + \beta \phi)} \quad (40)$$

$$\frac{\partial f_c}{\partial T^c} = \frac{p_T - \beta \frac{\partial \phi}{\partial T} \pi_2^*}{(1 - \alpha) p_c (1 + \beta \phi)} \quad (41)$$

We can see that on staple crops, the reduction in ϕ increases $\frac{\partial f_s}{\partial T^s}$, implying that optimal tree planting on staple crops will decrease. On cash crops, this impact of a reduction in ϕ increasing $\frac{\partial f_c}{\partial T^c}$ works against the increase in p_c , which decreases the equilibrium marginal productivity of trees; the net effect of rising land pressures on tree planting for cash crops on customary parcels will depend on the relative magnitude of these two mechanisms.

This final point is the key effect of the model. Contrary to responses on freehold land, which simply increase optimal long- and short-term investment in cash crops, on customary land the impacts are also affected by changing tenure security. On staple crops, optimal long-term investment actually decreases. On cash crops, the response is theoretically ambiguous, but certainly lower than on freehold parcels.

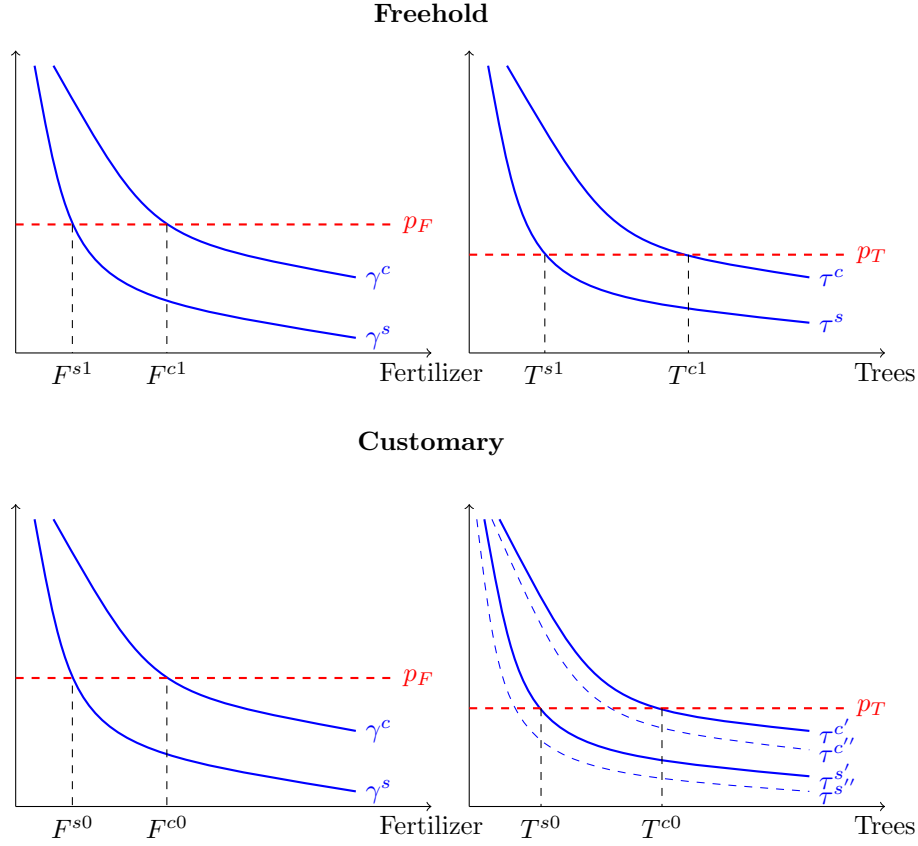


Figure 5: Equilibria conditions for fertilizer (left) and trees (right) investment on freehold (top) vs customary (bottom) land, for staple crops s and cash crops c . Expected total marginal productivity of fertilizer depicted by γ , and of trees depicted with τ . Optimal investment is where each curve crosses the dashed line, representing the price of the input. Optimal allocation of fertilizer is the same across tenure types, but on customary parcels, the risk of expropriation before the second period could lower the marginal productivity of trees to τ'' , implying lower investment in trees on customary parcels relative to freehold. If trees have a security-enhancing effect on customary parcels, this shift towards the axis is attenuated, represented by τ' .

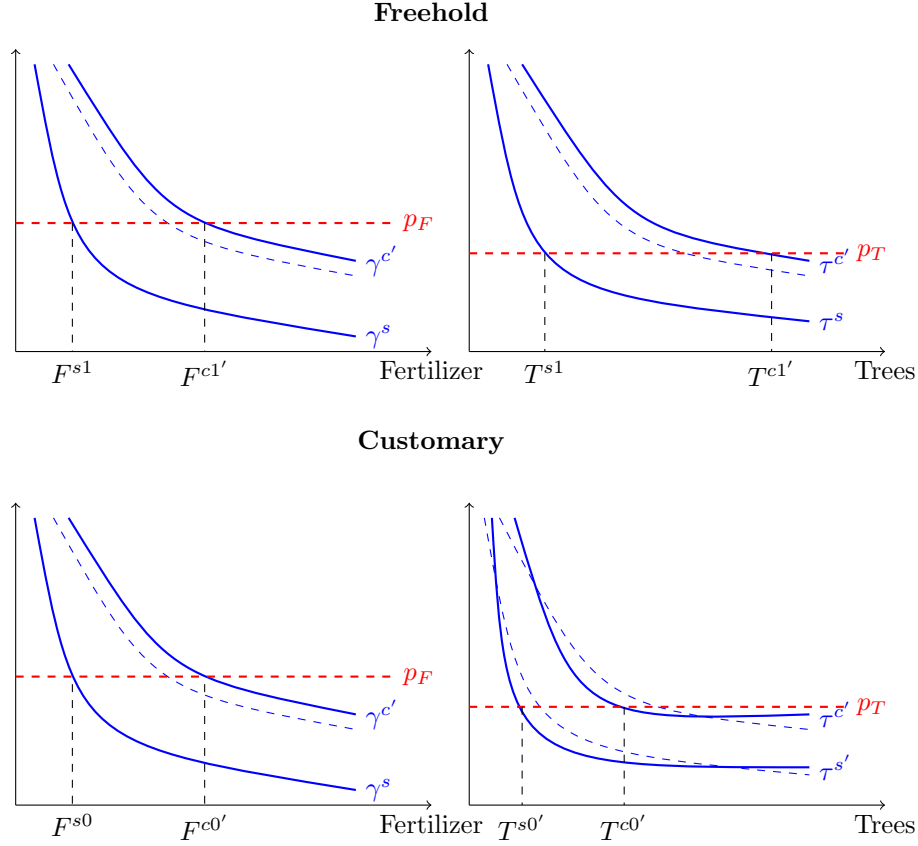


Figure 6: Equilibria conditions for fertilizer (left) and tree (right) investment on freehold (top) vs customary (bottom) land, for staple crops s and cash crops c , in a high land pressure environment. Expected total marginal productivity of fertilizer depicted by γ , and of trees depicted by τ . Optimal investment is where each curve crosses the dashed line, representing the price of the input. As land pressures increase, the relative output price of cash crops increases, raising the expected total marginal productivity of inputs on the cash crop from the dashed line to $\gamma^{c'}$ or $\tau^{c'}$, and thereby increasing optimal fertilizer and tree application to the cash crop on freehold land and fertilizer application to the cash crop on customary land. On customary land, however, increasing land pressures also increase the likelihood of expropriation, decreasing the expected total marginal productivity of trees on customary land. This results in the changed shape of the marginal productivity curves in the bottom right subfigure, with a lower optimal allocation of trees to both crops on customary land. This means that although optimal fertilizer use on customary parcels increases to $F^{c0'}$, optimal tree planting does not increase with higher land values and is instead depicted by $T^{s0'}$ and $T^{c0'}$.

B Continuous Measures of Input Use

One point at which many papers on tenure and investment allow their model and empirics to diverge is in the measurement of agricultural investment, the outcome of interest. Because many smallholders operate low-input, often subsistence farms, it is difficult to measure investment well. Therefore, many papers have used binary investment outcomes (Fenske, 2011). In the main text, I present binary outcomes using a linear probability model in order to allow for farmer group fixed effects (much as Deininger and Ali (2008) do).

However, the ACDP data also allows for continuous measures of investment, although these were measured with some error. Fenske (2011) has argued that measuring investment in a continuous way can make a substantial difference to the results. In particular, it may attenuate coefficients of interest by limiting identifying variation: in my model, although changing land pressures may increase optimal investment, the changes may not be large enough to lead a non-investing farmer to positive investment, particularly in a lumpy technology such as trees, which would be observationally equivalent to no impact. Fenske (2011) argues for the trimmed least absolute deviations estimator to deal with the prevalence of zeroes in types of investment. This is the panel equivalent of the Tobit estimator (Honore, 1992); as I only have a cross-section of data, I use the Tobit for continuous investment measures here. Coefficients on a Tobit model are interpreted as in a linear model; however, the linear effect is on the uncensored (non-zero) outcome. Broadly, these results are consistent with those in the binary outcomes case.

Above, I do find statistically significant relationships between tenure and binary investment outcomes, particularly when I explicitly consider the land pressure environment. Nevertheless, the ACDP survey's detailed agricultural production module allows me to construct continuous measures of long- and short-term investment: the value of fertilizer or pesticides applied. Unfortunately, I cannot construct a similar measure for the most long-term of my inputs, tree planting.

Tables A1 - A3 are generally consistent with the binary outcome results presented in the main text, with a greater divergence between customary and freehold parcels on longer-term inputs (organic fertilizer) than short-term inputs. However, I do not prioritize these results due to the lack of precision in reported valuations of inputs, which in the ACDP survey do not accord with those from other sources in Uganda.

VARIABLES	Value Organic Fertilizer Used			
	(1)	(2)	(3)	(4)
Rental Price	0.0948 (0.0925)	-0.0593*** (0.0146)		
Customary $\times$ Rental Price	0.416 (0.308)	-21.12*** (0.622)		
Inverse Travel Time			19.02** (9.309)	
Customary $\times$ Inverse Travel Time			4.714 (5.807)	
Decrease in Travel Time				85.78** (39.12)
Customary $\times$ Decrease in Travel Time				52.33** (24.63)
Customary	-1.034** (0.463)	0.0520 (0.142)	3.197 (2.919)	3.474 (3.234)
Wealth Index	0.200* (0.110)	0.117*** (0.00346)	0.179* (0.101)	0.141* (0.0814)
Landholdings	0.0231 (0.0268)	0.101*** (0.0131)	0.0365 (0.0241)	0.0533** (0.0219)
Cash Crop	3.720** (1.834)	0.394*** (0.117)	3.334* (1.772)	1.803* (1.069)
First Year Farming	0.0303 (0.0260)	0.0310*** (9.76e-05)	0.0268 (0.0246)	0.0256 (0.0233)
Observations	3,254	3,254	3,236	3,236
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

*** p<0.01, ** p<0.05, * p<0.1

Table A1: Tobit results with the outcome of the value of organic fertilizer applied to a parcel (a medium-term investment) regressed on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
	Value Inorganic Fertilizer Used			
Rental Price	0.0263*** (0.00540)	0.0255*** (0.000310)		
Customary $\times$ Rental Price	0.0188 (0.0341)	-0.532*** (0.0191)		
Inverse Travel Time			2.972*** (1.032)	
Customary $\times$ Inverse Travel Time			2.130* (1.125)	
Decrease in Travel Time				3.527 (2.520)
Customary $\times$ Decrease in Travel Time				1.599 (1.849)
Customary	0.103 (0.132)	0.0719*** (0.0127)	0.345 (0.342)	0.310 (0.237)
Wealth Index	0.0177*** (0.00626)	0.0105*** (0.000320)	0.0131** (0.00573)	0.0145** (0.00587)
Landholdings	-0.00290 (0.00450)	-0.00831*** (0.00156)	0.000238 (0.00547)	-0.000250 (0.00493)
Cash Crop	-0.0176 (0.0598)	0.0238 (0.0186)	-0.0154 (0.0574)	-0.0673 (0.0824)
First Year Farming	0.00612* (0.00321)	0.00531*** (7.70e-06)	0.00631* (0.00324)	0.00668** (0.00327)
Observations	3,254	3,254	3,236	3,236
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A2: Tobit results with the outcome of the value of inorganic fertilizer applied to a parcel (a short-term investment) regressed on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Value Pesticides Used		
Rental Price	0.0935 (0.128)	0.0520*** (0.00260)		
Customary $\times$ Rental Price	0.136 (0.137)	-0.497*** (0.0205)		
Inverse Travel Time			9.216 (5.914)	
Customary $\times$ Inverse Travel Time			19.33 (13.99)	
Decrease in Travel Time				22.47* (12.66)
Customary $\times$ Decrease in Travel Time				4.450 (8.374)
Customary	-0.327 (0.754)	-0.344*** (0.117)	-3.381 (2.837)	1.700 (1.297)
Wealth Index	0.0681*** (0.0249)	0.0517*** (0.00118)	0.0492*** (0.0158)	0.0497*** (0.0154)
Landholdings	-0.00741 (0.0179)	-0.0325*** (0.00466)	-0.00643 (0.0205)	-0.0116 (0.0237)
Cash Crop	1.773** (0.718)	0.874*** (0.0358)	1.707** (0.744)	1.393** (0.587)
First Year Farming	0.00469 (0.00559)	0.00880*** (3.18e-05)	0.00184 (0.00646)	0.00350 (0.00566)
Observations	3,254	3,254	3,236	3,236
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A3: Tobit results with the outcome of the value of pesticides applied to a parcel (a medium-term investment) regressed on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

C Robustness Checks

C.1 Expanded Geographic Sample

The baseline survey for the impact evaluation of ACDP was conducted in four districts in Uganda, with an extensive survey including all agricultural inputs and outputs. However, a more concise survey was additionally conducted for monitoring and evaluation (M&E) purposes in an additional 30 districts located around the country. This M&E survey only asks about agricultural investments on a single target crop in each district (rice, beans, maize, cassava, or coffee), and is therefore not well-suited for the main analysis conducted above. Nevertheless, it does contain parcel-level tenure characteristics, such as perceived insecurity and tenure system, as well as rental prices and GPS locations that allow me to match with travel time data as above. Therefore, I am able to repeat the exercise from panel A of table 1 using this larger, more geographically diverse sample in table A4. The results are, if anything, even stronger: tenure insecurity is unrelated to rising land pressures on freehold land, but customary parcels are felt more insecure in higher land pressure environments.

VARIABLES	Average Insecurity			
	(1)	(2)	(3)	(4)
Rental Price	-0.00121 (0.00110)	0.000555 (0.00128)		
Customary $\times$ Rental Price	5.639*** (0.687)	5.079*** (1.097)		
Inverse Travel Time			-0.306 (0.271)	
Customary $\times$ Inverse Travel Time			1.113*** (0.237)	
Decrease in Travel Time				0.453 (0.308)
Customary $\times$ Decrease in Travel Time				0.580** (0.258)
Customary	0.0766** (0.0345)	0.0660* (0.0358)	-0.302*** (0.0734)	0.00312 (0.0388)
Wealth Index	2.44e-05 (0.000973)	-0.000557 (0.000984)	-0.000797 (0.00105)	-0.000688 (0.000987)
Landholdings	2.83e-07 (2.49e-07)	2.54e-06*** (3.07e-07)	4.92e-07** (2.31e-07)	4.32e-07* (2.49e-07)
Observations	10,253	10,229	10,283	10,283
R-squared	0.256	0.331	0.256	0.251
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

*** p<0.01, ** p<0.05, * p<0.1

Table A4: OLS results of average tenure insecurity across all rightsholders (scale of 1-5, with higher values indicating more insecure) on a parcel on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time. Data comes not only from the Impact Evaluation sample used throughout the paper, but also from the more condensed ACDP Monitoring and Evaluation survey conducted in an additional 30 districts.

C.2 Spatial Correlation

In my primary specification above, I cluster standard errors at the parish level to account for any geographic relationships between households in the same parish. Here, I model the spatial dependencies more explicitly, using Conley (1999)’s spatially dependent standard errors. I allow for spatial dependence up to 50km, which would include all households within a district. These regressions, as in the main paper, include controls for how the parcel was acquired, plus the shown wealth index and total household landholdings. These results are the most different of all my robustness checks. In particular, in table A6, the coefficient on Customary $\times$ Inverse Travel Time becomes negative and statistically significant (at the 10% level), when in most other specifications (such as panel A of table 2) it is positive but insignificant. However, this is if anything more consistent with the theoretical model, where long-term input use is attenuated on customary parcels as land pressures rise. Similarly, the coefficient on Customary $\times$ Decrease in Travel Time switching from positive and significant in panel B of table 2 to negative and significant in table A7 is more in line with the model’s predictions of attenuation on customary parcels for this medium-term input. On the other hand, the same pattern is observed for pesticides in table A9, which is slightly more difficult to explain but may be due to spatial correlation in pest damage that is accounted for in this estimation.

VARIABLES	(1)	(2)	(3)	(4)
	Average Insecurity			
Rental Price	0.0213 (0.0227)	0.0247 (0.0170)		
Customary $\times$ Rental Price	0.0851*** (0.0203)	0.0953*** (0.00829)		
Inverse Travel Time			-1.397*** (0.368)	
Customary $\times$ Inverse Travel Time			2.164*** (0.630)	
Decrease in Travel Time				-2.651* (1.485)
Customary $\times$ Decrease in Travel Time				2.465 (2.230)
Customary	0.103 (0.0988)	0.0759 (0.0692)	-0.499*** (0.149)	-0.149 (0.159)
Wealth Index	-0.00221 (0.00141)	-0.00164 (0.00222)	-0.00143 (0.000916)	-0.000411 (0.000419)
Landholdings	-0.00440 (0.00380)	-0.00549* (0.00330)	-0.00544* (0.00326)	-0.00592* (0.00326)
Cash Crop	0.0564 (0.0680)	0.00479 (0.0727)	0.0764 (0.0845)	0.104 (0.0916)
First Year Farming	0.00149*** (0.000536)	0.00127** (0.000609)	0.00139*** (0.000489)	0.00130*** (0.000450)
Constant	-1.321 (1.146)	-1.228 (1.393)	-0.846 (1.065)	-0.811 (1.004)
Observations	3,240	3,240	3,222	3,222
R-squared	0.261	0.382	0.259	0.253
Fixed Effect	None	Group	None	None
SE	Spatial - 50km	Spatial - 50km	Spatial - 50km	Spatial - 50km
Standard errors in parentheses				
*** p<0.01, ** p<0.05, * p<0.1				

Table A5: GMM results of average tenure insecurity across all rightsholders (scale of 1-5, with higher values indicating more insecure) on a parcel on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are computed accounting for spatial dependence up to 50 kilometers (the size of a district) in all specifications (reported in parentheses) (Conley, 1999). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
			Trees	
Rental Price	0.0127 (0.00950)	0.00844 (0.00907)		
Customary $\times$ Rental Price	-0.0563*** (0.00731)	-0.0461*** (0.00923)		
Inverse Travel Time			0.886 (0.563)	
Customary $\times$ Inverse Travel Time			-0.886*** (0.289)	
Decrease in Travel Time				1.007 (1.286)
Customary $\times$ Decrease in Travel Time				-2.710** (1.183)
Customary	0.0283 (0.0758)	0.0233 (0.0739)	0.276*** (0.0319)	0.299*** (0.0985)
Wealth Index	-0.00162 (0.00171)	0.00167** (0.000722)	-0.00264 (0.00183)	-0.00139 (0.00142)
Landholdings	0.000321 (0.00169)	0.00147 (0.00126)	0.000742 (0.00175)	6.51e-05 (0.00171)
Cash Crop	0.146** (0.0595)	0.130*** (0.0404)	0.133*** (0.0372)	0.137*** (0.0477)
First Year Farming	-0.00285*** (0.00103)	-0.00324** (0.00163)	-0.00284** (0.00117)	-0.00287*** (0.00107)
Constant	6.208*** (1.980)	6.491** (3.270)	6.011*** (2.195)	6.143*** (1.986)
Observations	3,250	3,250	3,232	3,232
R-squared	0.061	0.242	0.071	0.066
Fixed Effect	None	Group	None	None
SE	Spatial - 50km	Spatial - 50km	Spatial - 50km	Spatial - 50km
Standard errors in parentheses				
*** p<0.01, ** p<0.05, * p<0.1				

Table A6: GMM results of a dummy equal to one for parcels with trees planted on them (a long term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are computed accounting for spatial dependence up to 50 kilometers (the size of a district) in all specifications (reported in parentheses) (Conley, 1999). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Organic Fertilizer		
Rental Price	-0.000986 (0.00105)	-0.00120 (0.000761)		
Customary $\times$ Rental Price	0.0147** (0.00607)	0.00428** (0.00170)		
Inverse Travel Time			0.541 (0.371)	
Customary $\times$ Inverse Travel Time			-0.245 (0.314)	
Decrease in Travel Time				3.411*** (0.885)
Customary $\times$ Decrease in Travel Time				-2.219*** (0.535)
Customary	-0.0661 (0.0544)	-0.0184 (0.0211)	0.000714 (0.0327)	0.170*** (0.0511)
Wealth Index	0.00398*** (0.00110)	0.000915*** (0.000341)	0.00318*** (0.000861)	0.00100** (0.000428)
Landholdings	0.00102 (0.00140)	0.00127 (0.00120)	0.00177 (0.00164)	0.00242* (0.00131)
Cash Crop	0.207*** (0.0189)	0.0368*** (0.00782)	0.194*** (0.0189)	0.137*** (0.0170)
First Year Farming	-1.78e-05 (0.000233)	-0.000113 (0.000662)	-0.000117 (0.000246)	-4.07e-05 (0.000265)
Constant	0.0102 (0.509)	0.197 (1.382)	0.0988 (0.458)	-0.159 (0.505)
Observations	3,248	3,248	3,230	3,230
R-squared	0.169	0.322	0.175	0.224
Fixed Effect	None	Group	None	None
SE	Spatial - 50km	Spatial - 50km	Spatial - 50km	Spatial - 50km
Standard errors in parentheses				
*** p<0.01, ** p<0.05, * p<0.1				

Table A7: GMM results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are computed accounting for spatial dependence up to 50 kilometers (the size of a district) in all specifications (reported in parentheses) (Conley, 1999). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Inorganic Fertilizer		
Rental Price	0.00330*** (0.000359)	0.00497*** (0.000932)		
Customary $\times$ Rental Price	0.00564 (0.00632)	-0.000753 (0.00164)		
Inverse Travel Time			0.750*** (0.0896)	
Customary $\times$ Inverse Travel Time			-0.162 (0.0994)	
Decrease in Travel Time				1.396*** (0.513)
Customary $\times$ Decrease in Travel Time				-0.739* (0.418)
Customary	0.00844 (0.0445)	0.00489 (0.0478)	0.0528** (0.0227)	0.0867*** (0.0328)
Wealth Index	0.00456*** (0.000850)	0.00203*** (0.000546)	0.00323*** (0.00106)	0.00332*** (0.00125)
Landholdings	0.00168*** (0.000557)	0.00148*** (0.000304)	0.00279*** (0.000742)	0.00269*** (0.000553)
Cash Crop	0.0151 (0.0144)	0.0143*** (0.00524)	0.00318 (0.0125)	-0.0126** (0.00598)
First Year Farming	0.00129** (0.000541)	0.000900 (0.000606)	0.00122** (0.000551)	0.00133*** (0.000475)
Constant	-2.677** (1.119)	-1.920 (1.238)	-2.703** (1.132)	-2.866*** (0.994)
Observations	3,245	3,245	3,227	3,227
R-squared	0.029	0.150	0.057	0.041
Fixed Effect	None	Group	None	None
SE	Spatial - 50km	Spatial - 50km	Spatial - 50km	Spatial - 50km
Standard errors in parentheses				
*** p<0.01, ** p<0.05, * p<0.1				

Table A8: GMM results of a dummy equal to one for parcels with inorganic fertilizer applied (a short term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are computed accounting for spatial dependence up to 50 kilometers (the size of a district) in all specifications (reported in parentheses) (Conley, 1999). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Pesticides		
Rental Price	-0.00142 (0.000955)	-0.00205*** (0.000473)		
Customary $\times$ Rental Price	0.0130* (0.00697)	-0.0103* (0.00530)		
Inverse Travel Time			0.855*** (0.0652)	
Customary $\times$ Inverse Travel Time			-0.271 (0.334)	
Decrease in Travel Time				2.155*** (0.453)
Customary $\times$ Decrease in Travel Time				-2.147*** (0.758)
Customary	-0.0315 (0.0973)	-0.0441 (0.0990)	0.0455** (0.0229)	0.192*** (0.0547)
Wealth Index	0.00590*** (0.000723)	0.00335** (0.00141)	0.00424*** (0.00106)	0.00424*** (0.00139)
Landholdings	0.00151 (0.00150)	0.000662* (0.000385)	0.00205* (0.00118)	0.00184** (0.000823)
Cash Crop	0.145*** (0.0201)	0.0876*** (0.00684)	0.130*** (0.0106)	0.106*** (0.0146)
First Year Farming	0.000623** (0.000257)	0.000547 (0.000445)	0.000533* (0.000276)	0.000606** (0.000304)
Constant	-1.241** (0.490)	-1.175 (0.874)	-1.218** (0.545)	-1.351** (0.580)
Observations	3,244	3,244	3,226	3,226
R-squared	0.079	0.203	0.103	0.096
Fixed Effect	None	Group	None	None
SE	Spatial - 50km	Spatial - 50km	Spatial - 50km	Spatial - 50km
Standard errors in parentheses				
*** p<0.01, ** p<0.05, * p<0.1				

Table A9: GMM results of a dummy equal to one for parcels with pesticides applied (a short term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are computed accounting for spatial dependence up to 50 kilometers (the size of a district) in all specifications (reported in parentheses) (Conley, 1999). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

C.3 Splitting Sample

The relationship between land pressures and input use on customary vs. freehold land may not be linear, as modeled in my primary specifications. Therefore, I also look at binary splits in the various land pressure measures, and compare outcomes on customary vs. freehold parcels in low vs. high land pressure environments using interactions.

First, in table A10, I create a dummy variable equal to one for all parcels that have an estimated rental price per acre above the median rental price in their district, and consider those parcels to be facing higher land pressures. The results accord with those above: input use is similar on customary and freehold parcels in low-land pressure areas; higher land pressure parcels are more likely to have inputs applied; but this effect is attenuated on customary parcels. I check, in table A11, that these results are not just driven by the distribution of customary vs freehold parcels across low vs. high land value areas, but the distribution is relatively similar.

Then, in table A12, I compare across districts, considering those close to Kampala (Masaka & Iganga) as facing high land pressures and those further away (Amuru & Ntungamo) having lower land pressures. This distinction is somewhat similar to the prior threshold of two hours travel time that I use in my threshold analysis. Here again, results are consistent with my model: input use is generally less common on customary parcels, more likely in districts near Kampala, but the difference between customary and freehold parcels is significantly different in districts near Kampala. Once again, customary and freehold parcels are distributed across both sets of districts, as seen in table A13.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Trees		Organic Fertilizer		Inorganic Fertilizer		Pesticides		Average Insecurity	
Customary	0.0888** (0.0382)	0.0695* (0.0394)	-0.0392* (0.0228)	0.00539 (0.0163)	0.0157 (0.0395)	0.0148 (0.0425)	0.0259 (0.0455)	0.0211 (0.0405)	0.154* (0.0791)	0.113 (0.0868)
Higher Price	0.0805** (0.0385)	0.0922** (0.0446)	0.0812*** (0.0222)	0.0193 (0.0161)	-0.0535*** (0.0165)	-0.0330* (0.0179)	0.0192 (0.0180)	0.0235* (0.0134)	0.0803 (0.104)	0.0644 (0.0943)
Customary $\times$ Higher Price	-0.131** (0.0539)	-0.105* (0.0606)	-0.0556** (0.0222)	-0.0542* (0.0277)	-0.0263 (0.0330)	-0.0282 (0.0296)	-0.123*** (0.0401)	-0.137*** (0.0431)	-0.0814 (0.113)	-0.0574 (0.1000)
Constant	6.355*** (1.655)	6.783*** (2.210)	0.154 (1.034)	0.317 (0.973)	-2.735*** (0.984)	-2.020* (1.131)	-1.146 (0.964)	-0.896 (1.162)	-0.989 (1.221)	-0.871 (1.707)
Observations	3,268	3,268	3,266	3,266	3,263	3,263	3,262	3,262	3,258	3,258
R-squared	0.063	0.244	0.174	0.323	0.037	0.152	0.084	0.206	0.252	0.372
Fixed Effect	None	Group	None	Group	None	Group	None	Group	None	Group
Cluster	Parish	Parish	Parish	Parish	Parish	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A10: OLS results where the outcome in columns (1) - (8) is a dummy equal to one if the given input was applied to the parcel, and in columns (9) - (10) is the average tenure insecurity across all rightsholders (scale of 1-5, with higher values indicating more insecure). Customary is a dummy equal to one for customary parcels and zero otherwise. Higher price is a dummy equal to one if the parcel has an estimated rental price per acre above the median rental price in the district. Customary $\times$ Higher price is the interaction of these two dummies. Controls (not shown) include the Progress out of Poverty wealth index, total household landholdings in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in columns (2), (4), (6), (8), and (10), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Differences in number of observations between outcomes is due to missing data for some outcomes.

Tenure Status	Low Rental Price	High Rental Price	Total
Freehold	2,003	1,905	3,908
Customary	1,203	900	2,103
Total	3,206	2,805	6,011

Table A11: Customary and Freehold parcels are found in relatively similar proportions in low and high rental value environments.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Trees		Organic Fertilizer	Inorganic Fertilizer			Pesticides		Average Insecurity	
Customary	0.190** (0.0825)	0.0884* (0.0509)	-0.0319 (0.0272)	-0.00359 (0.0261)	0.0419 (0.0317)	0.0180 (0.0276)	0.0209 (0.0369)	-0.0169 (0.0314)	-0.349** (0.160)	-0.453** (0.170)
Near Kampala	0.258*** (0.0571)	0.577*** (0.0203)	0.101*** (0.0296)	-0.00561 (0.0110)	0.152*** (0.0172)	-0.0294 (0.0208)	0.160*** (0.0314)	-0.0157 (0.0207)	-0.365*** (0.0864)	1.064*** (0.0389)
Customary $\times$ Near Kampala	-0.184* (0.0911)	-0.0774 (0.0650)	-0.0397 (0.0345)	-0.0185 (0.0218)	-0.0402* (0.0206)	-0.0191 (0.0284)	-0.0579 (0.0528)	-0.0299 (0.0511)	0.521*** (0.152)	0.611*** (0.170)
Constant	6.016*** (1.674)	6.415*** (2.068)	-0.0198 (1.028)	0.187 (0.942)	-2.786*** (0.966)	-2.068* (1.128)	-1.344 (0.906)	-1.174 (1.067)	-0.820 (1.343)	0.230 (1.836)
Observations	3,268	3,268	3,266	3,266	3,263	3,263	3,262	3,262	3,258	3,258
R-squared	0.087	0.240	0.174	0.322	0.055	0.149	0.099	0.201	0.266	0.380
Fixed Effect	None	Group	None	Group	None	Group	None	Group	None	Group
Cluster	Parish	Parish	Parish	Parish	Parish	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A12: OLS results where the outcome in columns (1) - (8) is a dummy equal to one if the given input was applied to the parcel, and in columns (9) - (10) is the average tenure insecurity across all rightsholders (scale of 1-5, with higher values indicating more insecure). Customary is a dummy equal to one for customary parcels and zero otherwise. Near Kampala is a dummy equal to one if the parcel is in Masaka or Iganga districts, which are closer to Kampala. Customary $\times$ Near Kampala is the interaction of these two dummies. Controls (not shown) include the Progress out of Poverty wealth index, total household landholdings in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in columns (2), (4), (6), (8), and (10), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Differences in number of observations between outcomes is due to missing data for some outcomes.

Tenure Status	Far from Kampala	Near Kampala	Total
Freehold	835	3,948	4,783
Customary	962	1,364	2,326
Total	1,797	5,312	7,109

Table A13: Customary and Freehold parcels are distributed across districts near and far from Kampala, although freehold is relatively more common near Kampala than further away.

C.4 Threshold Analysis

In this section, I present the regression results looking separately on either side of a threshold of 2 hours travel time from Kamapala (identified in interviews as being a reasonable radius for external demand for land).

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
			Average Insecurity			
Rental Price	0.0174 (0.0202)	-0.108 (0.464)				
Customary $\times$ Rental Price	0.101*** (0.0155)	10.34*** (1.463)				
Inverse Travel Time			-1.697*** (0.546)	-2.278 (5.110)		
Customary $\times$ Inverse Travel Time			0.709 (0.714)	-4.989 (6.236)		
Decrease in Travel Time					-2.093 (1.504)	-3.951* (1.902)
Customary $\times$ Decrease in Travel Time					-0.549 (1.394)	-4.084** (1.505)
Observations	2,387	817	2,405	817	2,405	817
R-squared	0.323	0.141	0.319	0.131	0.310	0.138
Fixed Effect	None	None	None	None	None	None
Cluster	Parish	Parish	Parish	Parish	Parish	Parish
Sample	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala
Robust standard errors in parentheses						
*** p<0.01, ** p<0.05, * p<0.1						

Table A14: Regression of average tenure insecurity across all rightholders (scale of 1-5, with higher values indicating more insecure) on a parcel on measures of land pressures. OLS regressions were run separately within and outside 2 hours travel time from Kampala, in paired sets of columns, with standard errors clustered at the parish level reported in parentheses. All specifications control for a dummy equal to one for customary parcels (and zero otherwise), the Progress out of Poverty wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
			Trees			
Rental Price	0.0123 (0.00797)	0.0785 (0.176)				
Customary $\times$ Rental Price	-0.0395*** (0.0113)	1.461 (0.902)				
Inverse Travel Time			1.382*** (0.438)	2.264 (3.986)		
Customary $\times$ Inverse Travel Time			0.302 (0.597)	-14.69*** (4.586)		
Decrease in Travel Time					1.224 (0.912)	1.000 (0.946)
Customary $\times$ Decrease in Travel Time					-1.508 (1.061)	-3.195* (1.735)
Observations	2,399	815	2,417	815	2,417	815
R-squared	0.061	0.099	0.086	0.116	0.066	0.105
Fixed Effect	None	None	None	None	None	None
Cluster	Parish	Parish	Parish	Parish	Parish	Parish
Sample	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table A15: Regression of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures. OLS regressions were run separately within and outside 2 hours travel time from Kampala, in paired sets of columns, with standard errors clustered at the parish level reported in parentheses. All specifications control for a dummy equal to one for customary parcels (and zero otherwise), the Progress out of Poverty wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
			Used Organic Fertilizer			
Rental Price	-0.000891 (0.00116)	0.0408 (0.127)				
Customary $\times$ Rental Price	0.0157*** (0.00412)	-0.265 (0.554)				
Inverse Travel Time			0.952*** (0.278)	11.30*** (1.503)		
Customary $\times$ Inverse Travel Time			0.403*** (0.126)	6.951*** (1.593)		
Decrease in Travel Time					3.960*** (0.562)	4.967*** (0.770)
Customary $\times$ Decrease in Travel Time					1.278*** (0.341)	2.156* (1.168)
Observations	2,396	816	2,414	816	2,414	816
R-squared	0.184	0.149	0.206	0.220	0.248	0.236
Fixed Effect	None	None	None	None	None	None
Cluster	Parish	Parish	Parish	Parish	Parish	Parish
Sample	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A16: Regression of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures. OLS regressions were run separately within and outside 2 hours travel time from Kampala, in paired sets of columns, with standard errors clustered at the parish level reported in parentheses. All specifications control for a dummy equal to one for customary parcels (and zero otherwise), the Progress out of Poverty wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
			Used Inorganic Fertilizer			
Rental Price	0.00353 (0.00289)	0.0632 (0.0990)				
Customary $\times$ Rental Price	0.00882 (0.00650)	-1.498** (0.536)				
Inverse Travel Time			0.694*** (0.135)	0.318 (1.792)		
Customary $\times$ Inverse Travel Time			0.559*** (0.152)	-4.182 (2.912)		
Decrease in Travel Time					1.388** (0.588)	-0.238 (0.608)
Customary $\times$ Decrease in Travel Time					0.561 (0.392)	-1.711 (1.237)
Constant	-2.529** (1.000)	-1.144 (2.063)	-2.796** (1.038)	-1.164 (2.191)	-2.815*** (0.966)	-1.248 (2.020)
Observations	2,396	813	2,414	813	2,414	813
R-squared	0.031	0.027	0.055	0.027	0.043	0.027
Fixed Effect	None	None	None	None	None	None
Cluster	Parish	Parish	Parish	Parish	Parish	Parish
Sample	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A17: Regression of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of land pressures. OLS regressions were run separately within and outside 2 hours travel time from Kampala, in paired sets of columns, with standard errors clustered at the parish level reported in parentheses. All specifications control for a dummy equal to one for customary parcels (and zero otherwise), the Progress out of Poverty wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
			Used Pesticides			
Rental Price	-0.00102 (0.00354)	0.116 (0.134)				
Customary $\times$ Rental Price	0.00811 (0.00574)	-0.211 (1.072)				
Inverse Travel Time			0.877*** (0.179)	4.075 (2.695)		
Customary $\times$ Inverse Travel Time			0.369 (0.293)	-10.94** (3.907)		
Decrease in Travel Time					2.062*** (0.618)	1.308 (1.020)
Customary $\times$ Decrease in Travel Time					-0.331 (0.558)	-5.531*** (1.383)
Constant	-0.732 (1.081)	-2.194 (2.782)	-1.095 (1.034)	-3.554 (2.890)	-1.006 (0.961)	-2.566 (2.652)
Observations	2,394	814	2,412	814	2,412	814
R-squared	0.103	0.060	0.123	0.076	0.120	0.084
Fixed Effect	None	None	None	None	None	None
Cluster	Parish	Parish	Parish	Parish	Parish	Parish
Sample	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala	> 2 hours from Kampala	< 2 hours from Kampala

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A18: Regression of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of land pressures. OLS regressions were run separately within and outside 2 hours travel time from Kampala, in paired sets of columns, with standard errors clustered at the parish level reported in parentheses. All specifications control for a dummy equal to one for customary parcels (and zero otherwise), the Progress out of Poverty wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition.

C.5 Parish Fixed Effects

My preferred specification makes use of the farmer group recruitment structure of the ACDP data to account for unobserved heterogeneity in farmer quality, which is likely to be shared among co-members of a self-selected farmer group. However, here I also present regressions using the more traditional administrative unit (parish in this case, given inconsistent identification of villages in the data) fixed effects. Results are consistent with those presented in the body of the paper.

VARIABLES	(1) Trees	(2) Used Organic	(3) Used Inorganic	(4) Used Pesticides	(5) Insecurity
Rental Price	0.0120 (0.00982)	3.30e-05 (0.000662)	0.00283 (0.00255)	-0.00274 (0.00290)	0.0218 (0.0203)
Customary $\times$ Rental Price	-0.0420*** (0.00716)	0.00338 (0.00214)	-0.00765*** (0.00260)	-0.00773** (0.00312)	0.113*** (0.0176)
Customary	0.0325 (0.0504)	-0.0220 (0.0241)	0.00680 (0.0403)	-0.0431 (0.0687)	0.104 (0.0814)
Wealth Index	0.000590 (0.000949)	0.00101* (0.000541)	0.00291*** (0.000951)	0.00393*** (0.000778)	-0.00124 (0.00248)
Landholdings	0.000376 (0.00168)	0.00137 (0.000988)	0.00189** (0.000725)	0.00169 (0.00128)	-0.00550 (0.00335)
Cash Crop	0.136*** (0.0396)	0.0390 (0.0276)	0.0210 (0.0216)	0.0980*** (0.0202)	0.0464 (0.0821)
First Year Farming	-0.00353*** (0.00100)	-8.47e-05 (0.000391)	0.00109** (0.000454)	0.000559 (0.000467)	0.00167** (0.000729)
Constant	7.936*** (2.025)	0.223 (0.816)	-2.266** (0.927)	-1.153 (0.935)	-1.527 (1.438)
Observations	3,250	3,248	3,245	3,244	3,240
R-squared	0.156	0.298	0.088	0.152	0.297
Fixed Effect	Parish	Parish	Parish	Parish	Parish
Cluster	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A19: OLS results of a dummy for the given input applied to the parcel (in columns (1) - (4) or average tenure security across all rightsholders on the parcel (scale of 1-5, with higher value indicating more insecure, reported in column (5)) on standardized rental prices for freehold and customary parcels, controlling for the Progress out of Poverty wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Parish fixed effects are included in all specifications, and standard errors are clustered at the parish level for all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise.

C.6 Alternate Measures

C.6.1 Validating Perceptions of Insecurity

In taking stock of the body of empirical evidence on tenure security and investment, several papers have noted the variety of definitions of tenure security as well as proxies used empirically (Arnot et al., 2011; Deininger and Ali, 2008). Common empirical measures include the existence of legal title, duration of tenure, method of acquisition, tenure type, existence of a conflict over the land, subjective perceptions of tenure security, and existence of particular rights (often transfer rights) (Arnot et al., 2011). It is often left implicit how these proxies are related to tenure *security*, which is the parameter modeled. Fenske (2011) notes that the choice of proxy does seem to be related to the effect found, and therefore this is a crucial choice. I do have a measure of subjective tenure security, which does respond to changing land pressures as in the model; here, I show in table A20 that alternate proxies such as documentation and past experiences of disputes have the expected correlations with perceived tenure security, and in table A21 that hypothetical elicitations of tenure security follow similar patterns.

My model addresses concerns with measures of tenure security by explicitly considering the relationship of the empirically observable tenure type to the fundamental parameter of interest, tenure security. The importance of this consideration is illustrated by the final hypothesis, that customary tenure exists in an institutional context that responds differently to changes in land value, and therefore the relationship between tenure type and security is different in high and low land value areas. Arnot et al. (2011) distinguish between the ‘content’ and ‘assurance’ aspects of tenure, and rather than eliding between those, I consider how the two are related.

Arnot et al. (2011) discusses how different proxies for tenure security have been used (to varying effect) in the literature. The ACDP survey directly elicits how worried respondents are (on a scale of 1-5) about losing their land in the next 5 years, which I average over all rightsholders on a given parcel as my primary measure of tenure insecurity. However, I also show in table A20 that these perceptions are related to common sources of insecurity. Having a formal document issued by a land board (either a title, lease, or certificate of customary ownership, in most cases) is associated with lower levels of insecurity, as household rights are more easily legally enforceable with such a document. On the other hand, parcels which have in the past been cause for concern over conflict or are currently involved in disputes are (unsurprisingly) held to be more insecure. Although these correlates of perceived insecurity are not the primary focus of this paper, they are consistent with findings in the literature and validate these survey measures.

Additionally, the survey included a vignette module to ask about the probability of a typical household in the respondents’ community losing a parcel of land under different hypothetical scenarios. Each respondent

VARIABLES	Average Insecurity					
	(1)	(2)	(3)	(4)	(5)	(6)
Formal Document	-0.289*** (0.0646)	-0.0901 (0.0631)				
Ever concerned dispute			0.936*** (0.140)	0.872*** (0.0860)		
Active conflicts					1.277*** (0.179)	1.120*** (0.138)
Wealth Index		-0.00178 (0.00215)		-0.00324 (0.00207)		-0.00193 (0.00236)
Landholdings		-0.00506* (0.00278)		-0.00658* (0.00335)		-0.00561* (0.00294)
First Year Farming		0.00119* (0.000673)		0.000985 (0.000795)		0.00104 (0.000738)
Constant	1.494*** (0.0483)	-0.632 (1.294)	1.255*** (0.0276)	-0.314 (1.554)	1.320*** (0.0292)	-0.458 (1.412)
Observations	5,589	3,293	5,591	3,296	5,596	3,302
R-squared	0.031	0.249	0.129	0.337	0.075	0.296
Controls	None	Parcel Access	None	Parcel Access	None	Parcel Access
Fixed Effect	None	None	None	None	None	None
Cluster	Parish	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A20: OLS results of an average of tenure security across all rightsholders (scale of 1-5, with higher values indicating more insecure) on a parcel of measures of land pressures. Columns (2), (4) and (6) control for the Progress out of Poverty wealth index, total household landholdings, the first year the parcel was farmed, and the method of parcel acquisition. Standard errors are clustered at the parish level in all specifications. Difference in number of observations between columns are due to missing data for some variables.

was explicitly prompted to think about land that the owner either had or had not made long-term investments in (the investment scenario was randomly chosen for each respondent, but common to all questions for a given respondent). They were then asked to use 20 stones to indicate the likelihood that a “typical household in your community” would have a parcel of land, either customary or freehold, taken away from them. The vignette allows me to abstract away from the specificities of tenure security, including parcel or individual characteristics. Instead, it elicits respondents’ understanding of the abstract factors of tenure security (although it does not ask about sources of tenure security, to directly test the mechanism of elite expropriation in my model).

In table A21, I construct the probability of losing a hypothetical parcel implied by the distribution of stones reported, and use this as the outcome variable. In columns (1) and (2), I regress this probability on the tenure type specified for the parcel, whether investments had been made (common for both parcels asked of a respondent), and the interaction of these experimentally varied factors. Across the full sample,

customary parcels are seen as more likely to be lost; this is true even in a within-respondent specification in column (2).⁶⁴ The interaction of investment and customary tenure is not statistically significant, but may suggest that investments may marginally reduce the likelihood of losing customary parcels (consistent with theoretical models of customary tenure).

In columns (3) - (6), I then regress this hypothetical likelihood of losing land on the vignette characteristics interacted with measures of land pressures the respondent faces, as in the main results. These results, with their more general framing, are very consistent with the insecurity respondents report on their own parcels. As land pressures increase, the likelihood of losing a parcel increases more for customary parcels than freehold ones. This responsiveness does not seem to differ by whether the land has been invested in. This suggests that the observed security and investment gaps I document in response to land pressures in the main results are understood by respondents to be general phenomena, not driven entirely by their own specific circumstances. These specific circumstances include individuals' own risk preferences or resource constraints, suggesting that differential selection into continuing to farm on customary or freehold parcels in high land-pressure environments is not the sole driver of my results.

⁶⁴Note that 'invested' is assigned at the respondent level, so is perfectly collinear with household fixed effects; its coefficient is therefore not interpretable in column (2).

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	Hypothetical Probability of Losing Typical Parcel					
Customary	0.141*** (0.0347)	0.141*** (0.0492)	0.140*** (0.0345)	0.140*** (0.0351)	0.00731 (0.0659)	0.313** (0.122)
Invested	0.00782 (0.0161)	0.207*** (0.00932)	0.00726 (0.0162)	0.0116 (0.0130)	-0.0281 (0.0566)	-0.0344 (0.0578)
Customary $\times$ Invested	-0.0134 (0.0133)	-0.0141 (0.0186)	-0.0125 (0.0136)	-0.0129 (0.0139)	0.0439 (0.0426)	0.0196 (0.0578)
Rental Price			-0.00286 (0.00179)	-0.00430*** (0.00107)		
Customary $\times$ Rental Price			0.0198*** (0.00364)	0.0198*** (0.00368)		
Invested $\times$ Rental Price			0.00688 (0.0129)	-0.0113 (0.0139)		
Customary $\times$ Invested $\times$ Rental Price			-0.0173 (0.0260)	-0.0176 (0.0266)		
Inverse Travel Time					0.459* (0.229)	
Customary $\times$ Inverse Travel Time					0.497* (0.262)	
Invested $\times$ Inverse Travel Time					0.153 (0.231)	
Customary $\times$ Invested $\times$ Inverse Travel Time					-0.203 (0.171)	
Decrease in Travel Time						1.707*** (0.573)
Customary $\times$ Decrease in Travel Time						-1.770 (1.156)
Invested $\times$ Decrease in Travel Time						0.503 (0.661)
Customary $\times$ Invested $\times$ Decrease in Travel Time						-0.379 (0.663)
Observations	3,653	3,657	3,645	3,645	3,613	3,613
R-squared	0.040	0.611	0.041	0.158	0.069	0.052
Fixed Effect	None	Household	None	Farmer Group	None	None
Cluster	Parish	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A21: OLS results of the hypothetical probability a parcel will be lost, regressed on experimentally varied tenure type, whether the parcel had been invested in, and their interaction. Columns (3) - (6) also include measures of land pressures. All specifications control for the Progress out of Poverty Index wealth index and total landholdings of the household in acres. Column (2) includes household fixed effects, and column (4) includes

Importantly, there are many potential sources of tenure insecurity in Uganda, especially in high land pressure environments. There is a wealth of qualitative evidence that more valuable land, especially facing urbanization pressures, is subject to more border encroachment, more inheritance disputes, and more government expropriation. These sources of insecurity, however, pose a problem for freehold and customary parcels alike. The differences between customary and freehold parcels are more likely to be driven by the threat of elite expropriation, which disproportionately affects customary parcels. The ACDP survey unfortunately does not provide any data on the sources of insecurity or threats people perceive, so I cannot directly test this mechanism. However, table A22 presents who outside the household has rights to a parcel, separately for customary and freehold parcels. Households are actually less likely to report any outsider holds rights to customary parcels (20% for sales rights versus 33% of freehold parcels; 22% for bequest rights versus 33% of freehold). At the same time, customary parcels are more likely than freehold to have customary or religious leaders who hold sales or bequest rights. The greater insecurity I document on customary parcels in high land pressure environments, then, seems likely to disproportionately stem from these customary leaders than any other rights holder outside the household.

Variable	(1) Freehold Mean/(SE)	(2) Customary Mean/(SE)	(1)-(2) Pairwise t-test Mean difference
Relative: sales right	0.121 (0.005)	0.107 (0.007)	0.014*
Local official: sales right	0.003 (0.001)	0.001 (0.001)	0.002
Customary leader: sales right	0.008 (0.001)	0.033 (0.004)	-0.025***
Religious leader: sales right	0.002 (0.001)	0.006 (0.002)	-0.004**
Non HH member other: sales right	0.191 (0.006)	0.046 (0.005)	0.144***
No non-HH member: sales right	0.671 (0.007)	0.798 (0.009)	-0.126***
Relative: bequeath right	0.128 (0.005)	0.127 (0.007)	0.001
Local official: bequeath right	0.003 (0.001)	0.001 (0.001)	0.003*
Customary leader: bequeath right	0.008 (0.001)	0.032 (0.004)	-0.024***
Religious leader: bequeath right	0.002 (0.001)	0.006 (0.002)	-0.004**
Non HH member other: bequeath right	0.184 (0.006)	0.046 (0.005)	0.139***
No non-HH member: bequeath right	0.670 (0.007)	0.778 (0.009)	-0.108***
Number of observations	4050	2130	6180

Significance: ***=.01, **=.05, *=.1.

Table A22: Balance table of rightsholders outside the household for freehold and customary parcels. Each variable is a dummy equal to one if the given rightsholder outside the household was indicated as having sales or bequest rights, respectively, to a given parcel; respondents could select multiple individuals as holding rights.

C.6.2 Alternate Measures of Land Values

Given that parcel rental prices may be endogenous to soil quality or past agricultural investment, I also explore alternate measures of land values. First, I predict per-acre rental prices using the probability of urbanization, latitude and longitude of the household, a dummy for customary parcels, a dummy for whether or not the parcel has trees, and village dummies. This should strip away noise in the estimated rental prices, and follows the intuition of a hedonic model for land values. Second, I use the median per-acre rental price in a village, to eliminate parcel-level heterogeneity in rental values. Finally, I use a non-standardized version of per-acre rental prices, which exhibits considerable skewness, but can be interpreted as the shilling value. Results with these three alternatives in tables A23 - A27 are by and large consistent with those in the main text, although are generally noisier and therefore less likely to be statistically significant.

VARIABLES	(1)	(2)	(3)	(4)	(5)
			Average Insecurity		
Predicted Price	2.17e-07 (1.91e-07)	4.85e-07** (2.37e-07)			
Customary $\times$ Predicted Price	3.32e-07 (4.69e-07)	7.72e-07 (5.44e-07)			
Village Price			-3.64e-06*** (8.59e-07)		
Customary $\times$ Village Price			1.22e-06 (9.75e-07)		
Rental Price (UGX)				4.10e-08 (4.16e-08)	4.74e-08 (3.00e-08)
Customary $\times$ Rental Price (UGX)				2.04e-07*** (2.62e-08)	2.30e-07*** (3.58e-08)
Customary	0.0731 (0.174)	-0.0129 (0.146)	-0.802*** (0.295)	0.0713 (0.0790)	0.0399 (0.0766)
Wealth Index	-0.00276 (0.00260)	-0.00145 (0.00284)	-0.00142 (0.00244)	-0.00221 (0.00248)	-0.00164 (0.00268)
Landholdings	-0.00503* (0.00273)	-0.00634* (0.00374)	-0.00630** (0.00249)	-0.00440 (0.00278)	-0.00549 (0.00363)
Cash Crop	0.0415 (0.0686)	0.0252 (0.0715)	0.0887 (0.0732)	0.0564 (0.0711)	0.00479 (0.0766)
First Year Farming	0.00135** (0.000641)	0.000752 (0.000836)	0.00138* (0.000696)	0.00149** (0.000641)	0.00127 (0.000861)
Observations	3,256	3,256	3,258	3,240	3,240
R-squared	0.253	0.376	0.267	0.261	0.382
Fixed Effect	None	Group	None	None	Group
Cluster	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A23: OLS results of average tenure insecurity across all rightsholders (scale of 1-5, with higher values indicating more insecure) on a parcel on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in columns (2) and (5), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Predicted price is estimated using the probability of urbanization, latitude and longitude of the household, a dummy for customary parcels, a dummy for whether or not the parcel has trees, and village dummies. Village price is the median per-acre rental price in the village. Rental price (UGX) is the per-acre Ugandan Shilling value of the land.

VARIABLES	(1)	(2)	(3) Trees	(4)	(5)
Predicted Price	3.97e-07** (1.74e-07)	5.56e-07** (2.69e-07)			
Customary $\times$ Predicted Price	-3.33e-07*** (1.03e-07)	-7.77e-08 (1.77e-07)			
Village Price			2.33e-06*** (5.81e-07)		
Customary $\times$ Village Price			4.25e-07 (8.27e-07)		
Rental Price (UGX)				2.44e-08 (1.57e-08)	1.62e-08 (1.83e-08)
Customary $\times$ Rental Price (UGX)				-8.36e-08*** (2.23e-08)	-7.24e-08*** (1.61e-08)
Customary	0.216*** (0.0557)	0.179*** (0.0521)	0.385*** (0.140)	0.0496 (0.0444)	0.0407 (0.0520)
Wealth Index	-0.00159 (0.00147)	0.00171** (0.000828)	-0.00285** (0.00130)	-0.00162 (0.00153)	0.00167** (0.000821)
Landholdings	0.000507 (0.00151)	0.00170 (0.00139)	0.00147 (0.00143)	0.000321 (0.00152)	0.00147 (0.00136)
Cash Crop	0.129*** (0.0334)	0.122*** (0.0373)	0.124*** (0.0290)	0.146*** (0.0344)	0.130*** (0.0385)
First Year Farming	-0.00290*** (0.000806)	-0.00346*** (0.00107)	-0.00279*** (0.000812)	-0.00285*** (0.000798)	-0.00324*** (0.00105)
Observations	3,266	3,266	3,268	3,250	3,250
R-squared	0.075	0.255	0.083	0.061	0.242
Fixed Effect	None	Group	None	None	Group
Cluster	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A24: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in columns (2) and (5), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Predicted price is estimated using the probability of urbanization, latitude and longitude of the household, a dummy for customary parcels, a dummy for whether or not the parcel has trees, and village dummies. Village price is the median per-acre rental price in the village. Rental price (UGX) is the per-acre Ugandan Shilling value of the land.

VARIABLES	(1)	(2)	(3)	(4)	(5)
			Used Organic Fertilizer		
Predicted Price	1.43e-07 (1.21e-07)	-6.49e-08 (6.39e-08)			
Customary $\times$ Predicted Price	1.24e-07** (5.66e-08)	-7.14e-08 (6.28e-08)			
Village Price			9.57e-07*** (2.94e-07)		
Customary $\times$ Village Price			5.14e-07*** (1.69e-07)		
Rental Price (UGX)				-1.89e-09 (2.47e-09)	-2.31e-09 (1.85e-09)
Customary $\times$ Rental Price (UGX)				2.63e-08*** (9.76e-09)	5.90e-09 (3.82e-09)
Customary	-0.0677 (0.0473)	-0.0158 (0.0386)	0.0162 (0.0541)	-0.0716*** (0.0260)	-0.0200 (0.0210)
Wealth Index	0.00397*** (0.000703)	0.000885 (0.000563)	0.00344*** (0.000658)	0.00398*** (0.000663)	0.000915 (0.000584)
Landholdings	0.00166 (0.00135)	0.00156 (0.00105)	0.00199 (0.00135)	0.00102 (0.00114)	0.00127 (0.000985)
Cash Crop	0.193*** (0.0296)	0.0276 (0.0253)	0.194*** (0.0312)	0.207*** (0.0308)	0.0368 (0.0254)
First Year Farming	-6.17e-05 (0.000529)	-8.66e-05 (0.000458)	-4.75e-06 (0.000510)	-1.78e-05 (0.000530)	-0.000113 (0.000471)
Observations	3,264	3,264	3,266	3,248	3,248
R-squared	0.168	0.323	0.174	0.169	0.322
Fixed Effect	None	Group	None	None	Group
Cluster	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A25: OLS results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in columns (2) and (5), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Predicted price is estimated using the probability of urbanization, latitude and longitude of the household, a dummy for customary parcels, a dummy for whether or not the parcel has trees, and village dummies. Village price is the median per-acre rental price in the village. Rental price (UGX) is the per-acre Ugandan Shilling value of the land.

VARIABLES	(1)	(2)	(3)	(4)	(5)
			Used Inorganic Fertilizer		
Predicted Price	-3.49e-08 (6.92e-08)	-7.30e-08 (8.26e-08)			
Customary $\times$ Predicted Price	-9.15e-08** (3.90e-08)	-1.62e-07** (6.99e-08)			
Village Price			1.47e-06*** (1.74e-07)		
Customary $\times$ Village Price			9.68e-07*** (1.78e-07)		
Rental Price (UGX)				6.34e-09 (5.12e-09)	9.54e-09** (3.57e-09)
Customary $\times$ Rental Price (UGX)				1.72e-08 (1.66e-08)	8.10e-09 (5.05e-09)
Customary	0.0226 (0.0327)	0.0289 (0.0355)	0.100*** (0.0344)	0.00631 (0.0377)	0.00518 (0.0409)
Wealth Index	0.00470*** (0.000740)	0.00192** (0.000897)	0.00347*** (0.000753)	0.00456*** (0.000756)	0.00203** (0.000878)
Landholdings	0.00205* (0.00102)	0.00191* (0.00106)	0.00304*** (0.000899)	0.00168** (0.000789)	0.00148* (0.000863)
Cash Crop	0.0174 (0.0145)	0.00289 (0.0222)	-0.000165 (0.0143)	0.0151 (0.0147)	0.0143 (0.0175)
First Year Farming	0.00131*** (0.000484)	0.00102* (0.000555)	0.00136*** (0.000470)	0.00129*** (0.000471)	0.000900* (0.000532)
Observations	3,261	3,261	3,263	3,245	3,245
R-squared	0.030	0.151	0.054	0.029	0.150
Fixed Effect	None	Group	None	None	Group
Cluster	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A26: OLS results of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in columns (2) and (5), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Predicted price is estimated using the probability of urbanization, latitude and longitude of the household, a dummy for customary parcels, a dummy for whether or not the parcel has trees, and village dummies. Village price is the median per-acre rental price in the village. Rental price (UGX) is the per-acre Ugandan Shilling value of the land.

VARIABLES	(1)	(2)	(3)	(4)	(5)
			Used Pesticides		
Predicted Price	1.81e-07** (7.12e-08)	6.75e-08 (8.47e-08)			
Customary $\times$ Predicted Price	-3.80e-08 (1.18e-07)	-1.37e-07 (1.12e-07)			
Village Price			1.57e-06*** (3.02e-07)		
Customary $\times$ Village Price			8.57e-07** (3.75e-07)		
Rental Price (UGX)				-2.73e-09 (6.56e-09)	-3.93e-09 (4.87e-09)
Customary $\times$ Rental Price (UGX)				2.23e-08 (1.56e-08)	-2.37e-08*** (6.40e-09)
Customary	0.0235 (0.0513)	0.0120 (0.0537)	0.104** (0.0483)	-0.0364 (0.0673)	-0.0402 (0.0679)
Wealth Index	0.00579*** (0.000762)	0.00344*** (0.00101)	0.00471*** (0.000842)	0.00590*** (0.000705)	0.00335*** (0.000995)
Landholdings	0.00158 (0.00133)	0.000513 (0.000953)	0.00236* (0.00139)	0.00151 (0.00144)	0.000662 (0.00100)
Cash Crop	0.135*** (0.0269)	0.0919*** (0.0212)	0.129*** (0.0248)	0.145*** (0.0261)	0.0876*** (0.0208)
First Year Farming	0.000597 (0.000493)	0.000524 (0.000561)	0.000680 (0.000460)	0.000623 (0.000483)	0.000547 (0.000552)
Observations	3,260	3,260	3,262	3,244	3,244
R-squared	0.083	0.202	0.100	0.079	0.203
Fixed Effect	None	Group	None	None	Group
Cluster	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A27: OLS results of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in columns (2) and (5), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Predicted price is estimated using the probability of urbanization, latitude and longitude of the household, a dummy for customary parcels, a dummy for whether or not the parcel has trees, and village dummies. Village price is the median per-acre rental price in the village. Rental price (UGX) is the per-acre Ugandan Shilling value of the land.

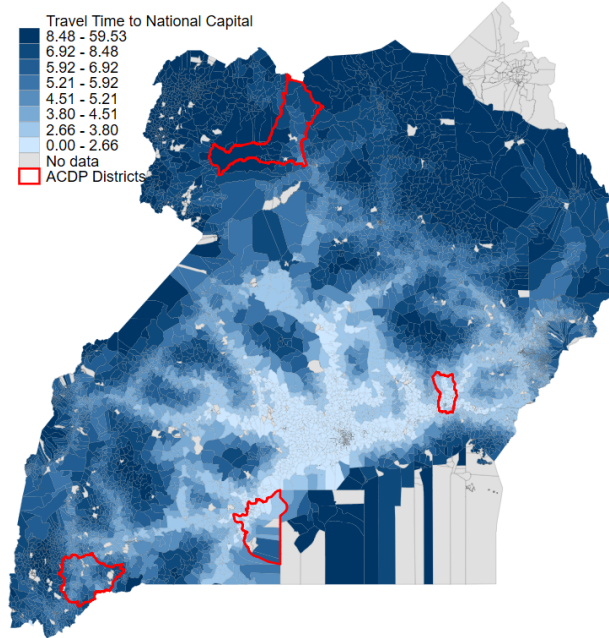


Figure 7: Map of average travel time in 2015 to Kampala for each Parish in Uganda, with ACDP districts highlighted in red. Administrative units in Uganda extend into Lake Victoria to the national border, to include several small islands.

C.6.3 Alternate Measures of Market Access

The external land pressures modeled above (and mapped in figures 7 and 8) are not exclusively coming from the national capital, Kampala. In fact, rapidly-growing regional capitals can create the sort of outsider demand for land that causes such insecurity over customary parcels. Therefore, in tables A28 - A32, I use four more regional measures of land pressures, all from Muller-Crepon (2021). First, I construct analogues to the inverse travel time and 10-year decrease in travel time, but to the regional capital rather than Kampala, the national capital and largest city in the country. However, these regional capitals have changed somewhat dramatically in the last 15 years; Uganda has experienced administrative unit proliferation (Grossman and Lewis, 2014). These newly-named regional capitals, which could drastically improve connections to a ‘regional capital’ in the data, may not reflect the demand for rural land by outsiders that I am modeling. Instead of capturing growing markets and real demand, this may be an artefact of politically-driven fragmentation.

I then also use a measure of market access that Muller-Crepon (2021) provides, stemming from Donaldson (2018). This ‘market access’ measure is a “reduced form expression derived from general equilibrium trade theory” (Donaldson and Hornbeck, 2016), which uses the within-country expansion of transport networks. I use this measure as provided in 2015, as well as the 10-year change in market access. This transport-focused market access, however, is more likely to affect agricultural total factor productivity and is less closely linked to the population-based drivers of external demand for land that I model.

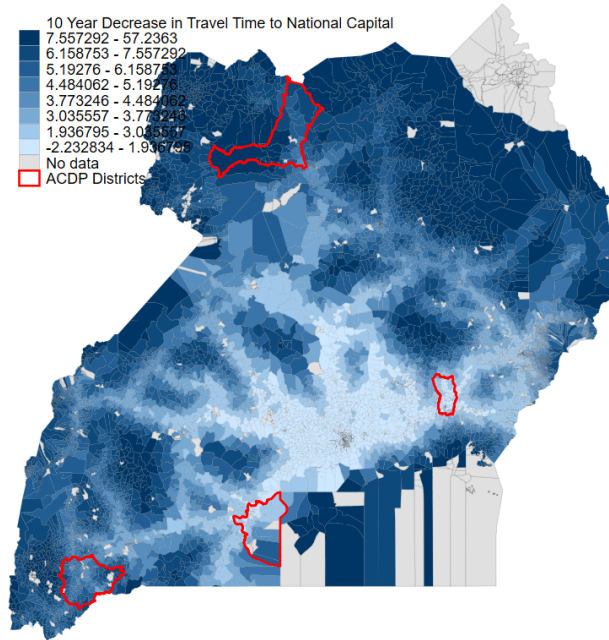


Figure 8: Map of the 10 year change in average travel time in 2015 to Kampala for each Parish in Uganda, with ACDP districts highlighted in red. Administrative units in Uganda extend into Lake Victoria to the national border, to include several small islands.

However, in tables A28 - A32, results with these alternate measures of land pressures are broadly consistent with those in the main text.

VARIABLES	(1)	(2)	(3)	(4)
		Average Insecurity		
Inverse Travel Time to Regional Capital	-0.446*			
	(0.228)			
Customary $\times$ Inverse Travel Time to Regional Capital	0.250			
	(0.234)			
Decrease in Travel Time to Regional Capital		-0.287		
		(0.426)		
Customary $\times$ Decrease in Travel Time to Regional Capital		-0.133		
		(0.185)		
2015 Market Access			-5.18e-07**	
			(2.18e-07)	
Customary $\times$ 2015 Market Access			4.92e-07	
			(3.49e-07)	
Change in Market Access				0.0333
				(0.111)
Customary $\times$ Change in Market Access				-0.358**
				(0.156)
Customary	-0.239	0.117	-0.0579	0.135**
	(0.200)	(0.0791)	(0.106)	(0.0663)
Wealth Index	-0.00141	-0.00207	-0.00157	-0.00226
	(0.00261)	(0.00256)	(0.00264)	(0.00242)
Landholdings	-0.00529**	-0.00474*	-0.00518*	-0.00561*
	(0.00258)	(0.00275)	(0.00269)	(0.00299)
Cash Crop	0.0647	0.0631	0.0505	0.0742
	(0.0764)	(0.0788)	(0.0747)	(0.0836)
First Year Farming	0.00141**	0.00151**	0.00146**	0.00125*
	(0.000649)	(0.000663)	(0.000623)	(0.000626)
Observations	3,222	3,169	3,222	3,222
R-squared	0.253	0.255	0.254	0.251
Fixed Effect	None	None	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A28: OLS results of average tenure security across all rightholders (scale of 1-5, with higher values indicating more insecure) on measures of regional land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Inverse travel time to regional capital is measured in 2015, and decrease in travel time to regional capital is the 10-year decrease; both are constructed analogue to the national measures in the main text but with the subnational capital. Market access and the change in market access are constructed from Donaldson (2018)'s measure.

VARIABLES	(1)	(2)	(3)	(4)
			Trees	
Inverse Travel Time to Regional Capital	0.255 (0.200)			
Customary $\times$ Inverse Travel Time to Regional Capital	0.0524 (0.239)			
Decrease in Travel Time To Regional Capital		0.523** (0.225)		
Customary $\times$ Decrease in Travel Time to Regional Capital		0.310* (0.154)		
2015 Market Access			6.95e-08 (2.00e-07)	
Customary $\times$ 2015 Market Access			-3.87e-08 (2.63e-07)	
Change in Market Access				0.0680 (0.0822)
Customary $\times$ Change in Market Access				-0.366*** (0.0651)
Customary	0.128 (0.107)	0.0194 (0.0429)	0.0437 (0.0627)	0.0546** (0.0246)
Wealth Index	-0.00235* (0.00139)	-0.000469 (0.00142)	-0.00170 (0.00136)	-0.00148 (0.00150)
Landholdings	0.000460 (0.00156)	-0.000282 (0.00169)	0.000435 (0.00152)	0.000248 (0.00156)
Cash Crop	0.142*** (0.0326)	0.121*** (0.0326)	0.147*** (0.0368)	0.156*** (0.0394)
First Year Farming	-0.00292*** (0.000903)	-0.00319*** (0.000881)	-0.00278*** (0.000865)	-0.00280*** (0.000795)
Observations	3,232	3,179	3,232	3,232
R-squared	0.063	0.073	0.058	0.067
Fixed Effect	None	None	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A29: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of regional land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Inverse travel time to regional capital is measured in 2015, and decrease in travel time to regional capital is the 10-year decrease; both are constructed analogue to the national measures in the main text but with the subnational capital. Market access and the change in market access are constructed from Donaldson (2018)'s measure.

VARIABLES	(1)	(2)	(3)	(4)
		Used Organic Fertilizer		
Inverse Travel Time to Regional Capital	0.170*			
	(0.0966)			
Customary $\times$ Inverse Travel Time to Regional Capital	0.126**			
	(0.0550)			
Decrease in Travel Time to Regional Capital		0.309**		
		(0.133)		
Customary $\times$ Decrease in Travel Time to Regional Capital		-0.00168		
		(0.0499)		
2015 Market Access			2.51e-08	
			(1.05e-07)	
Customary $\times$ 2015 Market Access			6.79e-08	
			(8.66e-08)	
Change in Market Access				0.312***
				(0.0605)
Customary $\times$ Change in Market Access				0.148**
				(0.0587)
Customary	-0.0456	-0.0745***	-0.0754***	-0.0365*
	(0.0386)	(0.0246)	(0.0274)	(0.0190)
Wealth Index	0.00349***	0.00451***	0.00398***	0.00319***
	(0.000700)	(0.000799)	(0.000761)	(0.000508)
Landholdings	0.00146	0.00137	0.00145	0.00137
	(0.00120)	(0.00134)	(0.00131)	(0.00114)
Cash Crop	0.200***	0.189***	0.203***	0.135***
	(0.0297)	(0.0322)	(0.0307)	(0.0346)
First Year Farming	-0.000206	8.14e-06	-7.55e-05	-0.000108
	(0.000533)	(0.000531)	(0.000524)	(0.000447)
Observations	3,230	3,177	3,230	3,230
R-squared	0.169	0.175	0.163	0.201
Fixed Effect	None	None	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A30: OLS results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of regional land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Inverse travel time to regional capital is measured in 2015, and decrease in travel time to regional capital is the 10-year decrease; both are constructed analogue to the national measures in the main text but with the subnational capital. Market access and the change in market access are constructed from Donaldson (2018)'s measure.

VARIABLES	(1)	(2)	(3)	(4)
		Used Inorganic Fertilizer		
Inverse Travel Time to Regional Capital	0.303*** (0.0622)			
Customary $\times$ Inverse Travel Time to Regional Capital	0.273*** (0.0803)			
Decrease in Travel Time to Regional Capital		0.00591 (0.134)		
Customary $\times$ Decrease in Travel Time to Regional Capital		-0.120** (0.0513)		
2015 Market Access			2.37e-07** (9.45e-08)	
Customary $\times$ 2015 Market Access			4.33e-07*** (1.20e-07)	
Change in Market Access				0.0153 (0.0704)
Customary $\times$ Change in Market Access				-0.140*** (0.0348)
Customary	0.0231 (0.0618)	0.00571 (0.0386)	-0.0209 (0.0487)	0.0154 (0.0277)
Wealth Index	0.00353*** (0.000740)	0.00445*** (0.000704)	0.00373*** (0.000747)	0.00473*** (0.000788)
Landholdings	0.00230** (0.000964)	0.00239** (0.00107)	0.00230** (0.000994)	0.00215** (0.000995)
Cash Crop	0.0108 (0.0124)	0.0145 (0.0146)	0.0229* (0.0132)	0.0205 (0.0232)
First Year Farming	0.00102** (0.000479)	0.00151*** (0.000466)	0.00111** (0.000476)	0.00130*** (0.000465)
Observations	3,227	3,174	3,227	3,227
R-squared	0.051	0.033	0.044	0.034
Fixed Effect	None	None	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A31: OLS results of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of regional land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Inverse travel time to regional capital is measured in 2015, and decrease in travel time to regional capital is the 10-year decrease; both are constructed analogue to the national measures in the main text but with the subnational capital. Market access and the change in market access are constructed from Donaldson (2018)'s measure.

VARIABLES	(1)	(2)	(3)	(4)
		Used Pesticides		
Inverse Travel Time to Regional Capital	0.353*** (0.0734)			
Customary $\times$ Inverse Travel Time to Regional Capital	0.341** (0.136)			
Decrease in Travel Time to Regional Capital		0.115 (0.161)		
Customary $\times$ Decrease in Travel Time to Regional Capital		-0.127 (0.0859)		
2015 Market Access			2.55e-07** (1.20e-07)	
Customary $\times$ 2015 Market Access			6.20e-07*** (1.93e-07)	
Change in Market Access				0.123 (0.0986)
Customary $\times$ Change in Market Access				-0.339*** (0.0498)
Customary	-0.0238 (0.0780)	-0.0391 (0.0654)	-0.0859 (0.0783)	0.00247 (0.0319)
Wealth Index	0.00445*** (0.000896)	0.00544*** (0.000802)	0.00467*** (0.000956)	0.00574*** (0.000622)
Landholdings	0.00155 (0.00118)	0.00149 (0.00148)	0.00159 (0.00137)	0.00123 (0.00127)
Cash Crop	0.139*** (0.0270)	0.147*** (0.0237)	0.153*** (0.0266)	0.142*** (0.0230)
First Year Farming	0.000281 (0.000477)	0.000681 (0.000507)	0.000384 (0.000483)	0.000573 (0.000505)
Constant	-0.673 (0.934)	-1.366 (0.993)	-0.774 (0.946)	-1.164 (0.983)
Observations	3,226	3,174	3,226	3,226
R-squared	0.100	0.081	0.094	0.096
Fixed Effect	None	None	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A32: OLS results of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of regional land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Inverse travel time to regional capital is measured in 2015, and decrease in travel time to regional capital is the 10-year decrease; both are constructed analogue to the national measures in the main text but with the subnational capital. Market access and the change in market access are constructed from Donaldson (2018)'s measure.

C.6.4 Probability of Urbanization

As an alternate proxy for the external pressures on land that I model, I considered using data from the Global Grid of Probabilities of Urban Expansion to 2030 (Seto et al., 2012, 2016). For each 2.5 arc-minute⁶⁵ grid cell that was non-urban in 2000, the data include an estimated probability of becoming urban by the year 2030, forecast using a population density driver map. This map is primarily driven by patterns of urbanization, and thus is reasonably exogenous to local agricultural patterns of land values; instead, it isolates the dimension of land values that will also create pressure on local elites to expropriate and sell customary land to outsiders.

However, this data is remarkably bimodal in Uganda, with sparse support of values of the probability of urbanization in the four districts where the baseline study occurred. Thirty-eight percent of parcels fall in a grid cell with a 100% probability of urbanization; another 19% have a 99% probability. There is another grouping at very low levels of predicted urbanization, including the entire northern district of Amuru which has a probability of zero, and then only a few parcels are assigned a probability between .5 and .8. This makes the probability of urbanization not my preferred proxy, but I present regression results with this measure in tables A33 and A34.

I match households to a grid cell using GPS locations collected during the survey (taken at the household residence, not at agricultural fields which could be at some remove). This measure is correlated with the per-acre rental values households reported for their land; the correlation is low across the whole sample (.0450), but when I winsorize several outliers in estimated rental price (among those who own their land and are estimating how much they could rent it out for), this correlation increases to .1492. When I look instead at the median rental price in a given district, the correlation with the probability of urbanization jumps to .6556,⁶⁶ implying (as expected) that this measure is more closely related to broader trends than individual parcel values which may be influenced by soil quality or other agricultural value considerations. Results in table A33 for binary outcomes and A34 for continuous Tobit estimation are relatively consistent with those obtained using either normalized rental values, inverse travel time to Kampala, or the 10-year reduction in travel times, as shown above.

⁶⁵2.5 arc-minutes contain approximately 5km at the equator, which Uganda spans

⁶⁶Similarly, the correlation with median village rental prices is .6697.

VARIABLES	(1) Trees	(2) Used Organic	(3) Used Inorganic	(4) Used Pesticides	(5) Insecurity
Urbanization	0.123** (0.0555)	0.0785** (0.0334)	0.0513 (0.0340)	0.101*** (0.0325)	-0.0787 (0.0765)
Customary $\times$ Urbanization	0.0507 (0.0763)	0.0185 (0.0207)	0.0246 (0.0328)	0.0245 (0.0303)	0.00713 (0.103)
Customary	0.0899 (0.0706)	-0.0165 (0.0250)	0.0297 (0.0407)	0.0340 (0.0529)	0.0432 (0.127)
Wealth Index	-0.00212 (0.00148)	0.00385*** (0.000692)	0.00439*** (0.000813)	0.00550*** (0.000727)	-0.00221 (0.00237)
Landholdings	0.000986 (0.00146)	0.00171 (0.00132)	0.00236** (0.000951)	0.00181 (0.00137)	-0.00562* (0.00291)
Cash Crop	0.137*** (0.0304)	0.198*** (0.0297)	0.0102 (0.0142)	0.137*** (0.0247)	0.0649 (0.0754)
First Year Farming	-0.00292*** (0.000826)	-9.28e-05 (0.000513)	0.00127** (0.000481)	0.000550 (0.000476)	0.00145** (0.000654)
Observations	3,266	3,264	3,261	3,260	3,256
R-squared	0.065	0.169	0.032	0.086	0.251
Fixed Effect	None	None	None	None	None
Cluster	Parish	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A33: OLS results where the outcome in columns (1) - (4) is a dummy equal to one if the given input was applied to the parcel, and in column (5) is the average tenure insecurity across all rightsholders. Customary is a dummy equal to one for customary parcels and zero otherwise. The probability of urbanization is a proxy for land pressures, and the regression controls for the Progress out of Poverty wealth index, total household landholdings in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Standard errors are clustered at the parish level in all specifications (reported in parentheses).

VARIABLES	(1) Value Organic	(2) Value Inorganic	(3) Value Pesticides
Urbanization	3.807 (2.458)	0.128 (0.158)	0.787** (0.383)
Customary $\times$ Urbanization	1.252 (1.537)	0.319* (0.177)	-0.854 (0.987)
Customary	1.369 (1.863)	-0.0662 (0.187)	0.970 (1.132)
Wealth Index	0.190* (0.103)	0.0163*** (0.00616)	0.0680** (0.0267)
Landholdings	0.0390 (0.0282)	0.000153 (0.00403)	-0.00949 (0.0216)
Cash Crop	3.349* (1.710)	-0.0163 (0.0537)	1.740** (0.728)
First Year Farming	0.0282 (0.0251)	0.00654** (0.00325)	0.00499 (0.00542)
Observations	3,270	3,270	3,270
Fixed Effect	None	None	None
Cluster	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A34: Tobit results where the outcome is the value of the given input applied to the parcel. Customary is a dummy equal to one for customary parcels and zero otherwise. The probability of urbanization is a proxy for land pressures, and the regression controls for the Progress out of Poverty wealth index, total household landholdings in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Standard errors are clustered at the parish level in all specifications (reported in parentheses).

C.7 Additional Controls

C.7.1 Parcel Characteristics

As can be seen in table A35, customary and freehold parcels are not observably identical. In particular, freehold parcels are more likely to have been purchased, more likely to be documented, and less likely to be primarily residential (none of which are surprising). Additionally, we can see different patterns of land use and tenure security: freehold parcels were acquired slightly more recently, are more likely to have formal documents, and are more likely to be irrigated – but are also more likely to have active conflicts and worries about ownership disputes, and less likely to be fallowed in the past two years or practice crop rotation.

Several of these characteristics may be responses to the insecurity incentives modeled in the paper, but I also run regressions controlling for residential land use and a dummy for whether the parcel was purchased (in addition to the controls used throughout the analysis). Although these characteristics are correlated with input use, the primary results of the paper (diverging responses to land pressures for long-term inputs) remain.

C.7.2 Crop Change

One notable divergence between the model and empirical results is the patterns of short-term input use increasing somewhat as land pressures increase, which I argued is likely due to changing in cropping patterns. In particular, many cash crops in Uganda are perennials, meaning that the likelihood of expropriation also affects the returns from these crops; at the same time, many cash crops require higher input use (pesticides in particular) in order to receive a market premium. However, I argue that the divergence in long-term input use between customary and freehold parcels as land pressures increase is not due entirely to changing cropping patterns; there is an additional tenure security effect. This additional marginal impact of tenure security is what is captured in the regressions in the main paper, which control for whether there is a cash crop on the parcel. However, because the shift to cash crops could be a result of the mechanisms of interest, I also run the regressions without this control in tables A42 - A46. This robustness check is important, as crop choice could be a bad control that absorbs part of the treatment effect; I prefer the more conservative specification which may underestimate the true effect but isolates the impact of tenure security on investment on the intensive margin rather than selection into different kinds of crops.

Results are extremely consistent with those in the main text when omitting this control, although very slightly larger in magnitude (as expected). This suggests that selection into more cash cropping as land pressures increase is not the primary mechanism, but instead farmers are adjusting their input use on the intensive margin as a result of tenure insecurity. Table A41 parallels panel B of table 1, except with the

Variable	(1) Freehold		(2) Customary		(1)-(2) Pairwise t-test	
	N	Mean/(SE)	N	Mean/(SE)	N	Difference
Residential Land Use	3854	0.080 (0.004)	2326	0.104 (0.006)	6180	-0.024***
Agricultural Land Use	3854	0.912 (0.005)	2326	0.891 (0.006)	6180	0.021***
Forest Land Use	3854	0.006 (0.001)	2326	0.003 (0.001)	6180	0.003*
Parcel was purchased or rented in	3863	0.739 (0.007)	2326	0.107 (0.006)	6189	0.632***
First Year Farmed, cleaned	2594	2007.114 (0.248)	1023	2001.629 (0.479)	3617	5.486***
Document issued by land board	4023	0.534 (0.008)	2109	0.359 (0.010)	6132	0.175***
Active conflicts	4029	0.040 (0.003)	2106	0.027 (0.004)	6135	0.014***
Ever concerned dispute ownership	4026	0.138 (0.005)	2105	0.066 (0.005)	6131	0.073***
Seasons parcel fallowed in last 2 years	3500	0.195 (0.009)	1947	0.388 (0.017)	5447	-0.193***
Crop rotation practiced on parcel	3540	0.567 (0.008)	2092	0.625 (0.011)	5632	-0.058***
Irrigation on parcel (binary)	3542	0.022 (0.002)	2093	0.013 (0.003)	5635	0.008**

Significance: ***=.01, **=.05, *=.1.

Table A35: Balance table of plot characteristics between freehold and customary parcels. Note that some of these, such as fallowing and crop rotation, are other kinds of investment in the land and therefore may be responding to the incentives described throughout. Residential, agricultural, and forest land use are dummies for the primary use of the parcel. Parcel was purchased or rented in is a dummy equal to one if the respondent reported accessing the parcel through purchase or a formal rental arrangement. Document issued by land board is a dummy equal to one if the parcel has a document (including a title deed, certificate of ownership, certificate of hereditary acquisition, lease or rental contract). Active conflicts is a dummy equal to one if the parcel has an active conflict or dispute ongoing. Ever concerned dispute ownership is a dummy equal to one if the respondent has ever been concerned that someone would dispute their ownership of the parcel. Seasons parcel fallowed in last 2 years is a count of the number of seasons the parcel has been left fallow, out of four potential seasons. Crop rotation and irrigation are dummies equal to one if the respondent reports rotating crops or irrigation on any plot within the parcel.

VARIABLES	(1)	(2)	(3)	(4)
		Average Insecurity		
Rental Price	0.0213 (0.0216)	0.0247 (0.0156)		
Customary $\times$ Rental Price	0.106*** (0.0136)	0.120*** (0.0187)		
Inverse Travel Time			-1.403*** (0.470)	
Customary $\times$ Inverse Travel Time			0.753 (0.502)	
Decrease in Travel Time				-2.702** (1.331)
Customary $\times$ Decrease in Travel Time				-0.198 (1.139)
Residential	0.0613 (0.0944)	0.00612 (0.0758)	0.0706 (0.0972)	0.0794 (0.0957)
Purchased	0.357 (0.470)	0.375 (0.466)	0.312 (0.486)	0.375 (0.493)
Customary	0.105 (0.0792)	0.0761 (0.0758)	-0.496** (0.222)	-0.151 (0.197)
Wealth Index	-0.00216 (0.00246)	-0.00158 (0.00268)	-0.00138 (0.00260)	-0.000329 (0.00243)
Landholdings	-0.00433 (0.00277)	-0.00551 (0.00362)	-0.00537** (0.00241)	-0.00585** (0.00266)
Cash Crop	0.0533 (0.0714)	0.00459 (0.0767)	0.0733 (0.0775)	0.101 (0.0876)
First Year Farming	0.00155** (0.000627)	0.00126 (0.000837)	0.00146** (0.000678)	0.00137** (0.000665)
o.d6_10	-	-	-	-
Constant	-1.514 (1.201)	-1.299 (1.677)	-1.040 (1.307)	-1.029 (1.254)
Observations	3,240	3,240	3,222	3,222
R-squared	0.262	0.382	0.259	0.253
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A36: OLS results of average tenure insecurity across all rightsholders on a parcel on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, dummies for the method of parcel acquisition, a dummy for if the parcel is primarily residential, a dummy for if the parcel was purchased or formally rented in, and the year the parcel was first farmed. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
			Trees	
Rental Price	0.0128 (0.00819)	0.00870 (0.00966)		
Customary $\times$ Rental Price	-0.0438*** (0.0117)	-0.0376*** (0.00839)		
Inverse Travel Time			0.869** (0.400)	
Customary $\times$ Inverse Travel Time			-0.0246 (0.432)	
Decrease in Travel Time				0.928 (0.886)
Customary $\times$ Decrease in Travel Time				-1.700** (0.797)
Residential	0.155** (0.0609)	0.141** (0.0616)	0.148** (0.0617)	0.149** (0.0647)
Purchased	-0.0647 (0.181)	-0.0745 (0.190)	-0.0571 (0.183)	-0.0702 (0.189)
Customary	0.0322 (0.0411)	0.0270 (0.0495)	0.282** (0.107)	0.294*** (0.0890)
Wealth Index	-0.00163 (0.00153)	0.00173** (0.000800)	-0.00259* (0.00131)	-0.00133 (0.00119)
Landholdings	0.000451 (0.00152)	0.00156 (0.00133)	0.000853 (0.00145)	0.000182 (0.00156)
Cash Crop	0.141*** (0.0340)	0.126*** (0.0370)	0.129*** (0.0314)	0.133*** (0.0353)
First Year Farming	-0.00271*** (0.000802)	-0.00309*** (0.00106)	-0.00271*** (0.000819)	-0.00274*** (0.000788)
Observations	3,250	3,250	3,232	3,232
R-squared	0.066	0.245	0.076	0.070
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A37: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, dummies for the method of parcel acquisition, a dummy for if the parcel is primarily residential, a dummy for if the parcel was purchased or formally rented in, and the year the parcel was first farmed. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2) Used Organic Fertilizer	(3)	(4)
Rental Price	-0.000960 (0.00128)	-0.00105 (0.00102)		
Customary $\times$ Rental Price	0.0135*** (0.00493)	0.00312 (0.00201)		
Inverse Travel Time			0.531** (0.200)	
Customary $\times$ Inverse Travel Time			0.280** (0.105)	
Decrease in Travel Time				3.364*** (0.435)
Customary $\times$ Decrease in Travel Time				1.190*** (0.315)
Residential	0.105*** (0.0384)	0.0798 (0.0503)	0.0975** (0.0387)	0.0839** (0.0379)
Purchased	0.0858 (0.0838)	-0.0505 (0.0697)	0.115 (0.103)	0.0662 (0.113)
Customary	-0.0636*** (0.0226)	-0.0163 (0.0193)	0.00499 (0.0406)	0.167*** (0.0446)
Wealth Index	0.00400*** (0.000697)	0.000951 (0.000601)	0.00322*** (0.000696)	0.00105 (0.000642)
Landholdings	0.00111 (0.00117)	0.00132 (0.000996)	0.00185 (0.00129)	0.00249** (0.00101)
Cash Crop	0.204*** (0.0303)	0.0347 (0.0265)	0.191*** (0.0298)	0.135*** (0.0274)
First Year Farming	7.30e-05 (0.000529)	-2.84e-05 (0.000453)	-2.93e-05 (0.000524)	3.37e-05 (0.000472)
Observations	3,248	3,248	3,230	3,230
R-squared	0.174	0.325	0.180	0.227
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A38: OLS results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, dummies for the method of parcel acquisition, a dummy for if the parcel is primarily residential, a dummy for if the parcel was purchased or formally rented in, and the year the parcel was first farmed. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Inorganic Fertilizer		
Rental Price	0.00328 (0.00267)	0.00484** (0.00191)		
Customary $\times$ Rental Price	0.00903 (0.00875)	0.00417 (0.00264)		
Inverse Travel Time			0.757*** (0.0833)	
Customary $\times$ Inverse Travel Time			0.598*** (0.111)	
Decrease in Travel Time				1.428*** (0.419)
Customary $\times$ Decrease in Travel Time				0.657* (0.328)
Residential	-0.0503 (0.0341)	-0.0695* (0.0348)	-0.0611* (0.0313)	-0.0586* (0.0343)
Purchased	0.0910 (0.183)	0.0281 (0.184)	0.0595 (0.180)	0.0312 (0.181)
Customary	0.00710 (0.0379)	0.00308 (0.0423)	0.0504 (0.0487)	0.0886** (0.0334)
Wealth Index	0.00457*** (0.000763)	0.00199** (0.000865)	0.00322*** (0.000750)	0.00329*** (0.000865)
Landholdings	0.00164** (0.000793)	0.00143 (0.000864)	0.00274*** (0.000903)	0.00265*** (0.000939)
Cash Crop	0.0165 (0.0149)	0.0162 (0.0178)	0.00493 (0.0130)	-0.0113 (0.0160)
First Year Farming	0.00125*** (0.000450)	0.000827 (0.000500)	0.00116** (0.000434)	0.00128*** (0.000434)
Observations	3,245	3,245	3,227	3,227
R-squared	0.030	0.152	0.059	0.043
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A39: OLS results of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, dummies for the method of parcel acquisition, a dummy for if the parcel is primarily residential, a dummy for if the parcel was purchased or formally rented in, and the year the parcel was first farmed. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Pesticides		
Rental Price	-0.00140 (0.00340)	-0.00193 (0.00255)		
Customary $\times$ Rental Price	0.0114 (0.00797)	-0.0123*** (0.00337)		
Inverse Travel Time			0.847*** (0.117)	
Customary $\times$ Inverse Travel Time			0.570** (0.220)	
Decrease in Travel Time				2.112*** (0.499)
Customary $\times$ Decrease in Travel Time				0.00511 (0.535)
Residential	0.0893*** (0.0308)	0.0655* (0.0365)	0.0772** (0.0289)	0.0757** (0.0319)
Purchased	0.0894 (0.110)	0.0607 (0.125)	0.127 (0.121)	0.0942 (0.129)
Customary	-0.0294 (0.0642)	-0.0424 (0.0660)	0.0490 (0.0495)	0.190*** (0.0423)
Wealth Index	0.00591*** (0.000731)	0.00340*** (0.00101)	0.00428*** (0.000855)	0.00429*** (0.000979)
Landholdings	0.00159 (0.00146)	0.000698 (0.00102)	0.00211 (0.00132)	0.00190* (0.00110)
Cash Crop	0.142*** (0.0260)	0.0858*** (0.0204)	0.128*** (0.0265)	0.104*** (0.0257)
First Year Farming	0.000702 (0.000487)	0.000614 (0.000561)	0.000603 (0.000477)	0.000674 (0.000491)
Observations	3,244	3,244	3,226	3,226
R-squared	0.082	0.204	0.105	0.098
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A40: OLS results of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, dummies for the method of parcel acquisition, a dummy for if the parcel is primarily residential, a dummy for if the parcel was purchased or formally rented in, and the year the parcel was first farmed. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

outcome being a dummy equal to one if there is a staple crop as the primary crop on at least on plot on the parcel (note that in an intercropped system, both cash crops and staple crops can be planted on a parcel). Table A47 focuses specifically on perennial crops as its outcome.

VARIABLES	(1)	(2) Planted Staple	(3)	(4)
Rental Price	-0.0120** (0.00520)	-0.0111*** (0.00392)		
Customary $\times$ Rental Price	-0.00106 (0.00417)	0.00329 (0.00336)		
Inverse Travel Time			-0.356 (0.273)	
Customary $\times$ Inverse Travel Time			0.00585 (0.217)	
Decrease in Travel Time				-3.102*** (0.641)
Customary $\times$ Decrease in Travel Time				-1.485* (0.747)
Customary	0.0423 (0.0286)	-0.00445 (0.0258)	-0.0578 (0.0657)	-0.140** (0.0667)
Wealth Index	-0.00348** (0.00159)	0.000491 (0.000526)	-0.00292** (0.00137)	2.44e-05 (0.000913)
Landholdings	-0.00312** (0.00124)	-0.00137* (0.000741)	-0.00300** (0.00113)	-0.00368*** (0.000950)
First Year Farming	-1.18e-05 (0.000683)	-0.000878 (0.000788)	2.54e-05 (0.000697)	-0.000149 (0.000696)
Observations	3,254	3,254	3,236	3,236
R-squared	0.060	0.284	0.062	0.105
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A41: OLS results of a dummy equal to one for parcels with at least one plot primarily devoted to a staple crop on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Average Insecurity		
Rental Price	0.0220 (0.0222)	0.0246 (0.0158)		
Customary $\times$ Rental Price	0.109*** (0.0138)	0.122*** (0.0193)		
Inverse Travel Time			-1.341*** (0.438)	
Customary $\times$ Inverse Travel Time			0.798 (0.492)	
Decrease in Travel Time				-2.162* (1.137)
Customary $\times$ Decrease in Travel Time				0.0894 (1.149)
Customary	0.0934 (0.0770)	0.0746 (0.0747)	-0.503** (0.219)	-0.139 (0.197)
Wealth Index	-0.00164 (0.00230)	-0.00141 (0.00269)	-0.000869 (0.00250)	-0.000146 (0.00236)
Landholdings	-0.00431 (0.00280)	-0.00578 (0.00377)	-0.00522** (0.00249)	-0.00552** (0.00270)
First Year Farming	0.00128* (0.000660)	0.000963 (0.000814)	0.00110 (0.000738)	0.00102 (0.000696)
Observations	3,284	3,284	3,265	3,265
R-squared	0.261	0.380	0.258	0.251
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A42: OLS results of average tenure insecurity across all rightsholders on a parcel on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
			Trees	
Rental Price	0.0145 (0.00901)	0.00979 (0.00991)		
Customary $\times$ Rental Price	-0.0418*** (0.0118)	-0.0380*** (0.00860)		
Inverse Travel Time			0.956** (0.428)	
Customary $\times$ Inverse Travel Time			0.0188 (0.439)	
Decrease in Travel Time				1.644* (0.888)
Customary $\times$ Decrease in Travel Time				-1.378* (0.747)
Customary	0.0114 (0.0464)	0.0199 (0.0500)	0.275** (0.112)	0.319*** (0.0930)
Wealth Index	-0.000725 (0.00157)	0.00166** (0.000800)	-0.00195 (0.00141)	-0.00128 (0.00129)
Landholdings	0.000974 (0.00144)	0.00173 (0.00128)	0.00133 (0.00138)	0.000802 (0.00153)
First Year Farming	-0.00311*** (0.000849)	-0.00335*** (0.00107)	-0.00309*** (0.000855)	-0.00308*** (0.000824)
Observations	3,250	3,250	3,232	3,232
R-squared	0.047	0.235	0.060	0.054
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A43: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2) Used Organic Fertilizer	(3)	(4)
Rental Price	0.00148 (0.00200)	-0.000819 (0.00109)		
Customary $\times$ Rental Price	0.0161** (0.00610)	0.00299 (0.00197)		
Inverse Travel Time			0.644** (0.261)	
Customary $\times$ Inverse Travel Time			0.323** (0.135)	
Decrease in Travel Time				4.049*** (0.498)
Customary $\times$ Decrease in Travel Time				1.516*** (0.340)
Customary	-0.0898*** (0.0258)	-0.0194 (0.0202)	-0.000372 (0.0461)	0.190*** (0.0521)
Wealth Index	0.00525*** (0.000984)	0.000912 (0.000589)	0.00418*** (0.000932)	0.00111 (0.000741)
Landholdings	0.00194 (0.00146)	0.00134 (0.00102)	0.00261* (0.00153)	0.00316*** (0.00109)
First Year Farming	-0.000385 (0.000598)	-0.000143 (0.000460)	-0.000476 (0.000586)	-0.000259 (0.000493)
Observations	3,248	3,248	3,230	3,230
R-squared	0.105	0.321	0.121	0.199
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A44: OLS results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Inorganic Fertilizer		
Rental Price	0.00348 (0.00262)	0.00512*** (0.00179)		
Customary $\times$ Rental Price	0.00912 (0.00873)	0.00419 (0.00261)		
Inverse Travel Time			0.752*** (0.0817)	
Customary $\times$ Inverse Travel Time			0.589*** (0.107)	
Decrease in Travel Time				1.338*** (0.398)
Customary $\times$ Decrease in Travel Time				0.628* (0.317)
Customary	0.00670 (0.0362)	0.00452 (0.0403)	0.0528 (0.0476)	0.0849** (0.0330)
Wealth Index	0.00465*** (0.000696)	0.00203** (0.000876)	0.00325*** (0.000727)	0.00331*** (0.000856)
Landholdings	0.00174** (0.000782)	0.00150* (0.000859)	0.00280*** (0.000913)	0.00262*** (0.000930)
First Year Farming	0.00127*** (0.000466)	0.000888 (0.000529)	0.00121** (0.000466)	0.00135*** (0.000466)
Observations	3,245	3,245	3,227	3,227
R-squared	0.028	0.150	0.057	0.041
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A45: OLS results of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2) Used Pesticides	(3)	(4)
Rental Price	0.000296 (0.00384)	-0.00113 (0.00260)		
Customary $\times$ Rental Price	0.0133 (0.00878)	-0.0126*** (0.00337)		
Inverse Travel Time			0.923*** (0.145)	
Customary $\times$ Inverse Travel Time			0.602*** (0.213)	
Decrease in Travel Time				2.647*** (0.561)
Customary $\times$ Decrease in Travel Time				0.259 (0.449)
Customary	-0.0486 (0.0661)	-0.0465 (0.0661)	0.0440 (0.0536)	0.207*** (0.0460)
Wealth Index	0.00680*** (0.000995)	0.00335*** (0.00102)	0.00492*** (0.00108)	0.00433*** (0.00105)
Landholdings	0.00215 (0.00165)	0.000837 (0.00103)	0.00262* (0.00146)	0.00241** (0.00115)
First Year Farming	0.000359 (0.000491)	0.000475 (0.000540)	0.000286 (0.000467)	0.000436 (0.000493)
Observations	3,244	3,244	3,226	3,226
R-squared	0.057	0.197	0.085	0.085
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A46: OLS results of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2) Perennial Crop	(3)	(4)
Rental Price	0.00210 (0.00210)	0.000365 (0.000703)		
Customary $\times$ Rental Price	0.0157** (0.00684)	0.00162 (0.00247)		
Inverse Travel Time			0.552 (0.335)	
Customary $\times$ Inverse Travel Time			0.104 (0.222)	
Decrease in Travel Time				4.653*** (0.651)
Customary $\times$ Decrease in Travel Time				2.108*** (0.661)
Customary	-0.0915*** (0.0281)	-0.00715 (0.0104)	0.0346 (0.0517)	0.196*** (0.0548)
Wealth Index	0.00634*** (0.00200)	0.000476 (0.000675)	0.00540*** (0.00186)	0.00117 (0.00141)
Landholdings	0.00251 (0.00201)	0.000306 (0.000712)	0.00253 (0.00185)	0.00345** (0.00145)
First Year Farming	-0.00260** (0.000989)	-0.00154* (0.000767)	-0.00269** (0.00100)	-0.00243*** (0.000870)
Observations	3,254	3,254	3,236	3,236
R-squared	0.172	0.532	0.183	0.282
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A47: OLS results of a dummy equal to one for parcels with at least one plot primarily devoted to a perennial crop on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

C.7.3 Distance to Markets

Changing patterns of input use could be responding not to rising land values, but increasing access to agricultural markets, either for inputs (agroinput dealers) or outputs. However, this should not impact customary parcels within the same local markets differently than freehold, so it seems unlikely to be driving our primary results. That is, if increasing access to Kampala reduces the price of inputs, or increases the price of outputs (as I model them doing for cash crops), these local market changes should be equally available for those farming freehold and customary parcels in the community. Improved market access may additionally be realized in non-price dimensions, such as availability or service quality, although these non-price constraints should be consistent across parcel types within the community, which is the variation I use to test the model.

However, for a subset of respondents we collected data on the distance to input and output markets, so in this appendix I control for those distances. In tables A49 - A53, I control for the one-way travel cost to the nearest agroinput dealer, to control for access to input markets. In tables A54 - A58, I control for the one-way travel cost to the nearest village in which crops can be sold, focusing on output markets. In results not reported, I also control for the average distance in kilometers at which crops were actually sold in the past season, but when crops were sold at the farm gate this was reported as a zero, making it a difficult to interpret measure (as more remote farms may be more likely to sell at the farm gate). These controls do not change the substance of my results in the main text, indicating access to markets is not driving them; however, I do not use this as a primary control given the 40% reduction in sample size when including these controls.⁶⁷

Given the high degree of missingness of these survey-based measures of market access, I do examine in table A48 whether parcels that have this data differ significantly from those without it. There are some statistically significant (although small in magnitude) differences between these two subsamples. The patterns of differences are not systematic: parcels with market access controls available are more customary, more insecure, and less likely to plant cash crops; but also more likely to use inorganic fertilizer and pesticides. Importantly, availability of these individual measures of market access are uncorrelated with my primary measures of land pressures used in the paper, rental prices and inverse travel time to Kampala.

C.7.4 Household Labor

One might be concerned that increasing land pressures, especially as driven by (new) proximity to urban markets, may tighten labor markets in addition to creating tenure insecurity on customary parcels. This does not present a major threat to my within-locale identification strategy, as co-located customary and

⁶⁷These questions were asked towards the end of the survey, and there was increasing nonresponse towards the end of the multi-hour survey.

Variable	(1) No Market Access Data Mean/(SE)	(2) Market Access Data Mean/(SE)	(1)-(2) Pairwise t-test Mean difference
Customary parcel	0.266 (0.009)	0.359 (0.007)	-0.093***
Rental price	0.003 (0.012)	-0.002 (0.017)	0.005
Inverse Travel Time	0.248 (0.003)	0.247 (0.001)	0.001
Average likelihood of losing land, 1-5	1.323 (0.019)	1.374 (0.012)	-0.051**
Cash crop	0.276 (0.013)	0.225 (0.006)	0.051***
Trees planted	0.649 (0.014)	0.634 (0.007)	0.015
Used organic fertilizer	0.108 (0.009)	0.117 (0.005)	-0.010
Inorganic fertilizer used	0.062 (0.007)	0.109 (0.005)	-0.047***
Pesticides used	0.104 (0.009)	0.159 (0.005)	-0.055***
Observations	2450	4659	7109

Significance: ***=.01, **=.05, *=.1.

Table A48: Descriptive statistics for the subsamples of respondents with and without data on one-way travel cost to agroinput dealers or the nearest village in which crops can be sold.

VARIABLES	(1)	(2)	(3)	(4)
	Average Insecurity			
Rental Price	0.0250 (0.0182)	0.0281** (0.0127)		
Customary $\times$ Rental Price	0.102*** (0.0155)	0.125*** (0.0234)		
Inverse Travel Time			-1.473*** (0.386)	
Customary $\times$ Inverse Travel Time			0.739 (0.527)	
Decrease in Travel Time				-3.221** (1.338)
Customary $\times$ Decrease in Travel Time				-0.443 (1.238)
Travel Cost to Agroinput Dealer	-1.88e-05* (9.47e-06)	-1.16e-05 (9.67e-06)	-1.15e-05 (8.04e-06)	-1.66e-05* (9.73e-06)
Customary	0.117 (0.0842)	0.0769 (0.0817)	-0.495** (0.213)	-0.169 (0.199)
Wealth Index	-0.00318 (0.00266)	-0.000852 (0.00244)	-0.00221 (0.00253)	-0.000763 (0.00238)
Landholdings	-0.00378 (0.00278)	-0.00678 (0.00419)	-0.00496** (0.00225)	-0.00557** (0.00252)
Cash Crop	0.0686 (0.0712)	0.0227 (0.0820)	0.0920 (0.0776)	0.128 (0.0854)
First Year Farming	0.00156** (0.000748)	0.000932 (0.00104)	0.00148* (0.000770)	0.00138* (0.000735)
Observations	1,724	1,724	1,715	1,715
R-squared	0.274	0.402	0.273	0.268
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A49: OLS results of average tenure insecurity across all rightsholders on a parcel on measures of land pressures, controlling for the one-way travel cost to the nearest agroinput dealer, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Trees		
Rental Price	0.0115 (0.00860)	0.00744 (0.00997)		
Customary $\times$ Rental Price	-0.0420*** (0.0138)	-0.0365*** (0.00887)		
Inverse Travel Time			0.922** (0.396)	
Customary $\times$ Inverse Travel Time			0.0648 (0.515)	
Decrease in Travel Time				1.210 (1.014)
Customary $\times$ Decrease in Travel Time				-1.530 (0.938)
Travel Cost to Agroinput dealer	1.97e-06 (6.81e-06)	-3.89e-06 (7.97e-06)	4.40e-07 (7.70e-06)	-1.51e-06 (7.62e-06)
Customary	0.0211 (0.0412)	0.0134 (0.0496)	0.261* (0.138)	0.296*** (0.104)
Wealth Index	-0.00163 (0.00164)	0.00162* (0.000820)	-0.00280** (0.00135)	-0.00176 (0.00126)
Landholdings	-0.000866 (0.00172)	0.000986 (0.00165)	-0.000336 (0.00167)	-0.000882 (0.00182)
Cash Crop	0.163*** (0.0354)	0.154*** (0.0480)	0.150*** (0.0343)	0.152*** (0.0411)
First Year Farming	-0.00293*** (0.000960)	-0.00363*** (0.00119)	-0.00295*** (0.000968)	-0.00297*** (0.000928)
Observations	1,728	1,728	1,719	1,719
R-squared	0.066	0.272	0.079	0.071
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A50: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures, controlling for the one-way travel cost to the nearest agroinput dealer, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2) Used Organic Fertilizer	(3)	(4)
Rental Price	0.000591 (0.00145)	-0.00123 (0.000822)		
Customary $\times$ Rental Price	0.0165*** (0.00610)	0.00688** (0.00270)		
Inverse Travel Time			0.607*** (0.216)	
Customary $\times$ Inverse Travel Time			0.397*** (0.123)	
Decrease in Travel Time				3.767*** (0.508)
Customary $\times$ Decrease in Travel Time				1.621*** (0.410)
Travel Cost to Agroinput Dealer	3.70e-06 (4.11e-06)	1.77e-06 (2.40e-06)	4.00e-06 (4.30e-06)	4.46e-06 (3.89e-06)
Customary	-0.0705** (0.0295)	-0.0124 (0.0258)	-0.0135 (0.0492)	0.159*** (0.0444)
Wealth Index	0.00538*** (0.000993)	0.00173** (0.000747)	0.00443*** (0.000904)	0.00184** (0.000755)
Landholdings	0.000967 (0.00138)	0.00148 (0.00110)	0.00185 (0.00161)	0.00266** (0.00130)
Cash Crop	0.236*** (0.0303)	0.0351 (0.0275)	0.221*** (0.0313)	0.159*** (0.0313)
First Year Farming	0.000577 (0.000513)	0.000531 (0.000466)	0.000479 (0.000534)	0.000591 (0.000485)
Observations	1,727	1,727	1,718	1,718
R-squared	0.207	0.404	0.216	0.271
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A51: OLS results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures, controlling for the one-way travel cost to the nearest agroinput dealer, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Inorganic Fertilizer		
Rental Price	-0.00500*** (0.00145)	-0.00222* (0.00116)		
Customary $\times$ Rental Price	0.0145 (0.0117)	0.00691** (0.00342)		
Inverse Travel Time			0.817*** (0.0922)	
Customary $\times$ Inverse Travel Time			0.827*** (0.188)	
Decrease in Travel Time				1.477*** (0.464)
Customary $\times$ Decrease in Travel Time				1.122*** (0.408)
Travel Cost to Agroinput Dealer	7.56e-06 (5.61e-06)	1.48e-05*** (4.01e-06)	8.59e-06* (4.94e-06)	8.30e-06 (5.22e-06)
Customary	0.00539 (0.0384)	0.00229 (0.0434)	0.00167 (0.0604)	0.0452 (0.0410)
Wealth Index	0.00570*** (0.000958)	0.00242** (0.00101)	0.00403*** (0.000901)	0.00404*** (0.00108)
Landholdings	0.00115 (0.00146)	0.00106 (0.00120)	0.00235 (0.00157)	0.00231 (0.00159)
Cash Crop	0.0260 (0.0198)	0.0205 (0.0234)	0.0122 (0.0191)	-0.00439 (0.0204)
First Year Farming	0.00125** (0.000542)	0.000757 (0.000559)	0.00116** (0.000528)	0.00132** (0.000537)
Observations	1,725	1,725	1,716	1,716
R-squared	0.041	0.189	0.078	0.055
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A52: OLS results of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of land pressures, controlling for the one-way travel cost to the nearest agroinput dealer, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Pesticides		
Rental Price	-0.00343 (0.00210)	-0.00614*** (0.00126)		
Customary $\times$ Rental Price	0.0208* (0.0114)	-0.00475 (0.00308)		
Inverse Travel Time			0.929*** (0.155)	
Customary $\times$ Inverse Travel Time			0.895*** (0.321)	
Decrease in Travel Time				2.521*** (0.662)
Customary $\times$ Decrease in Travel Time				0.440 (0.718)
Travel Cost to Agroinput Dealer	1.55e-05*** (5.51e-06)	1.82e-05*** (5.00e-06)	1.67e-05*** (5.22e-06)	1.43e-05** (6.05e-06)
Customary	-0.0451 (0.0815)	-0.0631 (0.0834)	-0.0322 (0.0759)	0.174*** (0.0518)
Wealth Index	0.00750*** (0.000850)	0.00408*** (0.00111)	0.00549*** (0.000880)	0.00525*** (0.00107)
Landholdings	0.00169 (0.00207)	0.000851 (0.00164)	0.00223 (0.00192)	0.00212 (0.00164)
Cash Crop	0.176*** (0.0375)	0.103*** (0.0268)	0.158*** (0.0363)	0.129*** (0.0357)
First Year Farming	0.00121** (0.000537)	0.00129* (0.000648)	0.00106* (0.000560)	0.00119** (0.000554)
Observations	1,724	1,724	1,715	1,715
R-squared	0.114	0.277	0.145	0.135
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A53: OLS results of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of land pressures, controlling for the one-way travel cost to the nearest agroinput dealer, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
	Average Insecurity			
Rental Price	0.0240 (0.0184)	0.0276** (0.0130)		
Customary $\times$ Rental Price	0.106*** (0.0159)	0.127*** (0.0239)		
Inverse Travel Time			-1.499*** (0.395)	
Customary $\times$ Inverse Travel Time			0.804 (0.519)	
Decrease in Travel Time				-3.289** (1.323)
Customary $\times$ Decrease in Travel Time				-0.212 (1.199)
Travel Cost to Output Market	8.36e-07 (7.03e-06)	-1.30e-06 (4.70e-06)	-1.16e-06 (6.08e-06)	-9.11e-07 (5.97e-06)
Customary	0.116 (0.0843)	0.0744 (0.0821)	-0.522** (0.211)	-0.200 (0.196)
Wealth Index	-0.00290 (0.00259)	-0.000864 (0.00247)	-0.00210 (0.00251)	-0.000627 (0.00237)
Landholdings	-0.00426 (0.00272)	-0.00681 (0.00424)	-0.00521** (0.00222)	-0.00591** (0.00246)
Cash Crop	0.0684 (0.0684)	0.0242 (0.0822)	0.0928 (0.0752)	0.128 (0.0820)
First Year Farming	0.00154** (0.000727)	0.000972 (0.00101)	0.00148* (0.000759)	0.00138* (0.000716)
Observations	1,721	1,721	1,712	1,712
R-squared	0.272	0.402	0.272	0.266
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A54: OLS results of average tenure insecurity across all rightsholders on a parcel on measures of land pressures, controlling for the one-way travel cost to the nearest village in which crops can be sold, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Trees		
Rental Price	0.0116 (0.00860)	0.00729 (0.0101)		
Customary $\times$ Rental Price	-0.0427*** (0.0132)	-0.0359*** (0.00841)		
Inverse Travel Time			0.918** (0.406)	
Customary $\times$ Inverse Travel Time			0.0683 (0.502)	
Decrease in Travel Time				1.145 (1.021)
Customary $\times$ Decrease in Travel Time				-1.530* (0.898)
Travel Cost to Output Market	-9.23e-07 (1.99e-06)	6.93e-07 (2.01e-06)	7.36e-07 (2.22e-06)	-1.13e-06 (1.94e-06)
Customary	0.0212 (0.0408)	0.0124 (0.0494)	0.260* (0.133)	0.289*** (0.102)
Wealth Index	-0.00169 (0.00162)	0.00164* (0.000832)	-0.00280*** (0.00136)	-0.00172 (0.00127)
Landholdings	-0.000814 (0.00166)	0.000966 (0.00168)	-0.000340 (0.00162)	-0.000924 (0.00175)
Cash Crop	0.162*** (0.0355)	0.155*** (0.0477)	0.150*** (0.0349)	0.153*** (0.0411)
First Year Farming	-0.00293*** (0.000956)	-0.00362*** (0.00119)	-0.00297*** (0.000973)	-0.00297*** (0.000931)
Observations	1,725	1,725	1,716	1,716
R-squared	0.066	0.271	0.079	0.071
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A55: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures, controlling for the one-way travel cost to the nearest village in which crops can be sold, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Organic Fertilizer		
Rental Price	0.000777 (0.00153)	-0.00109 (0.000793)		
Customary $\times$ Rental Price	0.0167*** (0.00617)	0.00690** (0.00263)		
Inverse Travel Time			0.624*** (0.212)	
Customary $\times$ Inverse Travel Time			0.419*** (0.120)	
Decrease in Travel Time				3.786*** (0.505)
Customary $\times$ Decrease in Travel Time				1.665*** (0.391)
Travel Cost to Output Market	4.44e-06*** (1.29e-06)	2.95e-06** (1.27e-06)	5.89e-06*** (1.41e-06)	6.83e-06*** (1.81e-06)
Customary	-0.0671** (0.0283)	-0.0119 (0.0258)	-0.0106 (0.0469)	0.161*** (0.0425)
Wealth Index	0.00536*** (0.000964)	0.00176** (0.000739)	0.00439*** (0.000890)	0.00183** (0.000765)
Landholdings	0.000985 (0.00137)	0.00148 (0.00108)	0.00188 (0.00160)	0.00270** (0.00130)
Cash Crop	0.233*** (0.0308)	0.0343 (0.0277)	0.216*** (0.0319)	0.154*** (0.0315)
First Year Farming	0.000515 (0.000524)	0.000498 (0.000478)	0.000401 (0.000541)	0.000513 (0.000490)
Observations	1,724	1,724	1,715	1,715
R-squared	0.207	0.398	0.216	0.271
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A56: OLS results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures, controlling for the one-way travel cost to the nearest village in which crops can be sold, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Inorganic Fertilizer		
Rental Price	-0.00462*** (0.00157)	-0.00172 (0.00132)		
Customary $\times$ Rental Price	0.0121 (0.0105)	0.00497 (0.00306)		
Inverse Travel Time			0.803*** (0.0975)	
Customary $\times$ Inverse Travel Time			0.761*** (0.195)	
Decrease in Travel Time				1.376*** (0.484)
Customary $\times$ Decrease in Travel Time				0.946** (0.416)
Travel Cost to Output Market	-3.44e-06** (1.60e-06)	-2.72e-06* (1.39e-06)	-1.43e-06 (1.23e-06)	-2.66e-06** (1.23e-06)
Customary	0.00490 (0.0384)	0.00291 (0.0436)	0.0165 (0.0615)	0.0517 (0.0424)
Wealth Index	0.00551*** (0.000922)	0.00255** (0.00103)	0.00394*** (0.000887)	0.00403*** (0.00109)
Landholdings	0.00136 (0.00143)	0.00105 (0.00129)	0.00252 (0.00156)	0.00245 (0.00158)
Cash Crop	0.0275 (0.0192)	0.0215 (0.0228)	0.0132 (0.0186)	-0.000511 (0.0196)
First Year Farming	0.00125** (0.000553)	0.000762 (0.000567)	0.00115** (0.000547)	0.00131** (0.000555)
Observations	1,722	1,722	1,713	1,713
R-squared	0.041	0.185	0.075	0.053
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A57: OLS results of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of land pressures, controlling for the one-way travel cost to the nearest village in which crops can be sold, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Pesticides		
Rental Price	-0.00258 (0.00194)	-0.00524*** (0.00138)		
Customary $\times$ Rental Price	0.0186* (0.0106)	-0.00716** (0.00298)		
Inverse Travel Time			0.996*** (0.151)	
Customary $\times$ Inverse Travel Time			0.838** (0.319)	
Decrease in Travel Time				2.652*** (0.660)
Customary $\times$ Decrease in Travel Time				0.312 (0.696)
Travel Cost to Output Market	3.49e-06** (1.34e-06)	5.20e-06*** (1.61e-06)	6.17e-06* (3.09e-06)	4.86e-06** (2.40e-06)
Customary	-0.0422 (0.0810)	-0.0574 (0.0835)	0.00651 (0.0728)	0.203*** (0.0533)
Wealth Index	0.00733*** (0.000857)	0.00402*** (0.00110)	0.00534*** (0.000895)	0.00510*** (0.00108)
Landholdings	0.00203 (0.00209)	0.000930 (0.00167)	0.00257 (0.00189)	0.00241 (0.00163)
Cash Crop	0.173*** (0.0371)	0.0980*** (0.0264)	0.154*** (0.0364)	0.125*** (0.0352)
First Year Farming	0.00114* (0.000574)	0.00112 (0.000696)	0.000974 (0.000597)	0.00111* (0.000592)
Observations	1,721	1,721	1,712	1,712
R-squared	0.110	0.275	0.143	0.133
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A58: OLS results of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of land pressures, controlling for the one-way travel cost to the nearest village in which crops can be sold, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

Variable	(1) Freehold Mean/(SE)	(2) Customary Mean/(SE)	(1)-(2) Pairwise t-test Mean difference
Any family labor used on parcel	0.992 (0.001)	0.996 (0.001)	-0.003
Any hired labor used on parcel	0.419 (0.008)	0.361 (0.010)	0.058***
Days of family labor used on parcel	46.646 (1.351)	50.892 (1.950)	-4.246*
Days of hired labor used on parcel	4.946 (0.248)	3.821 (0.217)	1.124***
Household Dependency Ratio	1.347 (0.018)	1.366 (0.024)	-0.020

Significance: ***=.01, **=.05, *=.1.

Table A59: Descriptive statistics on family and hired labor use between freehold and customary parcels.

freehold parcels share a labor market. If, however, customary parcels are systematically more reliant on hired labor than freehold parcels, higher wages in these markets would disproportionately increase the cost of (labor-intensive) investment in trees on customary parcels.

To see whether this is the case, I first compare labor usage on customary and freehold parcels, in table A59. Although there are systematic differences in use of hired and family labor across tenure types, they do not suggest a greater reliance on hired labor markets for customary parcels (in fact, the opposite appears to be true). This, if anything, would attenuate the tenure insecurity effects that I document in the main results.

As a further check that changing labor market conditions are not driving my results, I additionally re-run my primary specifications, controlling for the household dependency ratio (number of children under age 15 and elderly over 64, divided by the number of working age adults 15-64 in the household, calculated from the household roster) in tables A60 - A64. In tables A65 - A69, I instead control for the number of working-age adults in the household, as a direct measure of the household's ability to supply labor to their fields. The primary interactions of interest are quite similar to those in the main text.

C.7.5 Duration of Farming

Unfortunately, the ACDP data only contains information about the stock of trees on a parcel, rather than investments in planting new trees and the timing of planting. This differs from the model, in which farmers invest in planting trees in period one. In particular, the model predicts that farmers will increase their tree planting on freehold land in response to rising land pressures, but will do so less on customary parcels. If,

VARIABLES	(1)	(2)	(3)	(4)
		Average Insecurity		
Rental Price	0.0171 (0.0176)	0.0207* (0.0118)		
Customary $\times$ Rental Price	0.104*** (0.0138)	0.117*** (0.0168)		
Inverse Travel Time			-1.258** (0.489)	
Customary $\times$ Inverse Travel Time			0.757 (0.507)	
Decrease in Travel Time				-2.604* (1.295)
Customary $\times$ Decrease in Travel Time				-0.264 (1.161)
Dependency Ratio	0.00504 (0.0167)	0.00783 (0.0192)	0.00953 (0.0152)	0.0108 (0.0153)
Customary	0.102 (0.0786)	0.0731 (0.0756)	-0.460* (0.228)	-0.138 (0.201)
Wealth Index	-0.00140 (0.00261)	-0.000964 (0.00261)	-0.000691 (0.00265)	0.000550 (0.00243)
Landholdings	-0.00420 (0.00274)	-0.00563 (0.00376)	-0.00526** (0.00242)	-0.00579** (0.00267)
Cash Crops	0.0580 (0.0743)	0.000748 (0.0762)	0.0726 (0.0799)	0.101 (0.0897)
First Year Farming	0.00149** (0.000585)	0.00148 (0.000925)	0.00136** (0.000629)	0.00129** (0.000611)
Observations	3,190	3,190	3,174	3,174
R-squared	0.258	0.379	0.254	0.250
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A60: OLS results of average tenure insecurity across all rightsholders on a parcel on measures of land pressures, controlling for the household dependency ratio, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Trees		
Rental Price	0.0127 (0.00798)	0.00726 (0.00856)		
Customary $\times$ Rental Price	-0.0415*** (0.0121)	-0.0347*** (0.00848)		
Inverse Travel Time			0.889** (0.403)	
Customary $\times$ Inverse Travel Time			-0.0122 (0.442)	
Decrease in Travel Time				0.990 (0.916)
Customary $\times$ Decrease in Travel Time				-1.739** (0.827)
Dependency Ratio	-0.0113 (0.0103)	-0.0170* (0.00945)	-0.0126 (0.0111)	-0.00991 (0.0113)
Customary	0.0229 (0.0424)	0.0194 (0.0497)	0.275** (0.110)	0.296*** (0.0921)
Wealth Index	-0.00148 (0.00164)	0.00165* (0.000917)	-0.00256* (0.00150)	-0.00118 (0.00139)
Landholdings	0.000352 (0.00153)	0.00144 (0.00133)	0.000801 (0.00146)	7.67e-05 (0.00158)
Cash Crops	0.144*** (0.0342)	0.126*** (0.0389)	0.132*** (0.0316)	0.135*** (0.0359)
First Year Farming	-0.00294*** (0.000842)	-0.00325*** (0.00106)	-0.00291*** (0.000834)	-0.00298*** (0.000825)
Observations	3,200	3,200	3,184	3,184
R-squared	0.061	0.245	0.071	0.066
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A61: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures, controlling for the household dependency ratio, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2) Used Organic Fertilizer	(3)	(4)
Rental Price	-0.000991 (0.00127)	-0.00136 (0.000935)		
Customary $\times$ Rental Price	0.0137** (0.00569)	0.00457* (0.00246)		
Inverse Travel Time			0.537** (0.208)	
Customary $\times$ Inverse Travel Time			0.300*** (0.105)	
Decrease in Travel Time				3.479*** (0.478)
Customary $\times$ Decrease in Travel Time				1.213*** (0.308)
Dependency Ratio	0.00198 (0.00628)	-0.0111 (0.00733)	0.000699 (0.00684)	-0.00488 (0.00689)
Customary	-0.0664** (0.0257)	-0.0188 (0.0206)	-0.00203 (0.0439)	0.174*** (0.0489)
Wealth Index	0.00403*** (0.000589)	0.000501 (0.000660)	0.00317*** (0.000581)	0.000722 (0.000618)
Landholdings	0.000885 (0.00109)	0.00130 (0.00100)	0.00166 (0.00123)	0.00238** (0.000965)
Cash Crops	0.207*** (0.0329)	0.0381 (0.0268)	0.195*** (0.0320)	0.140*** (0.0284)
First Year Farming	-0.000264 (0.000564)	-0.000282 (0.000517)	-0.000339 (0.000557)	-0.000250 (0.000518)
Observations	3,198	3,198	3,182	3,182
R-squared	0.172	0.326	0.177	0.227
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A62: OLS results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures, controlling for the household dependency ratio, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2) Used Inorganic Fertilizer	(3)	(4)
Rental Price	0.00358 (0.00238)	0.00494** (0.00189)		
Customary $\times$ Rental Price	0.0105 (0.00779)	0.00556** (0.00242)		
Inverse Travel Time			0.746*** (0.0876)	
Customary $\times$ Inverse Travel Time			0.603*** (0.106)	
Decrease in Travel Time				1.372*** (0.421)
Customary $\times$ Decrease in Travel Time				0.686** (0.321)
Dependency Ratio	-0.00127 (0.00523)	0.000519 (0.00589)	-0.00374 (0.00518)	-0.00374 (0.00499)
Customary	0.00588 (0.0359)	0.000453 (0.0404)	0.0437 (0.0465)	0.0776** (0.0323)
Wealth Index	0.00471*** (0.000770)	0.00214** (0.000853)	0.00324*** (0.000769)	0.00334*** (0.000874)
Landholdings	0.00161* (0.000797)	0.00138 (0.000876)	0.00275*** (0.000909)	0.00264*** (0.000946)
Cash Crops	0.0127 (0.0149)	0.0147 (0.0175)	0.00244 (0.0135)	-0.0131 (0.0163)
First Year Farming	0.00116** (0.000495)	0.000719 (0.000547)	0.00113** (0.000467)	0.00123** (0.000470)
Observations	3,195	3,195	3,179	3,179
R-squared	0.030	0.151	0.057	0.041
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A63: OLS results of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of land pressures, controlling for the household dependency ratio, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Pesticides		
Rental Price	-0.000995 (0.00375)	-0.00209 (0.00254)		
Customary $\times$ Rental Price	0.0122 (0.00759)	-0.0118*** (0.00364)		
Inverse Travel Time			0.874*** (0.119)	
Customary $\times$ Inverse Travel Time			0.602** (0.228)	
Decrease in Travel Time				2.269*** (0.483)
Customary $\times$ Decrease in Travel Time				0.0682 (0.537)
Dependency Ratio	-0.00447 (0.00481)	-0.00580 (0.00886)	-0.00762 (0.00469)	-0.00848* (0.00482)
Customary	-0.0405 (0.0651)	-0.0530 (0.0661)	0.0365 (0.0489)	0.189*** (0.0460)
Wealth Index	0.00604*** (0.000690)	0.00328*** (0.00106)	0.00422*** (0.000845)	0.00413*** (0.000958)
Landholdings	0.00146 (0.00143)	0.000489 (0.000992)	0.00205 (0.00129)	0.00186* (0.00105)
Cash Crops	0.146*** (0.0253)	0.0861*** (0.0206)	0.132*** (0.0256)	0.107*** (0.0259)
First Year Farming	0.000388 (0.000484)	0.000275 (0.000538)	0.000326 (0.000462)	0.000385 (0.000489)
Observations	3,195	3,195	3,179	3,179
R-squared	0.083	0.206	0.107	0.100
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A64: OLS results of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of land pressures, controlling for the household dependency ratio, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Average Insecurity		
Rental price	0.0209 (0.0219)	0.0247 (0.0157)		
Customary $\times$ Rental Price	0.105*** (0.0137)	0.119*** (0.0183)		
Inverse Travel Time			-1.349*** (0.457)	
Customary $\times$ Inverse Travel Time			0.839* (0.497)	
Decrease in Travel Time				-2.591* (1.307)
Customary $\times$ Decrease in Travel Time				-0.0672 (1.134)
Number Adults	-0.0212* (0.0119)	-0.0119 (0.0137)	-0.0164 (0.0103)	-0.0183* (0.0108)
Customary	0.104 (0.0787)	0.0749 (0.0751)	-0.506** (0.223)	-0.154 (0.198)
Wealth Index	-0.00292 (0.00254)	-0.00218 (0.00277)	-0.00210 (0.00256)	-0.00112 (0.00239)
Landholdings	-0.00396 (0.00256)	-0.00519 (0.00355)	-0.00505** (0.00223)	-0.00549** (0.00247)
Cash Crops	0.0596 (0.0716)	0.00926 (0.0758)	0.0780 (0.0777)	0.105 (0.0882)
First Year Farming	0.00148** (0.000638)	0.00124 (0.000869)	0.00138* (0.000686)	0.00130* (0.000657)
Observations	3,240	3,240	3,222	3,222
R-squared	0.263	0.382	0.260	0.254
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A65: OLS results of average tenure insecurity across all rightsholders on a parcel on measures of land pressures, controlling for the number of adults in the household, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Trees		
Rental Price	0.0131 (0.00809)	0.00851 (0.00951)		
Customary $\times$ Rental Price	-0.0423*** (0.0120)	-0.0359*** (0.00846)		
Inverse Travel Time			0.842** (0.395)	
Customary $\times$ Inverse Travel Time			-0.0658 (0.433)	
Decrease in Travel Time				0.939 (0.873)
Customary $\times$ Decrease in Travel Time				-1.834** (0.783)
Number of Adults	0.0187** (0.00843)	0.0181*** (0.00526)	0.0149** (0.00725)	0.0195** (0.00807)
Customary	0.0283 (0.0429)	0.0250 (0.0502)	0.282** (0.109)	0.305*** (0.0909)
Wealth Index	-0.00100 (0.00160)	0.00247*** (0.000847)	-0.00204 (0.00129)	-0.000635 (0.00121)
Landholdings	-6.92e-05 (0.00170)	0.00101 (0.00147)	0.000380 (0.00162)	-0.000401 (0.00180)
Cash Crops	0.144*** (0.0338)	0.123*** (0.0389)	0.132*** (0.0314)	0.136*** (0.0358)
First Year Farming	-0.00285*** (0.000810)	-0.00321*** (0.00105)	-0.00284*** (0.000811)	-0.00287*** (0.000792)
Observations	3,250	3,250	3,232	3,232
R-squared	0.065	0.245	0.074	0.070
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A66: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures, controlling for the number of adults in the household, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2) Used Organic Fertilizer	(3)	(4)
Rental Price	-0.000913 (0.00131)	-0.00117 (0.000967)		
Customary $\times$ Rental Price	0.0139** (0.00520)	0.00396 (0.00247)		
Inverse Travel Time			0.541*** (0.192)	
Customary $\times$ Inverse Travel Time			0.297** (0.117)	
Decrease in Travel Time				3.410*** (0.436)
Customary $\times$ Decrease in Travel Time				1.190*** (0.317)
Number of Adults	0.00340 (0.00693)	0.00880 (0.00665)	-1.96e-05 (0.00697)	0.000268 (0.00644)
Customary	-0.0661** (0.0247)	-0.0176 (0.0200)	0.000707 (0.0433)	0.170*** (0.0460)
Wealth Index	0.00410*** (0.000831)	0.00131* (0.000733)	0.00318*** (0.000862)	0.00101 (0.000786)
Landholdings	0.000945 (0.00112)	0.00105 (0.000935)	0.00177 (0.00126)	0.00242** (0.000952)
Cash Crops	0.207*** (0.0307)	0.0336 (0.0255)	0.194*** (0.0300)	0.137*** (0.0271)
First Year Farming	-1.73e-05 (0.000537)	-9.53e-05 (0.000480)	-0.000117 (0.000526)	-4.08e-05 (0.000476)
Observations	3,248	3,248	3,230	3,230
R-squared	0.169	0.324	0.175	0.224
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A67: OLS results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures, controlling for the number of adults in the household, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Inorganic Fertilizer		
Rental Price	0.00363 (0.00278)	0.00500** (0.00187)		
Customary $\times$ Rental Price	0.00991 (0.00771)	0.00494* (0.00261)		
Inverse Travel Time			0.719*** (0.0821)	
Customary $\times$ Inverse Travel Time			0.540*** (0.105)	
Decrease in Travel Time				1.348*** (0.416)
Customary $\times$ Decrease in Travel Time				0.561* (0.316)
Number Adults	0.0151*** (0.00518)	0.00713 (0.00453)	0.0109** (0.00462)	0.0144*** (0.00519)
Customary	0.00842 (0.0349)	0.00559 (0.0399)	0.0573 (0.0471)	0.0911** (0.0348)
Wealth Index	0.00506*** (0.000771)	0.00235** (0.000935)	0.00367*** (0.000750)	0.00387*** (0.000871)
Landholdings	0.00136* (0.000785)	0.00129 (0.000926)	0.00252*** (0.000891)	0.00235** (0.000944)
Cash Crops	0.0129 (0.0142)	0.0116 (0.0175)	0.00219 (0.0129)	-0.0133 (0.0155)
First Year Farming	0.00130*** (0.000462)	0.000914* (0.000525)	0.00122*** (0.000452)	0.00133*** (0.000449)
Observations	3,245	3,245	3,227	3,227
R-squared	0.035	0.151	0.060	0.046
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A68: OLS results of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of land pressures, controlling for the number of adults in the household, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Pesticides		
Rental Price	-0.00113 (0.00346)	-0.00202 (0.00256)		
Customary $\times$ Rental Price	0.0125 (0.00747)	-0.0115*** (0.00370)		
Inverse Travel Time			0.831*** (0.124)	
Customary $\times$ Inverse Travel Time			0.548** (0.240)	
Decrease in Travel Time				2.114*** (0.495)
Customary $\times$ Decrease in Travel Time				-0.0741 (0.546)
Number Adults	0.0135** (0.00552)	0.00873 (0.00684)	0.00809 (0.00600)	0.0123** (0.00581)
Customary	-0.0315 (0.0644)	-0.0433 (0.0661)	0.0488 (0.0503)	0.196*** (0.0436)
Wealth Index	0.00634*** (0.000765)	0.00374*** (0.00111)	0.00456*** (0.000949)	0.00472*** (0.00106)
Landholdings	0.00123 (0.00134)	0.000437 (0.00103)	0.00185 (0.00124)	0.00154 (0.00100)
Cash Crops	0.143*** (0.0265)	0.0844*** (0.0214)	0.130*** (0.0268)	0.106*** (0.0265)
First Year Farming	0.000629 (0.000499)	0.000567 (0.000566)	0.000538 (0.000478)	0.000605 (0.000499)
Observations	3,244	3,244	3,226	3,226
R-squared	0.083	0.204	0.104	0.099
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A69: OLS results of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of land pressures, controlling for the number of adults in the household, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

however, those customary parcels already have trees planted on them as remaining stock from a prior low land pressure environment, I may not be able to empirically capture the dynamics of the model. However, the direction of this bias should underestimate the true divergence between customary and freehold parcels (as the observed stock of trees on customary parcels in high pressure environments would be higher than predicted by the model), making my results a conservative estimate of the true effect.

Although I am limited by the data in my ability to see flows of trees, I can see whether the divergence between customary and freehold parcels in response to land pressures is greater on more recently farmed parcels. That is, for parcels which were first farmed after 2011 (the median date of first year farming in my data), is the divergence between customary and freehold parcels greater than for parcels first farmed earlier? These more recently-farmed parcels should have less residual stock of trees from earlier land pressure environments, and so should capture more recent investment. Indeed, when comparing table A70 to table A71, more recently farmed parcels show a stronger differential in tree planting responses to land pressures across customary and freehold parcels. This is consistent with my main results underestimating the true impact of land pressures on tree planting.

C.7.6 Risk Preferences

The structural transformation literature would suggest an alternate mechanism behind my results: differential selection into who continues farming on customary versus freehold parcels in higher land pressure environments. I suggested that differential resource constraints of these remaining farmers could be one explanation for the divergence in short-term inputs that was not predicted by the model; this would not explain the larger divergence in long-term inputs that I attribute to higher tenure insecurity as a result of the potential for elite expropriation.

There could also be differential selection on the risk aversion of farmers, however. In the model, a risk-neutral farmer will under-invest in long-term inputs on customary parcels in high land pressure environments due to tenure insecurity. That could be exacerbated if the risk of losing the land weighs more on more risk-averse farmers who remain on customary parcels as land values increase. The underlying tenure insecurity is still driving the divergence, but the compositional channel amplifies its effect. I can, however, control for risk aversion of the respondents, measured using a modified Binswanger (1980) elicitation task. Results, in tables A72 - A76, are robust to the inclusion of this control. This suggests that the insecurity does not operate entirely through the differential risk aversion of farmers, but instead is a response to changing risk of expropriation itself.

Additionally, even if there exist differential aversion to the risk presented by customary tenure in high land-pressure environments, these risks are perceived by the full sample. This can be seen in the hypothetical

VARIABLES	(1)	(2)	(3)	(4)
		Trees: Recent Parcels		
Rental Price	0.0120*	0.00784		
	(0.00684)	(0.00863)		
Customary $\times$ Rental Price	-0.419	-0.613*		
	(0.344)	(0.315)		
Inverse Travel Time			0.958**	
			(0.439)	
Customary $\times$ Inverse Travel Time			-0.0671	
			(0.485)	
Decrease in Travel Time				0.707
				(1.020)
Customary $\times$ Decrease in Travel Time				-1.975**
				(0.905)
Customary	-0.00558	-0.0263	0.289**	0.256**
	(0.0709)	(0.0719)	(0.140)	(0.114)
Wealth Index	-0.00243	0.00187	-0.00441**	-0.00276*
	(0.00173)	(0.00124)	(0.00175)	(0.00152)
Landholdings	-0.00249	0.000179	-0.000702	-0.00251
	(0.00518)	(0.00356)	(0.00439)	(0.00500)
Cash Crops	0.0785	0.0911*	0.0703	0.0724
	(0.0481)	(0.0506)	(0.0480)	(0.0438)
First Year Farming	-0.0212*	-0.0250*	-0.0207	-0.0212*
	(0.0120)	(0.0136)	(0.0123)	(0.0122)
Observations	1,589	1,589	1,590	1,590
R-squared	0.040	0.303	0.055	0.043
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A70: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time. Sample restricted to parcels first farmed after 2011.

VARIABLES	(1)	(2)	(3)	(4)
		Trees: Older Parcels		
Rental Price	0.0847 (0.134)	0.0104 (0.0815)		
Customary $\times$ Rental Price	-0.0411** (0.0153)	-0.0342*** (0.0103)		
Inverse Travel Time			0.818** (0.352)	
Customary $\times$ Inverse Travel Time			0.0265 (0.448)	
Decrease in Travel Time				1.303 (0.958)
Customary $\times$ Decrease in Travel Time				-1.694* (0.915)
customary	0.0439 (0.0386)	0.0283 (0.0513)	0.266** (0.131)	0.350*** (0.117)
Wealth Index	-0.000207 (0.00193)	0.00286* (0.00159)	-0.000401 (0.00182)	0.000638 (0.00183)
Landholdings	0.000710 (0.00132)	0.00292* (0.00168)	0.000891 (0.00146)	0.000531 (0.00164)
Cash Crops	0.175*** (0.0390)	0.153*** (0.0532)	0.158*** (0.0409)	0.159*** (0.0471)
First Year Farming	-0.00267** (0.00101)	-0.00252* (0.00137)	-0.00262** (0.000965)	-0.00268** (0.000999)
Constant	5.747*** (2.036)	5.505* (2.759)	5.480*** (1.988)	5.623*** (2.048)
Observations	1,661	1,661	1,642	1,642
R-squared	0.085	0.306	0.091	0.093
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A71: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time. Sample restricted to parcels first farmed prior to 2011.

VARIABLES	(1)	(2)	(3)	(4)
		Average Insecurity		
Rental Price	0.112 (0.0688)	0.0709 (0.0617)		
Customary $\times$ Rental Price	0.0988*** (0.0167)	0.112*** (0.0234)		
Inverse in Travel Time			-1.727*** (0.600)	
Customary $\times$ Inverse Travel Time			0.855 (0.561)	
Decrease in Travel Time				-3.245* (1.746)
Customary $\times$ Decrease in Travel Time				-0.0925 (1.241)
Risk Category 2	-0.00809 (0.0790)	0.0582 (0.0775)	0.0174 (0.0839)	0.0212 (0.0794)
Risk Category 3	-0.218* (0.112)	-0.151 (0.106)	-0.220* (0.117)	-0.218* (0.119)
Risk Category 4	0.243 (0.198)	0.367* (0.191)	0.272 (0.202)	0.256 (0.203)
Risk Category 5	0.384 (0.368)	0.357 (0.448)	0.399 (0.383)	0.307 (0.388)
Risk Category 6	0.140 (0.348)	0.229 (0.307)	0.210 (0.344)	0.205 (0.339)
Customary	0.122 (0.0973)	0.0795 (0.0879)	-0.598** (0.258)	-0.202 (0.243)
Wealth Index	-0.00270 (0.00257)	-0.00224 (0.00307)	-0.00205 (0.00278)	-0.000974 (0.00262)
Landholdings	-0.00442 (0.00293)	-0.00583 (0.00425)	-0.00586** (0.00246)	-0.00642** (0.00282)
Cash Crop	0.0606 (0.0752)	0.0192 (0.0665)	0.0936 (0.0866)	0.126 (0.102)
First Year Farming	0.00132 (0.000941)	2.33e-05 (0.00150)	0.00135 (0.000957)	0.00119 (0.000940)
Constant	-0.944 (1.957)	1.337 (3.078)	-0.561 (1.969)	-0.413 (1.949)
Observations	2,738	2,738	2,727	2,727
R-squared	0.272	0.418	0.267	0.259
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A72: OLS results of average tenure insecurity across all rightsholders on a parcel on measures of land pressures, controlling for a categorical measure of risk aversion, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Trees		
Rental Price	0.0456*** (0.0149)	0.0504*** (0.0133)		
Customary $\times$ Rental Price	-0.0396*** (0.0106)	-0.0293*** (0.00499)		
Inverse Travel Time			0.844* (0.424)	
Customary $\times$ Inverse Travel Time			0.0353 (0.459)	
Decrease in Travel Time				1.044 (0.908)
Customary $\times$ Decrease in Travel Time				-1.445* (0.820)
Risk Category 2	0.0474 (0.0509)	0.0494 (0.0549)	0.0336 (0.0512)	0.0232 (0.0492)
Risk Category 3	0.0803 (0.0525)	0.0764 (0.0538)	0.0844 (0.0563)	0.0673 (0.0545)
Risk Category 4	0.117 (0.0779)	0.0507 (0.0711)	0.108 (0.0791)	0.0859 (0.0797)
Risk Category 5	-0.0968 (0.135)	-0.0889 (0.0783)	-0.0760 (0.145)	-0.128 (0.149)
Risk Category 6	-0.0589 (0.139)	-0.0401 (0.130)	-0.0896 (0.143)	-0.0889 (0.142)
Customary	0.00677 (0.0498)	-0.000781 (0.0576)	0.232* (0.131)	0.257** (0.0961)
Wealth Index	-0.00193 (0.00170)	0.00116 (0.000927)	-0.00285* (0.00157)	-0.00181 (0.00144)
Landholdings	0.000490 (0.00148)	0.00196 (0.00165)	0.000849 (0.00152)	0.000260 (0.00165)
Cash Crop	0.163*** (0.0343)	0.134*** (0.0416)	0.153*** (0.0328)	0.156*** (0.0363)
First Year Farming	-0.00228** (0.000890)	-0.00270*** (0.000955)	-0.00229** (0.000879)	-0.00238** (0.000892)
Constant	5.083*** (1.773)	5.385*** (1.937)	4.923*** (1.734)	5.167*** (1.772)
Observations	2,749	2,749	2,738	2,738
R-squared	0.067	0.260	0.075	0.070
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A73: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures, controlling for the a categorical measure of risk aversion, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Organic Fertilizer		
Rental Price	0.00470 (0.00798)	0.00512 (0.00455)		
Customary $\times$ Rental Price	0.0135*** (0.00359)	0.00681** (0.00270)		
Inverse Travel Time			0.577** (0.223)	
Customary $\times$ Inverse Travel Time			0.259** (0.119)	
Decrease in Travel Time				3.727*** (0.512)
Customary $\times$ Decrease in Travel Time				1.152*** (0.335)
Risk Category 2	-0.00975 (0.0384)	-0.0506 (0.0379)	-0.0153 (0.0406)	-0.0351 (0.0419)
Risk Category 3	0.0731 (0.0533)	0.00851 (0.0477)	0.0783 (0.0543)	0.0746 (0.0536)
Risk Category 4	-0.0727** (0.0309)	-0.121** (0.0503)	-0.0686* (0.0343)	-0.0653 (0.0412)
Risk Category 5	-0.0436** (0.0214)	-0.00502 (0.0124)	-0.0158 (0.0240)	0.0665* (0.0354)
Risk Category 6	0.0538 (0.160)	0.0436 (0.133)	0.0321 (0.160)	-0.0146 (0.160)
Customary	-0.0563** (0.0245)	-0.0187 (0.0226)	0.0313 (0.0502)	0.218*** (0.0539)
Wealth Index	0.00427*** (0.000782)	0.00114 (0.000719)	0.00355*** (0.000772)	0.00135* (0.000776)
Landholdings	0.000975 (0.00116)	0.00144 (0.00114)	0.00174 (0.00129)	0.00246** (0.000976)
Cash Crop	0.208*** (0.0276)	0.0345 (0.0226)	0.195*** (0.0266)	0.136*** (0.0242)
First Year Farming	0.000106 (0.000606)	-2.59e-05 (0.000516)	3.88e-05 (0.000597)	6.78e-05 (0.000546)
Constant	-0.241 (1.234)	0.0420 (1.074)	-0.230 (1.193)	-0.418 (1.095)
Observations	2,747	2,747	2,736	2,736
R-squared	0.178	0.331	0.183	0.236
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A74: OLS results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures, controlling for a categorical measure of risk aversion, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Inorganic Fertilizer		
Rental Price	-0.00992** (0.00468)	-0.00311 (0.00367)		
Customary × Rental Price	0.00807 (0.00598)	0.00445* (0.00242)		
Inverse Travel Time			0.784*** (0.0968)	
Customary × Inverse Travel Time			0.500*** (0.112)	
Decrease in Travel Time				1.385*** (0.441)
Customary × Decrease in Travel Time				0.457 (0.351)
Risk Category 2	-0.0287 (0.0174)	-0.0198 (0.0237)	-0.0330* (0.0172)	-0.0366* (0.0182)
Risk Category 3	0.0608 (0.0656)	0.0605 (0.0645)	0.0732 (0.0679)	0.0623 (0.0687)
Risk Category 4	-0.0557 (0.0334)	-0.0928** (0.0365)	-0.0423 (0.0294)	-0.0518 (0.0322)
Risk Category 5	-0.111*** (0.0174)	-0.0554 (0.0390)	-0.0589*** (0.0180)	-0.0690** (0.0272)
Risk Category 6	-0.132*** (0.0474)	-0.0970 (0.0594)	-0.156*** (0.0484)	-0.157*** (0.0435)
Customary	0.00467 (0.0380)	0.00299 (0.0412)	0.0844 (0.0539)	0.103** (0.0389)
Wealth Index	0.00475*** (0.000773)	0.00231** (0.000918)	0.00365*** (0.000739)	0.00378*** (0.000836)
Landholdings	0.00179** (0.000879)	0.00146 (0.00101)	0.00295*** (0.00102)	0.00282*** (0.00104)
Cash Crop	0.0140 (0.0162)	0.0216 (0.0214)	-0.00126 (0.0154)	-0.0141 (0.0188)
First Year Farming	0.00138** (0.000542)	0.000988 (0.000627)	0.00139** (0.000525)	0.00142** (0.000531)
Constant	-2.900** (1.091)	-2.142* (1.263)	-3.077*** (1.065)	-3.082*** (1.070)
Observations	2,744	2,744	2,733	2,733
R-squared	0.037	0.166	0.065	0.048
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A75: OLS results of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of land pressures, controlling for a categorical measure of risk aversion, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Pesticides		
Rental Price	0.0150 (0.0123)	0.0127 (0.00755)		
Customary $\times$ Rental Price	0.0122* (0.00616)	-0.00554* (0.00314)		
Inverse Travel Time			0.813*** (0.147)	
Customary $\times$ Inverse Travel Time			0.509** (0.245)	
Decrease in Travel Time				1.970*** (0.531)
Customary $\times$ Decrease in Travel Time				-0.203 (0.528)
Risk Category 2	0.0120 (0.0654)	0.0195 (0.0617)	0.00720 (0.0657)	-0.00587 (0.0620)
Risk Category 3	-0.0110 (0.0451)	0.0225 (0.0478)	0.00115 (0.0464)	-0.0150 (0.0459)
Risk Category 4	0.0318 (0.0662)	0.0335 (0.0636)	0.0495 (0.0671)	0.0307 (0.0615)
Risk Category 5	-0.107*** (0.0261)	-0.0719** (0.0284)	-0.0556** (0.0223)	-0.0790** (0.0359)
Risk Category 6	-0.0107 (0.0432)	-0.0594 (0.0471)	-0.0342 (0.0411)	-0.0490 (0.0416)
Customary	-0.0436 (0.0701)	-0.0587 (0.0707)	0.0426 (0.0667)	0.183*** (0.0497)
Wealth Index	0.00556*** (0.000795)	0.00305*** (0.000966)	0.00411*** (0.000955)	0.00427*** (0.00110)
Landholdings	0.00103 (0.00123)	0.000273 (0.000773)	0.00149 (0.00109)	0.00122 (0.000854)
Cash Crop	0.149*** (0.0230)	0.0953*** (0.0199)	0.136*** (0.0239)	0.118*** (0.0240)
First Year Farming	0.000552 (0.000580)	0.000439 (0.000619)	0.000543 (0.000543)	0.000499 (0.000578)
Constant	-1.129 (1.142)	-0.945 (1.228)	-1.266 (1.060)	-1.172 (1.123)
Observations	2,742	2,742	2,731	2,731
R-squared	0.082	0.215	0.101	0.095
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A76: OLS results of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of land pressures, controlling for a categorical measure of risk aversion, the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

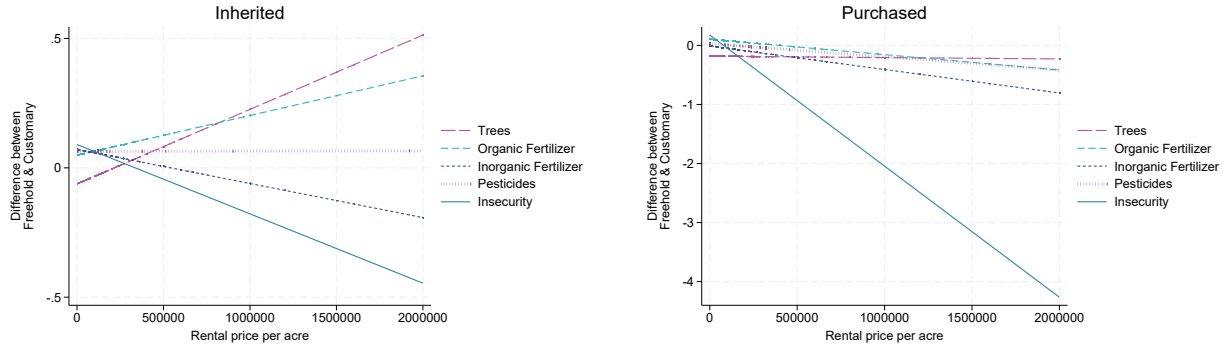


Figure 9:

vignettes analyzed in table A21, which were asked of all respondents about a typical household in their community. Customary land is generally understood to be more likely to be lost, and this differential is greater in high land pressure areas.

C.8 Alternate Samples

C.8.1 Acquisition Mode

My identification strategy rests in part on the exogenous determination of tenure type across parcels, an assumption which has been argued by Deininger and Ali (2008) to hold in the Ugandan context due to historical features. However, this assumption is a strong one, particularly given the fact that freehold parcels are much more likely to have been purchased than customary ones, as seen in table A35. Individual households have much more ability to select the tenure type of parcels they purchase than those they inherit or are allocated. Therefore, I rerun my main analysis separately for parcels purchased (where the exogenous assignment of tenure is less credible) and inherited or allocated (where it is more likely to hold), in figures 9 - 11. Splitting the sample like this makes results quite noisy, although the increasing divergence of tree planting between customary and freehold parcels with rising land pressures holds on both inherited and purchased parcels. This can be seen in the positive slope for trees in all figures. In figure 9, although the negative slope on the difference in insecurity is greater on purchased parcels, tree planting responses to rental prices diverge more strongly on inherited parcels (consistent with my model holding for inherited parcels more than purchased ones). By contrast, tree planting responses to distance to Kampala are stronger on purchased parcels in figure 10 and to the decrease in travel time in figure 11, although the direction is consistent in all figures.

I additionally present regression results for insecurity and tree planting in tables A77 - A78, interacting the coefficients of interest with a dummy variable equal to one for purchased parcels and zero otherwise, to

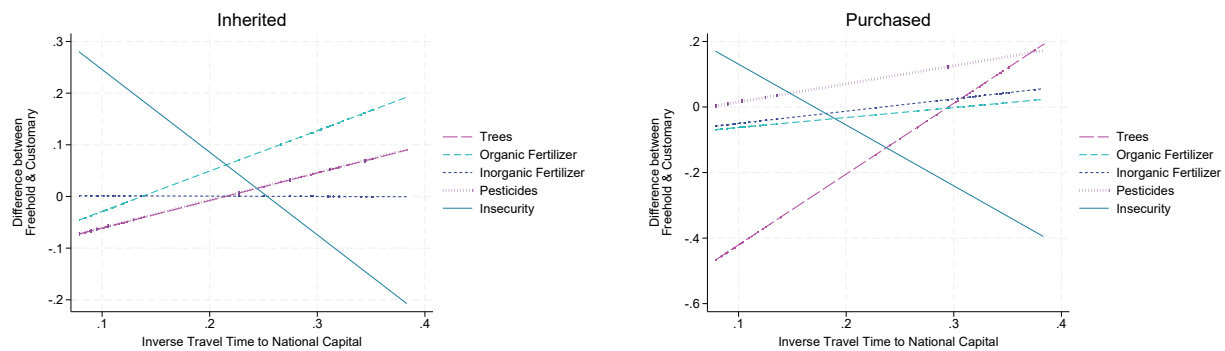


Figure 10:

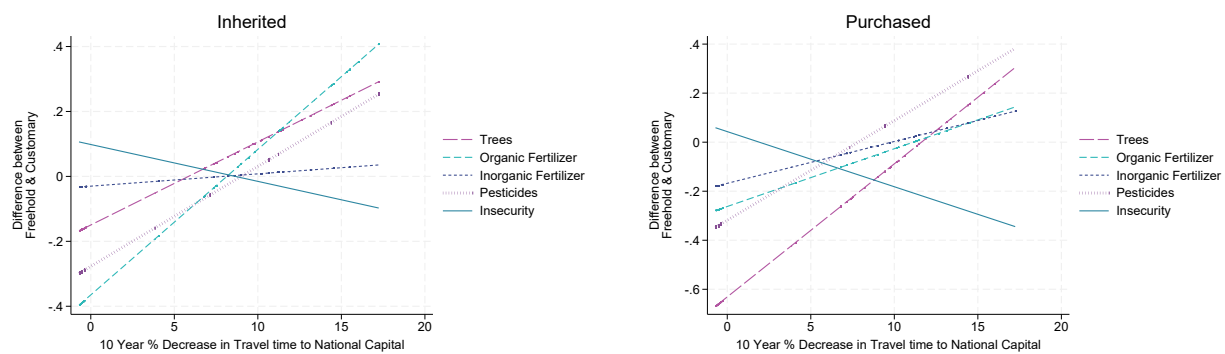


Figure 11:

see if effects of land pressures are significantly different across acquisition types. These results are equally noisy and do not tell a consistent story: impacts consistent with the model are not concentrated entirely among purchased or inherited parcels, but without a strong pattern as to which gradients are stronger among which acquisition mode. Although this is not entirely reassuring about the exogeneity of parcels, the results do not seem to be concentrated entirely among purchased parcels, which are more likely to have their tenure system selected. These results are therefore not particularly illuminating about the credibility of exogenous assignment of tenure across parcels.

Given this limited support for the key assumption that would identify the empirical exercise as causal, the paper's empirical contribution may be more cautiously understood as comparative statics consistent with the predictions of the model.

C.8.2 Excluding Mailo

One notable feature of the land rights environment in Uganda is the presence of *mailo* land in the central region. This legacy of British colonialism allocated square mile tracts to Buganda elites for their ownership as absentee landlords; these landlords in turn sub-leased rights to actual land users (Deininger and Castagnini, 2004). Post-independence Uganda has attempted to balance the tension between these competing rightsholders (Coldham, 2000), identifying elite owners' rights as freehold while also recognizing the rights of 'lawful occupants' of the land who had used it for at least 12 years. These users were granted tenure security, including the right to bequeath their parcels and sub-let with consent of the owner, conditional on continuous possession and payment of a nominal rent to the owner. However, mailo tenants still face substantial insecurity, with particular concern about expropriation by owners.

Only one district in my sample contains a substantial share (20%) of mailo parcels. I categorized mailo parcels reported by tenants as customary land, given the concerns about elite expropriation that parallel those of customary rightsholders. However, there may be some concern that this categorization is incorrect. I therefore check that mailo parcels are not driving my main results by excluding these parcels entirely for tables A82 - A86. Results are remarkably similar to those in the main text, both in magnitude and statistical significance.

VARIABLES	(1)	(2)	(3)	(4)
		Average Insecurity		
Rental Price	0.106*** (0.0137)	0.128*** (0.0248)		
Purchased × Rental Price	0.567*** (0.173)	2.263*** (0.570)		
Customary × Rental Price	-0.0850*** (0.0279)	-0.105*** (0.0238)		
Purchased × Customary × Rental Price	0.986* (0.550)	-0.426 (1.162)		
Inverse Travel Time			0.612 (0.504)	
Purchased × Inverse Travel Time			4.052*** (1.016)	
Customary × Inverse Travel Time			-3.617*** (1.036)	
Purchased × Customary × Inverse Travel Time			0.148 (0.397)	
Decrease in Travel Time				-0.476 (1.166)
Purchased × Decrease in Travel Time				3.262 (2.558)
Customary × Decrease in Travel Time				-3.949 (2.596)
Purchased × Decrease in Travel Time				0.148 (1.074)
Customary	0.0972 (0.0793)	0.0407 (0.0813)	-0.956*** (0.322)	-0.327 (0.287)
Wealth Index	-0.00241 (0.00253)	-0.00159 (0.00269)	-0.000882 (0.00259)	3.06e-05 (0.00243)
Landholdings	-0.00424 (0.00287)	-0.00472 (0.00413)	-0.00536* (0.00295)	-0.00583* (0.00298)
Cash Crop	0.0480 (0.0706)	-0.00126 (0.0719)	0.0390 (0.0639)	0.0816 (0.0777)
First Year Farming	0.00152** (0.000641)	0.00145 (0.000924)	0.00122* (0.000691)	0.00139** (0.000660)
Purchased	-0.427** (0.182)	-0.345* (0.182)	-1.351*** (0.337)	-0.761** (0.338)
Constant	-1.361 (1.265)	-1.497 (1.866)	-0.0216 (1.424)	-0.810 (1.297)
Observations	3,240	3,240	3,222	3,222
R-squared	0.262	0.386	0.273	0.256
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A77: OLS results of average tenure insecurity across all rightsholders on a parcel on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Purchased is a dummy equal to one for parcels acquired through purchase and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Trees		
Rental Price	-0.0425*** (0.0114)	-0.0337*** (0.00688)		
Purchased × Rental Price	1.786*** (0.386)	1.023*** (0.286)		
Customary × Rental Price	0.0539*** (0.0109)	0.0412*** (0.00890)		
Purchased × Customary × Rental Price	-3.174*** (0.559)	-0.522 (0.550)		
Inverse Travel Time			-0.0339 (0.433)	
Purchased × Inverse Travel Time			1.012* (0.547)	
Customary × Inverse Travel Time			0.438 (0.403)	
Purchased × Customary × Inverse Travel Time			-0.105 (0.299)	
Decrease in Travel Time				-1.847** (0.783)
Purchased × Decrease in Travel Time				0.238 (1.187)
Customary × Decrease in Travel Time				1.879** (0.881)
Purchased × Customary × Decrease in Travel Time				-0.0191 (0.846)
Customary	0.00333 (0.0479)	0.00603 (0.0532)	0.147 (0.120)	0.203** (0.0863)
Wealth Index	-0.00214 (0.00140)	0.00168** (0.000783)	-0.00245* (0.00130)	-0.00115 (0.00118)
Landholdings	0.000982 (0.00176)	0.00187 (0.00149)	0.000750 (0.00158)	0.000124 (0.00172)
Cash Crop	0.112*** (0.0277)	0.126*** (0.0376)	0.123*** (0.0325)	0.124*** (0.0385)
First Year Farming	-0.00290*** (0.000785)	-0.00315*** (0.000992)	-0.00290*** (0.000804)	-0.00282*** (0.000794)
Purchased	0.210*** (0.0671)	0.180*** (0.0443)	-0.139 (0.124)	-0.0718 (0.117)
Constant	6.354*** (1.586)	6.365*** (2.005)	6.266*** (1.590)	6.140*** (1.588)
Observations	3,250	3,250	3,232	3,232
R-squared	0.088	0.246	0.077	0.069
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A78: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Purchased is a dummy equal to one for parcels acquired through purchase and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Organic Fertilizer		
Rental Price	0.0137** (0.00510)	0.00356 (0.00216)		
Purchased × Rental Price	0.784*** (0.209)	0.129 (0.102)		
Customary × Rental Price	-0.0153*** (0.00539)	-0.00487** (0.00229)		
Purchased × Customary × Rental Price	-0.325 (0.225)	-0.0980 (0.214)		
Inverse Travel Time			0.263** (0.103)	
Purchased × Inverse Travel Time			0.693*** (0.174)	
Customary × Inverse Travel Time			0.0731 (0.153)	
Purchased × Customary × Inverse Travel Time			0.182* (0.101)	
Decrease in Travel Time				1.022*** (0.316)
Purchased × Decrease in Travel Time				2.019*** (0.575)
Customary × Decrease in Travel Time				1.927*** (0.505)
Purchased × Customary × Decrease in Travel Time				0.607** (0.254)
Customary	-0.0766** (0.0285)	-0.0205 (0.0205)	-0.0659** (0.0308)	0.116** (0.0470)
Wealth Index	0.00372*** (0.000594)	0.000916 (0.000584)	0.00324*** (0.000658)	0.00112* (0.000597)
Landholdings	0.00129 (0.000961)	0.00132 (0.000956)	0.00180 (0.00118)	0.00246*** (0.000867)
Cash Crop	0.193*** (0.0311)	0.0364 (0.0254)	0.187*** (0.0306)	0.128*** (0.0254)
First Year Farming	-1.80e-05 (0.000508)	-0.000104 (0.000472)	-0.000125 (0.000517)	-2.43e-06 (0.000480)
Purchased	-0.00246 (0.0298)	-0.0253 (0.0397)	-0.162*** (0.0484)	-0.148** (0.0676)
Constant	0.0296 (1.038)	0.184 (0.975)	0.187 (1.035)	-0.175 (0.952)
Observations	3,248	3,248	3,230	3,230
R-squared	0.180	0.322	0.178	0.227
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A79: OLS results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Purchased is a dummy equal to one for parcels acquired through purchase and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Inorganic Fertilizer		
Rental Price	0.00902 (0.00875)	0.00415 (0.00259)		
Purchased $\times$ Rental Price	0.711*** (0.107)	-0.0120 (0.108)		
Customary $\times$ Rental Price	-0.00626 (0.00871)	0.000837 (0.00284)		
Purchased $\times$ Customary $\times$ Rental Price	-0.487 (0.344)	0.0531 (0.439)		
Inverse Travel Time			0.607*** (0.107)	
Purchased $\times$ Inverse Travel Time			0.550* (0.318)	
Customary $\times$ Inverse Travel Time			0.176 (0.199)	
Purchased $\times$ Customary $\times$ Inverse Travel Time			-0.185 (0.172)	
Decrease in Travel Time				0.802** (0.304)
Purchased $\times$ Decrease in Travel Time				-0.0493 (0.670)
Customary $\times$ Decrease in Travel Time				0.989** (0.483)
Purchased $\times$ Customary $\times$ Decrease in Travel Time				-0.515 (0.412)
Customary	-0.00112 (0.0375)	0.00515 (0.0409)	0.0713 (0.0471)	0.133*** (0.0443)
Wealth Index	0.00433*** (0.000784)	0.00203** (0.000884)	0.00324*** (0.000735)	0.00322*** (0.000866)
Landholdings	0.00193** (0.000808)	0.00147 (0.000884)	0.00276*** (0.000884)	0.00266*** (0.000928)
Cash Crop	0.00206 (0.0142)	0.0143 (0.0176)	0.00627 (0.0135)	-0.00456 (0.0159)
First Year Farming	0.00129*** (0.000470)	0.000900* (0.000534)	0.00121** (0.000455)	0.00130*** (0.000449)
Purchased	0.0293 (0.0772)	0.0164 (0.0833)	0.0649 (0.133)	0.135 (0.152)
Constant	-2.655*** (0.947)	-1.920* (1.074)	-2.703*** (0.935)	-2.853*** (0.914)
Observations	3,245	3,245	3,227	3,227
R-squared	0.037	0.150	0.058	0.043
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A80: OLS results of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Purchased is a dummy equal to one for parcels acquired through purchase and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Used Pesticides		
Rental Price	0.0118 (0.00837)	-0.0114*** (0.00390)		
Purchased × Rental Price	1.004*** (0.174)	0.237 (0.241)		
Customary × Rental Price	-0.0140 (0.00990)	0.00911* (0.00503)		
Purchased × Customary × Rental Price	-0.913 (0.724)	-0.443 (0.749)		
Inverse Travel Time			0.610** (0.230)	
Purchased × Inverse Travel Time			0.905*** (0.254)	
Customary × Inverse Travel Time			0.121 (0.206)	
Purchased × Customary × Inverse Travel Time			-0.432* (0.251)	
Decrease in Travel Time				0.141 (0.487)
Purchased × Decrease in Travel Time				0.935 (0.676)
Customary × Decrease in Travel Time				1.632** (0.644)
Purchased × Customary × Decrease in Travel Time				-1.160** (0.521)
Customary	-0.0451 (0.0686)	-0.0481 (0.0692)	0.0333 (0.0591)	0.170*** (0.0569)
Wealth Index	0.00557*** (0.000706)	0.00335*** (0.000990)	0.00432*** (0.000886)	0.00435*** (0.000985)
Landholdings	0.00186 (0.00121)	0.000757 (0.000974)	0.00199 (0.00122)	0.00183* (0.00100)
Cash Crop	0.127*** (0.0229)	0.0867*** (0.0205)	0.133*** (0.0292)	0.108*** (0.0291)
First Year Farming	0.000615 (0.000452)	0.000561 (0.000548)	0.000483 (0.000469)	0.000595 (0.000479)
Purchased	-0.0738 (0.0457)	-0.0814** (0.0357)	-0.146* (0.0808)	-0.146* (0.0824)
Constant	-1.200 (0.891)	-1.192 (1.093)	-1.114 (0.905)	-1.322 (0.912)
Observations	3,244	3,244	3,226	3,226
R-squared	0.092	0.203	0.106	0.099
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A81: OLS results of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Purchased is a dummy equal to one for parcels acquired through purchase and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time.

VARIABLES	(1)	(2)	(3)	(4)
		Average Insecurity		
Rental Price	0.0213 (0.0216)	0.0247 (0.0156)		
Customary $\times$ Rental Price	0.105*** (0.0136)	0.119*** (0.0189)		
Inverse Travel Time			-1.383*** (0.489)	
Customary $\times$ Inverse Travel Time			0.780 (0.501)	
Decrease in Travel Time				-2.991** (1.416)
Customary $\times$ Decrease in Travel Time				-0.215 (1.167)
Customary	0.131* (0.0662)	0.105 (0.0630)	-0.469** (0.224)	-0.148 (0.204)
Wealth Index	-0.00222 (0.00248)	-0.00162 (0.00263)	-0.00152 (0.00260)	-0.000196 (0.00237)
Landholdings	-0.00467* (0.00276)	-0.00561 (0.00368)	-0.00572** (0.00249)	-0.00627** (0.00270)
Cash Crop	0.0411 (0.0802)	0.0190 (0.0813)	0.0608 (0.0862)	0.0935 (0.0971)
First Year Farming	0.00162** (0.000726)	0.000823 (0.000837)	0.00154* (0.000765)	0.00144* (0.000744)
Observations	3,118	3,118	3,100	3,100
R-squared	0.272	0.391	0.269	0.263
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A82: OLS results of average tenure insecurity across all rightsholders on a parcel on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time. Parcels held under mailo are excluded.

VARIABLES	(1)	(2)	(3)	(4)
			Trees	
Rental Price	0.0126 (0.00813)	0.00833 (0.00957)		
Customary $\times$ Rental Price	-0.0436*** (0.0114)	-0.0378*** (0.00845)		
Inverse Travel Time			0.922** (0.384)	
Customary $\times$ Inverse Travel Time			0.0116 (0.428)	
Decrease in Travel Time				1.090 (0.870)
Customary $\times$ Decrease in Travel Time				-1.698** (0.832)
Customary	0.0324 (0.0412)	0.0260 (0.0489)	0.286** (0.107)	0.310*** (0.0900)
Wealth Index	-0.00163 (0.00153)	0.00188** (0.000789)	-0.00269** (0.00131)	-0.00142 (0.00119)
Landholdings	8.41e-05 (0.00157)	0.00127 (0.00144)	0.000528 (0.00151)	-0.000163 (0.00166)
Cash Crops	0.152*** (0.0376)	0.145*** (0.0436)	0.138*** (0.0340)	0.142*** (0.0388)
First Year Farming	-0.00284*** (0.000762)	-0.00337*** (0.00108)	-0.00284*** (0.000792)	-0.00287*** (0.000747)
Observations	3,129	3,129	3,111	3,111
R-squared	0.055	0.242	0.067	0.061
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A83: OLS results of a dummy equal to one for parcels with trees planted on them (a long-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time. Parcels held under mailo are excluded.

VARIABLES	(1)	(2)	(3)	(4)
		Used Organic Fertilizer		
Rental Price	-0.000977 (0.00127)	-0.00107 (0.00108)		
Customary $\times$ Rental Price	0.0130*** (0.00477)	0.00316* (0.00186)		
Inverse Travel Time			0.509** (0.196)	
Customary $\times$ Inverse Travel Time			0.276** (0.105)	
Decrease in Travel Time				3.337*** (0.504)
Customary $\times$ Decrease in Travel Time				1.128*** (0.292)
Customary	-0.0550** (0.0256)	-0.0220 (0.0240)	0.00757 (0.0386)	0.175*** (0.0464)
Wealth Index	0.00408*** (0.000724)	0.00102* (0.000583)	0.00334*** (0.000754)	0.00120* (0.000652)
Landholdings	0.000999 (0.00113)	0.00115 (0.000929)	0.00174 (0.00128)	0.00238** (0.000999)
Cash Crops	0.204*** (0.0249)	0.0287 (0.0199)	0.191*** (0.0237)	0.137*** (0.0224)
First Year Farming	1.89e-05 (0.000558)	6.28e-05 (0.000458)	-8.26e-05 (0.000567)	3.16e-06 (0.000522)
Observations	3,126	3,126	3,108	3,108
R-squared	0.162	0.319	0.168	0.218
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A84: OLS results of a dummy equal to one for parcels with organic fertilizer applied (a medium-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time. Parcels held under mailo are excluded.

VARIABLES	(1)	(2)	(3)	(4)
		Used Inorganic Fertilizer		
Rental Price	0.00320 (0.00272)	0.00487** (0.00186)		
Customary $\times$ Rental Price	0.00899 (0.00874)	0.00432 (0.00287)		
Inverse Travel Time			0.756*** (0.0776)	
Customary $\times$ Inverse Travel Time			0.592*** (0.105)	
Decrease in Travel Time				1.349*** (0.464)
Customary $\times$ Decrease in Travel Time				0.652* (0.324)
Customary	0.0122 (0.0341)	0.0102 (0.0356)	0.0560 (0.0405)	0.0840** (0.0318)
Wealth Index	0.00464*** (0.000734)	0.00223** (0.000845)	0.00330*** (0.000711)	0.00343*** (0.000858)
Landholdings	0.00160* (0.000815)	0.00140 (0.000863)	0.00273*** (0.000907)	0.00261*** (0.000951)
Cash Crops	0.0140 (0.0138)	0.0232 (0.0162)	0.00230 (0.0123)	-0.0121 (0.0150)
First Year Farming	0.00142** (0.000544)	0.00105* (0.000619)	0.00135** (0.000553)	0.00148*** (0.000541)
Observations	3,123	3,123	3,105	3,105
R-squared	0.030	0.162	0.061	0.042
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A85: OLS results of a dummy equal to one for parcels with inorganic fertilizer applied (a short-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time. Parcels held under mailo are excluded.

VARIABLES	(1)	(2)	(3)	(4)
		Used Pesticides		
Rental Price	-0.00146 (0.00334)	-0.00201 (0.00254)		
Customary $\times$ Rental Price	0.0107 (0.00813)	-0.0132*** (0.00286)		
Inverse Travel Time			0.838*** (0.119)	
Customary $\times$ Inverse Travel Time			0.591*** (0.216)	
Decrease in Travel Time				2.115*** (0.480)
Customary $\times$ Decrease in Travel Time				0.0240 (0.485)
Customary	-0.0226 (0.0626)	-0.0266 (0.0557)	0.0465 (0.0489)	0.193*** (0.0439)
Wealth Index	0.00569*** (0.000608)	0.00311*** (0.000889)	0.00404*** (0.000693)	0.00407*** (0.000810)
Landholdings	0.00117 (0.00123)	0.000290 (0.000834)	0.00172 (0.00109)	0.00149* (0.000870)
Cash Crops	0.131*** (0.0190)	0.0812*** (0.0194)	0.117*** (0.0194)	0.0941*** (0.0195)
First Year Farming	0.000821** (0.000392)	0.000749 (0.000527)	0.000738* (0.000382)	0.000818** (0.000382)
Observations	3,122	3,122	3,104	3,104
R-squared	0.069	0.200	0.094	0.086
Fixed Effect	None	Group	None	None
Cluster	Parish	Parish	Parish	Parish

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A86: OLS results of a dummy equal to one for parcels with pesticides applied (a short-term investment) on measures of land pressures, controlling for the Progress out of Poverty Index wealth index, total landholdings of the household in acres, a dummy equal to one if a cash crop is the primary crop on any plot, the first year the parcel was farmed, and dummies for the method of parcel acquisition. Farmer Group fixed effects included in column (2), and standard errors are clustered at the parish level in all specifications (reported in parentheses). Customary is a dummy equal to one for customary parcels and zero otherwise. Difference in number of observations between columns (1) and (2) vs. (3) and (4) is due to missing GPS locations for some households, so they could not be matched to spatial measures of travel time. Parcels held under mailo are excluded.